Optimization Techniques Laboratory

LAB ASSIGNMENT 7
MULTIOBJECTIVE OPTIMIZATION

Student detail	
Name	Sofia Abidi
Registration number	241071072
Roll suffix	72
Course	B.Tech Computer Engineering
Batch	D

AIM:
To implement, verify and compare classical, evolutionary, swarm, indicator, preference and gradient-based multiobjective optimization methods on the specified P1-P14 problem bank.

OBJECTIVES:
Implement all 37 assigned mechanisms; verify shared numerical utilities and constraints; study A35 R-NSGA-II, A10 VEGA and A22 MOPSO in depth; compare methods with reproducible metrics; and document suitable applications and limitations.

GitHub repository: https://github.com/sofiaabidi/OT_LAB07

This report presents tested Python implementations of all 37 catalogue mechanisms, their measured tradeoffs on the assignment problems, and the three assigned study cards. It includes Tasks A to F, raw-result references and reproducible commands. The main finding is that selection geometry, constraints and preferences determine useful choices: no algorithm leads across every problem and metric.

Ten-trial stochastic studies use a 10,000-evaluation cap, with P9 feature-selection experiments explicitly reduced to 200. Classical scalar solvers often consume fewer calls. Measured quality is separated from theoretical mechanism claims; some advantages and disadvantages needing direct experimental support still need targeted ablations.

Contents

1 Overview and completion status 2
2 Methodology and reproducibility 2
3 Problem formulations 3
4 Shared tools and verification 10
5 Tasks A and B algorithm implementations and comparison 10
6 Task C assigned study cards and hand traces 40
35. R-NSGA-II — Reference Point-Based NSGA-II 40
10. VEGA — Vector Evaluated Genetic Algorithm 41
22. MOPSO — Multi-Objective Particle Swarm Optimization 42
7 Task D head to head comparisons 51
8 Task E algorithm selection cases 59
9 Task F viva preparation 60
10 Findings and limitations 62
11 Reproducibility and repository instructions 63
12 References 63
13 Appendix: Task B master comparison 64

1 Overview and completion status

The project has 37 separately named algorithm modules, 37 defining-mechanism tests and 62 passing automated tests in total. The main manifest contains 1196 optimizer experiment rows and 60 additional network learning curves. Every front figure is drawn from persisted arrays of decisions, objectives and violations, not mock data.

Roll assignment calculation for R=72: ((72-1) mod 37)+1=35; ((72+11) mod 37)+1=10; ((72+23) mod 37)+1=22.

2 Methodology and reproducibility

Implementations use NumPy and the standard library. Matplotlib creates plots. SciPy SLSQP is used only in algorithms 1 to 7 for permitted scalar subproblems; scalarization, reference/epsilon sweeps, deviation elimination and priorities are handwritten. Scikit-learn loads only the Wisconsin dataset; five-neighbour classification and five-fold evaluation are handwritten. Neural-network forward and backward passes use NumPy without autograd.

Stochastic optimizer seeds are 720 to 729, ten independent streams. Benchmark construction and Monte Carlo metric seeds are separate fixed configuration values. Population and output archive cap are 40 except methods whose archive naturally shrinks. Candidate evaluation budgets include local-search proposals. Objective-vector evaluation counts one candidate irrespective of objective count. Finite-difference solver probes count in classical methods, including objective evaluations invoked by constraints. Reported runtime is wall-clock and should not be treated as a machine-independent algorithm complexity.

All natural maximization objectives are negated before dominance. Feasible candidates beat infeasible ones; smaller normalized positive constraint residual wins among infeasible

candidates. Feasibility threshold is 1e-10, with a 1e-8 numerical verification tolerance on scalar-solver outputs. This prevents roundoff from being confused with a materially infeasible design.

Quality normalization uses fixed problem constants listed in Section 3. Two-objective HV is an exact staircase area. Higher-dimensional HV uses 8192 identical uniform samples per comparison, seed 991, in [0,1.1]^M. Its absolute Monte Carlo uncertainty is bounded by box_volume/(2*sqrt(8192)), about 0.00735 for M=3 and 0.00890 for M=5; differences smaller than that require cautious interpretation. SD across ten optimizer seeds is distinct from HV sampling error. IGD uses 301 exact-front samples where an analytic front is available; no true-front IGD is invented for P4 P8 or P9. Spacing is the sample SD of nearest L1 distances; a singleton scores zero without offering coverage.

Early nonelitist algorithms use a passive recording archive for comparable outputs. The final breeding population is saved separately in NPZ files. Thus a good recorded front does not prove that VEGA MOGA NSGA or NPGA preserved its parents. Classical algorithms use a common cap but naturally different consumed counts, so these are equal-cap rather than equal-consumed-budget studies.

The five supplied lecture decks were checked against the code. The teaching NPGA, MOSA, IBEA, RVEA and HypE conventions are stated in the appendix. Algorithms changed during that check were rerun; current figures and tables use refreshed rows.

3 Problem formulations

P1 Factory speed

Decision variables and encoding x in [1,10], one real variable

$$min\, F(x) \;=\; [(x \;-\; 3)^2,\; (x \;-\; 7)^2]$$

Constraints and handling Bounds only; Pareto x in [3,7].

Encoding: one-element real vector. Equal weight solution x=7-4w. Convex front; uniform weight steps are uniform x steps, not uniform arc lengths.

Numerical sanity check x=3 gives (f1,f2)=(0,16), while x=7 gives (16,0); both are Pareto endpoints.

Normalization: (F − ideal)/(nadir − ideal); ideal = [0.0, 0.0]; nadir = [16.0, 16.0]; HV reference = [1.1, 1.1] (normalized).

Termination Classical scalar sweeps stop at solver tolerance under the 10,000-call cap; stochastic methods use at most 10,000 evaluations.

Evaluated methods: Benson reference improvement, Weight Metric, Weighted Sum. See the corresponding Task A pages and raw files.

P2 Fleet size

Decision variables and encoding x in [1,10], real or integer

$$min\, F(x) \;=\; [100\,/\,x,\; x^2]$$

Constraints and handling Bounds; every x is Pareto optimal.

Integer experiment has at most ten distinct objective points. VEGA runs and exact integer problem definitions retain integer repair.

Numerical sanity check x=1 gives (100,1) and x=10 gives (10,100), confirming the opposing monotone objectives.

Normalization: (F − ideal)/(nadir − ideal); ideal = [10.0, 1.0]; nadir = [100.0, 100.0]; HV reference = [1.1, 1.1] (normalized).

Termination Scalar or stochastic runs use the shared 10,000-evaluation cap; integer repair is applied after variation.

Evaluated methods: Epsilon-Constraint, VEGA. See the corresponding Task A pages and raw files.

P3 Specified ZDT2

Decision variables and encoding 10 real variables in [0,1]

$$g(x) = 1 + x_2 + x_3 + \cdots + x_{10}$$

$$min\, F(x) = [x_1,\ g(x)\,(1 - (x_1 / g(x))^2)]$$

Constraints and handling Bounds; exact front [t,1-t^2].

The factor matches the assignment at n=10; no alternate benchmark is substituted. Weighted Sum cannot select the unsupported interior with a global scalar solve.

Numerical sanity check With x2..x10=0, g=1; x1=0.5 gives (0.5,0.75) on f2=1-f1^2.

Normalization: (F − ideal)/(nadir − ideal); ideal = [0.0, 0.0]; nadir = [1.0, 1.0]; HV reference = [1.1, 1.1] (normalized).

Termination Task A and Task D stochastic comparisons stop at 10,000 evaluations; scalar sweeps use the same cap.

Evaluated methods: Epsilon-Constraint, IBEA, MO-TLBO, MOABC, MOBA, MODE, MOEA/D, MOEA/D-DE, MOGA, MOSA, NPGA, NSGA-II, PESA-II, SMS-EMOA, VEGA, Weight Metric, Weighted Sum. See the corresponding Task A pages and raw files.

P4 Portfolio

Decision variables and encoding four nonnegative real weights

$$min\, F(w) = [w^{\mathrm{T}}\Sigma w,\ -\mu^{\mathrm{T}}w]$$

Constraints and handling sum(w)=1. Repair positive entries by sum; scalar solvers use equality.

mu=(.08,.12,.15,.10). Sigma rows (.010,.002,.001,.003),(.002,.040,.006,.004),(.001,.006,.090,.005),(.003,.004,.005,.020). Scaling risk lower bound 0 is conservative, not true attainable ideal. Repair/penalty investigation is a random-design ablation.

Numerical sanity check w=(0.25,0.25,0.25,0.25) sums to one, has risk 0.012625 and expected return 0.1125.

Normalization: (F − ideal)/(nadir − ideal); ideal = [0.0, -0.15]; nadir = [0.09, -0.08]; HV reference = [1.1, 1.1] (normalized).

Termination Portfolio optimizers stop at 10,000 evaluations; scalar solvers stop at tolerance under that cap.

Evaluated methods: MOBA, MOPSO, SPEA2, Value Function. See the corresponding Task A pages and raw files.

P5 Cantilever beam

Decision variables and encoding b in [.01,.10]m, h in [.02,.20]m

$$min\ F(b,h) \ = \ [7850bh,\ 4000\ /\ (200 \times 10^9\ bh^3)]$$

Constraints and handling CV=max((2.4e-5-b*h^2)/2.4e-5,0).

P=1000N,L=1m,E=200GPa,rho=7850kg/m^3. Feasibility first. For given area A=b*h, maximize h=min(.2,A/.01) to derive true tradeoff; smallest feasible area is .01*sqrt(2.4e-5/.01).

Numerical sanity check At (b,h)=(0.10,0.02), b*h^2=4e-5 exceeds 2.4e-5, mass=15.7 kg and deflection=0.025.

Normalization: (F − ideal)/(nadir − ideal); ideal = [0.0, 0.0]; nadir = [157.0, 0.25]; HV reference = [1.1, 1.1] (normalized).

Termination Constrained stochastic runs stop at 10,000 evaluations with Deb feasibility applied at every comparison.

Evaluated methods: Constrained NSGA-II, MO-TLBO, MODE, MOEA/D-DE, MOGA, MOGWO, MOMA, MOPSO, MOSA, NPGA, NSGA, NSGA-II, PESA-II, R-NSGA-II, SPEA2. See the corresponding Task A pages and raw files.

P6 Binh-Korn

Decision variables and encoding x1 in [0,5], x2 in [0,3]

$$min\ F(x_1,x_2) \ = \ [4({x_1}^2 \ + \ {x_2}^2),\ (x_1 \ - \ 5)^2 \ + \ (x_2 \ - \ 5)^2]$$

Constraints and handling (x1-5)^2+x2^2<=25; (x1-8)^2+(x2+3)^2>=7.7.

Residuals divided by 25 and 7.7 before summing positive parts. Second constraint inactive: minimum 18. Front x1=x2 from 0 to 3, then x2=3 and x1 from 3 to 5.

Numerical sanity check A=(0,0) gives (0,50), B=(1,1) gives (8,32), and both are feasible; the report retains the full four-point Deb-rule calculation.

Normalization: (F − ideal)/(nadir − ideal); ideal = [0.0, 4.0]; nadir = [136.0, 50.0]; HV reference = [1.1, 1.1] (normalized).

Termination Stochastic runs stop at 10,000 evaluations; feasibility is checked before objective dominance.

Evaluated methods: Constrained NSGA-II, MOABC. See the corresponding Task A pages and raw files.

Point	Decision	f1	f2	Circle lhs	CV
A	(0,0)	0	50	25	0
B	(1,1)	8	32	17	0
C	(0,3)	36	29	34	0.36
D	(.2,3)	36.16	27.04	32.04	0.2816

Manual Deb comparison: A and B are feasible and trade off objectives; both outrank C and D. D outranks C because .2816 < .36. The layers are [A,B], [D], [C]. The second constraint is inactive for all four points; infeasible objective values do not override violation order.

P7 Water allocation

Decision variables and encoding x1 agriculture [10,70], x2 industry [20,80]

$$min\ F(x_1, x_2) \ = \ [-\ 0.8x_1, \ -\ 1.2x_2]$$

Constraints and handling x1+x2<=90; CV=max((sum(x)-90)/90,0).

Natural maximized outputs preserve interpretation after sign conversion. Exact front sum=90. Output reference (40,50) requires 91.6667 units, so it is unattainable exactly.

Numerical sanity check (x1,x2)=(40,50) exactly uses the capacity 90 and gives minimized objectives (-32,-60).

Normalization: (F − ideal)/(nadir − ideal); ideal = [-56.0, -96.0]; nadir = [-8.0, -24.0]; HV reference = [1.1, 1.1] (normalized).

Termination Scalar and stochastic runs use the shared 10,000-evaluation cap.

Evaluated methods: Benson reference improvement, Goal Programming, Lexicographic, NSGA. See the corresponding Task A pages and raw files.

P8 Delivery routing

Decision variables and encoding permutation of 15 city indices, closed tour

$$min\ F(\pi) \ = \ [\sum_{(i,\ j)\in tour} d_{i,\ j},\ \sum_{(i,\ j)\in tour} t_{i,\ j}]$$

Constraints and handling Each city once. Invalid permutations CV=1.

Legacy RandomState(7), coordinates uniform [0,100]^2; independent upper-triangle symmetric traffic multipliers uniform [.5,2]. Deposits include closing edge. OX, swap and inversion remain valid tours. The wording 15 customers/depot is resolved as the specified 15-city permutation.

Numerical sanity check Every stored decision is checked to contain each city index 0..14 exactly once; nearest-neighbour distance and time baselines are reported separately.

Normalization: (F − ideal)/(nadir − ideal); ideal = [0.0, 0.0]; nadir = [2200.0, 3300.0]; HV reference = [1.1, 1.1] (normalized).

Termination Routing methods stop at 10,000 evaluated tours, always after permutation repair or construction.

Evaluated methods: MOACO, MOEA/D-ACO. See the corresponding Task A pages and raw files.

City	x	y
0	7.631	77.99
1	43.84	72.35
2	97.8	53.85
3	50.11	7.205
4	26.84	49.99
5	67.92	80.37
6	38.09	6.594

City	x	y
7	28.81	90.96
8	21.34	45.21
9	93.12	2.49
10	60.05	95.01
11	23.03	54.85
12	90.91	13.32
13	52.34	75.04
14	66.9	46.78

P9 Feature selection

Decision variables and encoding 30-bit mask with at least one selected feature

$$min\ F(m) \ = \ [1 \ - \ accuracy_5 NN(m),\ \sum_i m_i]$$

Constraints and handling Empty mask CV=1; repair selects feature 0.

569 Wisconsin observations from sklearn dataset loader only. Fixed shuffled five folds seed 63; fold-local standardization avoids leakage. Squared distances and majority of five neighbours coded with NumPy. 200 candidate evaluations, ten trials: exploratory, full 10,000 budget incomplete.

Numerical sanity check The all-features five-fold 5NN baseline error is 0.04042; an empty mask has positive violation and is repaired before optimization.

Normalization: (F − ideal)/(nadir − ideal); ideal = [0.0, 1.0]; nadir = [0.2, 30.0]; HV reference = [1.1, 1.1] (normalized).

Termination Feature-selection runs intentionally stop at 200 candidate evaluations; the suggested 10,000 budget remains incomplete.

Evaluated methods: MOGWO, MOMA. See the corresponding Task A pages and raw files.

Most frequent features in the pooled retained feature-selection fronts are listed below. Frequencies pool the two methods and ten seeds; the reduced budget does not establish globally optimal masks.

Feature	Front frequency
mean smoothness	0.7143
mean concave points	0.7143
fractal dimension error	0.7143
worst radius	0.7143
worst area	0.7143

P10 DTLZ2

Decision variables and encoding n=M+9 variables in [0,1], M=3 or 5

$$g(x) = (x_{\square} - 0.5)^2 + \cdots + (x_{\square} - 0.5)^2$$

$$f_1 = (1 + g)\, cos(\pi x_1/2) \cdots cos(\pi x_{(}M - 1_{)}/2)$$

$$f_{\square} = (1 + g)\,[product\ of\ the\ first\ M - m\ cosine\ terms] \times sin(\pi x_{(}M - m + 1_{)}/2)$$

For m = 2,…,M, the product covers i=1,…,M−m. The variable index in the final sine is M−m+1.

Constraints and handling Bounds; at g=0 sum(f_j^2)=1.

Python zero-based tail starts at M-1. f1 uses cos indices 0..M-2; later objectives shorten product and use sine at M-j-1. IGD reference: 301 fixed absolute Gaussian unit directions seed 100+M, an exact-front sample with finite sampling resolution.

Numerical sanity check When every tail variable is 0.5, g=0 and the implementation test confirms sum(f_j^2)=1.

Normalization: (F − ideal)/(nadir − ideal); ideal = [0.0, 0.0, 0.0, 0.0, 0.0]; nadir = [1.0, 1.0, 1.0, 1.0, 1.0]; HV reference = [1.1, 1.1, 1.1, 1.1, 1.1] (normalized).

Termination Three- and five-objective runs stop at 10,000 evaluations.

Evaluated methods: HypE, MOEA/D, NSGA-III, RVEA. See the corresponding Task A pages and raw files.

P11 Production planning

Decision variables and encoding xA,xB>=0; finite bounds [0,400]^2

$$min\ F(x_{\alpha}, x_{\beta}) = [-4x_{\alpha} - 6x_{\beta},\ -x_{\alpha} - x_{\beta}]$$

Constraints and handling 2*xA+3*xB<=1200; 4*xA+2*xB<=1600.

Corners (0,0),(400,0),(300,200),(0,400). Profit=2400 ties on an edge; units priority selects (300,200), units=500. Goals profit>=2400,units>=600 leave at least 100 units shortfall. Unit and normalized deviations both solved.

Numerical sanity check (xA,xB)=(300,200) makes both resource left sides 1200 and 1600, with profit 2400 and 500 total units.

Normalization: (F − ideal)/(nadir − ideal); ideal = [-2400.0, -500.0]; nadir = [0.0, 0.0]; HV reference = [1.1, 1.1] (normalized).

Termination Goal and lexicographic scalar solves terminate at solver tolerance under the shared cap.

Evaluated methods: Goal Programming, Lexicographic. See the corresponding Task A pages and raw files.

P12 Supplier ranking

Decision variables and encoding index 0..4 or supplied finite alternatives table

$$min\ F(supplier) = [cost,\ -quality,\ delivery,\ -reliability]$$

Constraints and handling Five supplied alternatives; no manufactured supplier data.

S1=(120,8,10,92),S2=(100,7,14,88),S3=(140,9,7,95),S4=(90,6,18,80),S5=(130,8,9,90). Weights (.3,.3,.2,.2); shift cost +.1 and quality -.1 to preserve sum. TOPSIS vector scales; VIKOR range scales.

Numerical sanity check Supplier S1 is encoded as (120,-8,10,-92) in the common minimization convention; the supplied five-row table is not synthesized.

Normalization: (F − ideal)/(nadir − ideal); ideal = [90.0, -9.0, 7.0, -95.0]; nadir = [140.0, -6.0, 18.0, -80.0]; HV reference = [1.1, 1.1, 1.1, 1.1] (normalized).

Termination TOPSIS and VIKOR terminate after scoring the five supplied alternatives or the stored NSGA-II front.

Evaluated methods: TOPSIS and VIKOR on supplier data. See the corresponding Task A pages and raw files.

All five supplied suppliers are nondominated under cost minimization and quality delivery reliability directions. Ranking imposes an additional preference order on these tradeoffs.

P13 Differentiable multi-task toy and network

Decision variables and encoding toy x in R^2, start (0,5); network 5->8 tanh->two heads

$$min\, F(x) \;=\; [\|x \;-\; (3,3)\|^2,\; \|x \;-\; (-\,3,-\,3)\|^2]$$

$$Network\ objectives\colon\ [binary\ cross\ -\ entropy,\ mean\ squared\ error]$$

Constraints and handling Toy unconstrained. Numerical plotting scale [0,72], no claim of bounding all R^2.

Pareto set x1=x2 in [-3,3]. MGDA minimizes norm of convex gradient combination. GradNorm balances weighted shared-body gradient norms. Synthetic 128x5 data seed 63, class sign(x0-x1), regression 3*x0+2*x2+noise. Network H=tanh(XW+b), class=sigmoid(H*wc+bc), regression=H*wr+br; 66 real parameters, 48 shared. BCE=-mean(y*log(class)+(1-y)*log(1-class)); MSE=mean((regression-target)^2). Regression scale 1 vs 10; all training derivatives handwritten.

Numerical sanity check For the toy at x=(0,0), both squared-distance losses equal 18; finite-difference tests check the network gradients.

Normalization: (F − ideal)/(nadir − ideal); ideal = [0.0, 0.0]; nadir = [72.0, 72.0]; HV reference = [1.1, 1.1] (normalized).

Termination Toy gradient runs use 10,000 objective evaluations; network experiments use 1000 full-batch training steps.

Evaluated methods: GradNorm, MGDA. See the corresponding Task A pages and raw files.

P14 Laptop preference

Decision variables and encoding screen [13,17]in, battery [40,90]Wh

$$min\, F(s,b) \;=\; [0.4 \;+\; 0.05s \;+\; 0.012b,\;\; -\,0.2b\,/\,(0.5 \;+\; 0.03s)]$$

Constraints and handling Bounds; Pareto screen=13.

Every screen increase worsens both objectives. Natural reference (1.8kg,12h) becomes (1.8,-12). At 12h front weight=1.6908kg; reference is dominated, so preference focuses near a attainable compromise rather than exact equal target.

Numerical sanity check At screen=13 and runtime=12 h, battery=53.4 Wh and weight=1.6908 kg; increasing screen size worsens both objectives.

Normalization: (F − ideal)/(nadir − ideal); ideal = [1.53, -20.224719101123597]; nadir = [2.33, -7.920792079207921]; HV reference = [1.1, 1.1] (normalized).

Termination Preference and scalar runs stop at 10,000 evaluations or solver tolerance under the same cap.

Evaluated methods: IBEA, R-NSGA-II, SMS-EMOA, Value Function. See the corresponding Task A pages and raw files.

4 Shared tools and verification

moo_utils.py centralizes dominance, pairwise front sorting, crowding distance, archive filtering, normalization, positive-residual violation, exact 2D HV, Monte Carlo HV, IGD, Spacing and reproducible reference directions. Pairwise arrays expose the calculations directly and cost O(M*N^2); vectorization improves execution time without changing asymptotic work.

Check	Input	Expected
Dominance	(1,2) versus (2,3)	First dominates
Tradeoff	(1,3) versus (2,2)	Neither dominates
HV	(1,3) (2,2) (3,1); reference (4,4)	6
IGD identical fronts	same point sets	0
Crowding	(0,2) (1,1) (2,0)	infinity 2 infinity
Lecture car front	A B C D E F table	A B C D
Lecture Deb layers	A D feasible; B CV2; C CV5	A,D then B then C
DTLZ2 g0	tail variables all .5	squared objective sum 1
Network derivatives	central finite difference +/-1e-6	analytical match to 5 decimal places

Final automated result: 62 tests passed. Tests include seed reproducibility for all 26 stochastic modules and bounds for recorded solutions. For P8 the crossover and both mutation variants are checked across 20 seeds; cities occur exactly once. See results/roll72/test_output.txt for the named checks. Tests verify defining mechanisms, not a global-optimality guarantee.

5 Tasks A and B algorithm implementations and comparison

The consolidated Task A log below is generated from verified base summaries. Each algorithm/problem combination has its own row so budgets and metrics cannot be misaligned. Ranking methods report N/A for optimizer metrics and untimed runtime because those quantities were not recorded; no values are invented.

No.	Algorithm	Problem	Evaluations / cap	HV	IGD	Runtime s	Status
1	Weighted Sum	P1	168 / 10,000	1.026 ± 0	0.02014 ± 0	1.106 ± 0	works
1	Weighted Sum	P3	847 / 10,000	0.21 ± 0	0.3538 ± 0	1.177 ± 0	works
2	Epsilon-Constraint	P2	870 / 10,000	1.087 ± 0	0.03713 ± 0	0.2331 ± 0	works
2	Epsilon-Constraint	P3	3488 / 10,000	0.5205 ± 0	0.02658 ± 0	0.5931 ± 0	works
3	Benson reference improvement	P1	714 / 10,000	1.026 ± 0	0.02014 ± 0	0.1737 ± 0	works
3	Benson reference improvement	P7	3947 / 10,000	0.4563 ± 0	0.1788 ± 0	0.9187 ± 0	works
4	Weight Metric	P1	1857 / 10,000	1.035 ± 0	0.01234 ± 0	0.6479 ± 0	works
4	Weight Metric	P3	9021 / 10,000	0.5237 ± 0	0.01616 ± 0	2.638 ± 0	works
5	Value Function	P14	45 / 10,000	0.3291 ± 0	0.353 ± 0	0.02234 ± 0	works
5	Value Function	P4	156 / 10,000	0.6662 ± 0	N/A	0.06295 ± 0	works
6	Goal Programming	P11	1219 / 10,000	1.21 ± 0	9.635e-11 ± 0	0.5276 ± 0	works
6	Goal Programming	P7	304 / 10,000	0.3407 ± 0	0.3819 ± 0	0.1776 ± 0	works
7	Lexicographic	P11	219 / 10,000	1.21 ± 0	1.631e-11 ± 0	0.08428 ± 0	works
7	Lexicographic	P7	47 / 10,000	0.11 ± 0	0.7071 ± 0	0.02369 ± 0	works
8	TOPSIS	P12 alternatives	ranking only	N/A	N/A	N/A (not timed)	works
8	TOPSIS	P4 NSGA-II front	ranking only	N/A	N/A	N/A (not timed)	works
9	VIKOR	P12 alternatives	ranking only	N/A	N/A	N/A (not timed)	works
9	VIKOR	P4 NSGA-II front	ranking only	N/A	N/A	N/A (not timed)	works

No.	Algorithm	Problem	Evaluations / cap	HV	IGD	Runtime s	Status
10	VEGA	P2	10,000 / 10,000	1.111 ± 0.0009254	0.01436 ± 0.001558	3.039 ± 0.2736	works
10	VEGA	P3	10,000 / 10,000	0.173 ± 0.04582	0.3587 ± 0.07168	3.063 ± 0.2203	works
11	MOGA	P3	10,000 / 10,000	0.5238 ± 0.001085	0.0127 ± 0.0004731	3.767 ± 0.4383	works
11	MOGA	P5	10,000 / 10,000	1.182 ± 0.0001344	0.01403 ± 0.00126	3.546 ± 0.3147	works
12	NSGA	P5	10,000 / 10,000	1.182 ± 6.707e-05	0.01358 ± 0.001486	3.644 ± 0.2729	works
12	NSGA	P7	10,000 / 10,000	0.6887 ± 0.001488	0.01228 ± 0.0007608	3.873 ± 0.3127	works
13	NPGA	P3	10,000 / 10,000	0.4572 ± 0.07144	0.05291 ± 0.06393	6.464 ± 0.3814	works
13	NPGA	P5	10,000 / 10,000	1.181 ± 0.0008078	0.01335 ± 0.00159	6.78 ± 0.4436	works
14	NSGA-II	P3	10,000 / 10,000	0.5286 ± 0.001001	0.01208 ± 0.001244	3.528 ± 0.3362	works
14	NSGA-II	P5	10,000 / 10,000	1.182 ± 3.963e-05	0.01406 ± 0.001287	3.567 ± 0.2887	works
15	NSGA-III	P10 M=3	10,000 / 10,000	0.6611 ± 0.005896	0.108 ± 0.00582	5.127 ± 0.4593	works
15	NSGA-III	P10 M=5	10,000 / 10,000	0.9883 ± 0.01881	0.3083 ± 0.006507	5.266 ± 0.3087	works
16	SPEA2	P4	10,000 / 10,000	0.9332 ± 0.001707	N/A	6.122 ± 0.088	works
16	SPEA2	P5	10,000 / 10,000	1.182 ± 0.0001209	0.01608 ± 0.002603	8.238 ± 0.3582	works
17	PESA-II	P3	10,000 / 10,000	0.511 ± 0.007104	0.02193 ± 0.003303	5.236 ± 0.2394	works
17	PESA-II	P5	10,000 / 10,000	1.181 ± 0.0005926	0.01697 ± 0.002015	6.328 ± 0.459	works
18	MOEA/D	P10 M=5	10,000 / 10,000	0.9954 ± 0.03858	0.3212 ± 0.02067	19.58 ± 0.8841	works

No.	Algorithm	Problem	Evaluations / cap	HV	IGD	Runtime s	Status
18	MOEA/D	P3	10,000 / 10,000	0.5318 ± 8.619e-05	0.009567 ± 0.0001113	14.44 ± 0.7004	works
19	MOEA/D-DE	P3	10,000 / 10,000	0.2124 ± 0.2742	1.042 ± 0.9331	11.79 ± 2.149	works
19	MOEA/D-DE	P5	10,000 / 10,000	1.182 ± 3.231e-05	0.0491 ± 0.03155	12.84 ± 0.5549	works
20	MOEA/D-ACO	P8	10,000 / 10,000	0.9235 ± 0.0007362	N/A	26.42 ± 0.9056	works
21	Constrained NSGA-II	P5	10,000 / 10,000	1.182 ± 3.963e-05	0.01406 ± 0.001287	3.317 ± 0.3422	works
21	Constrained NSGA-II	P6	10,000 / 10,000	1.011 ± 0.001179	0.01327 ± 0.0007729	3.39 ± 0.3077	works
22	MOPSO	P4	10,000 / 10,000	0.9259 ± 0.00455	N/A	2.179 ± 0.2485	works
22	MOPSO	P5	10,000 / 10,000	1.182 ± 0.0006447	0.01421 ± 0.002302	2.031 ± 0.3814	works
23	MOGWO	P5	10,000 / 10,000	1.182 ± 0.0001194	0.01341 ± 0.001289	4.102 ± 0.3726	partly works
23	MOGWO	P9	200 / 200	0.7818 ± 0.04271	N/A	30.45 ± 2.014	partly works
24	MOABC	P3	10,000 / 10,000	0.5085 ± 0.008993	0.02247 ± 0.00449	9.72 ± 0.7229	works
24	MOABC	P6	10,000 / 10,000	1.016 ± 0.0002737	0.01101 ± 0.0003537	13.04 ± 1.245	works
25	MOACO	P8	10,000 / 10,000	0.9239 ± 0	N/A	18.14 ± 2.17	works
26	MOSA	P3	10,000 / 10,000	0.5135 ± 0.006066	0.0204 ± 0.003815	1.47 ± 0.1376	works
26	MOSA	P5	10,000 / 10,000	1.181 ± 0.0003664	0.01318 ± 0.0007081	1.523 ± 0.1193	works
27	MODE	P3	10,000 / 10,000	0.11 ± 3.023e-10	0.6098 ± 1.147e-10	1.77 ± 0.06894	works
27	MODE	P5	10,000 / 10,000	1.181 ± 0.002083	0.01381 ± 0.003286	2.19 ± 0.2773	works

No.	Algorithm	Problem	Evaluations / cap	HV	IGD	Runtime s	Status
28	MO-TLBO	P3	10,000 / 10,000	0.5318 ± 7.088e-05	0.009509 ± 9.074e-05	11.39 ± 1.237	works
28	MO-TLBO	P5	10,000 / 10,000	1.182 ± 1.659e-05	0.01234 ± 0.0004422	11.36 ± 1.012	works
29	MOBA	P3	10,000 / 10,000	0.3968 ± 0.1443	0.1289 ± 0.1631	4.604 ± 0.2289	works
29	MOBA	P4	10,000 / 10,000	0.9332 ± 0.001286	N/A	4.796 ± 0.3789	works
30	MOMA	P5	10,000 / 10,000	1.182 ± 5.827e-05	0.01291 ± 0.001555	6.331 ± 0.652	partly works
30	MOMA	P9	200 / 200	0.8736 ± 0.03211	N/A	39.46 ± 6.746	partly works
31	RVEA	P10 M=3	10,000 / 10,000	0.6403 ± 0.01143	0.1088 ± 0.003233	5.642 ± 0.5761	works
31	RVEA	P10 M=5	10,000 / 10,000	1.012 ± 0.03024	0.3111 ± 0.008554	4.995 ± 0.5221	works
32	IBEA	P14	10,000 / 10,000	0.8194 ± 0.01225	0.02618 ± 0.007653	11.83 ± 0.6681	works
32	IBEA	P3	10,000 / 10,000	0.1928 ± 0.01701	0.3581 ± 0.02807	11.45 ± 1.023	works
33	SMS-EMOA	P14	10,000 / 10,000	0.8587 ± 1.992e-05	0.007577 ± 1.879e-05	30.59 ± 1.685	works
33	SMS-EMOA	P3	10,000 / 10,000	0.5318 ± 0.000109	0.009565 ± 0.0001278	31.09 ± 1.972	works
34	HypE	P10 M=3	10,000 / 10,000	0.6611 ± 0.008224	0.1171 ± 0.004252	20.2 ± 0.6013	works
34	HypE	P10 M=5	10,000 / 10,000	0.8057 ± 0.1473	0.4362 ± 0.07524	23.72 ± 1.897	works
35	R-NSGA-II	P14	10,000 / 10,000	0.8556 ± 0.0008798	0.01004 ± 0.0008287	4.152 ± 0.3333	works
35	R-NSGA-II	P5	10,000 / 10,000	1.176 ± 0.01429	0.01448 ± 0.003167	4.811 ± 0.3217	works
36	MGDA	P13	10,000 / 10,000	0.2839 ± 0	0.6279 ± 0	2.352 ± 0	works

No.	Algorithm	Problem	Evaluations / cap	HV	IGD	Runtime s	Status
37	GradNorm	P13	10,000 / 10,000	0.6052 ± 0	0.433 ± 0	2.667 ± 0	works

The 37 algorithm summaries below present implementation mechanism, selected numerical results and recorded fronts. The complete four advantages and four disadvantages for each method appear once in the Task B master table, avoiding duplicated text. These metrics compare achieved quality under the stated caps; they do not establish causal superiority.

Classical methods convert preferences or bounds into scalar solves, making their numerical accuracy depend on the scalar solver. Early GAs use objective selection, rank or sharing without parent elitism; the passive archive records discoveries but does not prevent the breeding population from losing them. Elitist methods spend quadratic comparisons to retain high-ranked parents and use crowding, density or grids for diversity. Decomposition gives each direction its own scalar pressure, although a fixed lattice may not match disconnected fronts. Swarms and local heuristics rely on archive leaders and neighbourhood moves; their schedules can exploit quickly or stagnate. Indicators replace explicit density with epsilon or hypervolume fitness but require scales and, for HV, a reference box. Reference methods remain useful when dominance loses discrimination in five dimensions. Preferences intentionally trade full-front coverage for a useful region. Gradient methods optimize differentiable tasks toward stationarity and do not produce a full front from one start. Feasibility rules are applied before Pareto comparisons, and comparisons use common normalized metric scales.

1 Weighted Sum

Mechanism Linear scalar solve and a sweep

Implementation Weighted Sum: minimize dot(w,F_normalized), sweep simplex weights. Positive weights yield supported efficient points; zeros may give weak efficiency.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P1	1	1.026	0.02013	1.535e-08	1.106	168 / 10,000
P3	1	0.21	0.3538	1.155	1.177	847 / 10,000

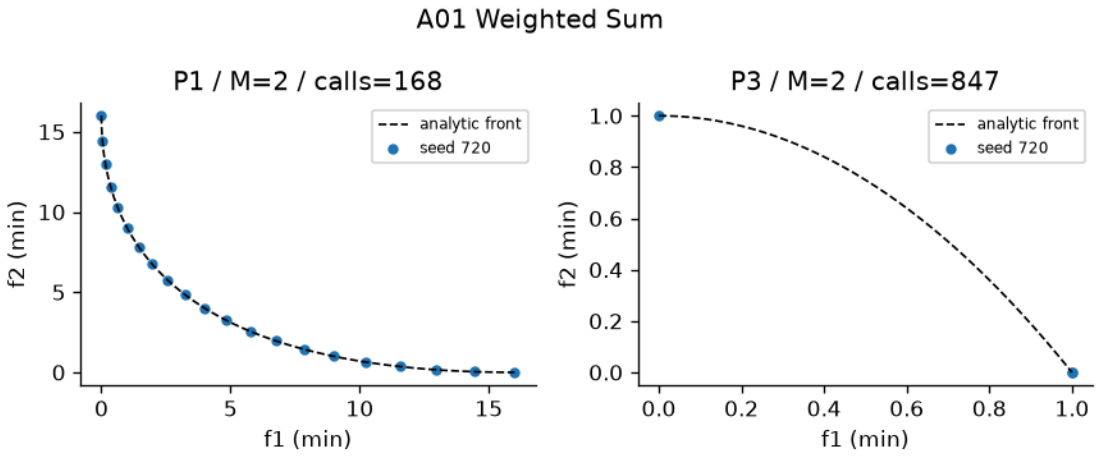


Figure A01. Recorded output on the designated problems.

Observation P1: HV 1.026 ± 0, IGD 0.02013 ± 0; P3: HV 0.21 ± 0, IGD 0.3538 ± 0. Scalar subproblems may consume fewer evaluations than the shared cap; this is not a guarantee of global optimality.

Useful for: Convex smooth two-objective design with interpretable weights. Less suitable for: Dented fronts and an unknown preference distribution.

2 Epsilon-Constraint

Mechanism One objective minimized and others bounded

Implementation Epsilon-Constraint: minimize F1 with normalized F2<=epsilon. Default epsilon range=[0,1].

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P2	1	1.086	0.03713	0.1418	0.2331	870 / 10,000
P3	1	0.5206	0.02658	0.04354	0.5931	3,488 / 10,000

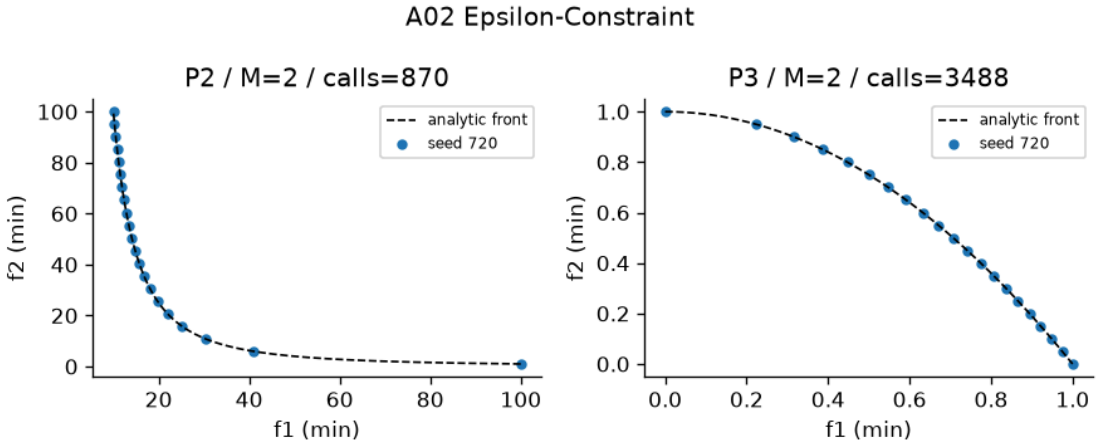


Figure A02. Recorded output on the designated problems.

Observation P2: HV 1.086 ± 0, IGD 0.03713 ± 0; P3: HV 0.5206 ± 0, IGD 0.02658 ± 0. Scalar subproblems may consume fewer evaluations than the shared cap; this is not a guarantee of global optimality.

Useful for: Nonconvex constrained design with hard limits. Less suitable for: High-dimensional objective grids.

3 Benson reference improvement

Mechanism Weighted improvement below a reference

Implementation Benson reference improvement: maximize weighted nonnegative improvements from a dominated reference r=[1.1,1.1], require F<=r. This follows Foundations slide 35 reference achievement.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P1	1	1.026	0.02013	2.735e-08	0.1737	714 / 10,000
P7	1	0.4563	0.1788	0.5859	0.9187	3,947 / 10,000

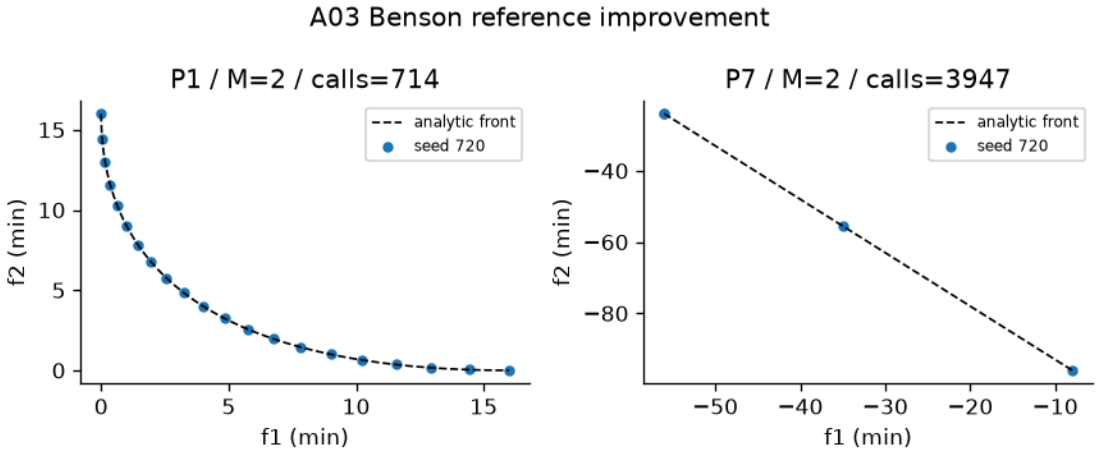


Figure A03. Recorded output on the designated problems.

Observation P1: HV 1.026 ± 0, IGD 0.02013 ± 0; P7: HV 0.4563 ± 0, IGD 0.1788 ± 0. Scalar subproblems may consume fewer evaluations than the shared cap; this is not a guarantee of global optimality.

Useful for: Teaching reference improvement on P1/P7. Less suitable for: Claiming a general Benson convergence theorem.

4 Weight Metric

Mechanism Weighted distance to fixed ideal

Implementation Weight Metric: distance from fixed ideal, weighted p=1,2,infinity. Sweep weights for all three p values.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P1	1	1.035	0.01234	0.02381	0.6479	1,857 / 10,000
P3	1	0.5237	0.01616	0.03474	2.638	9,021 / 10,000

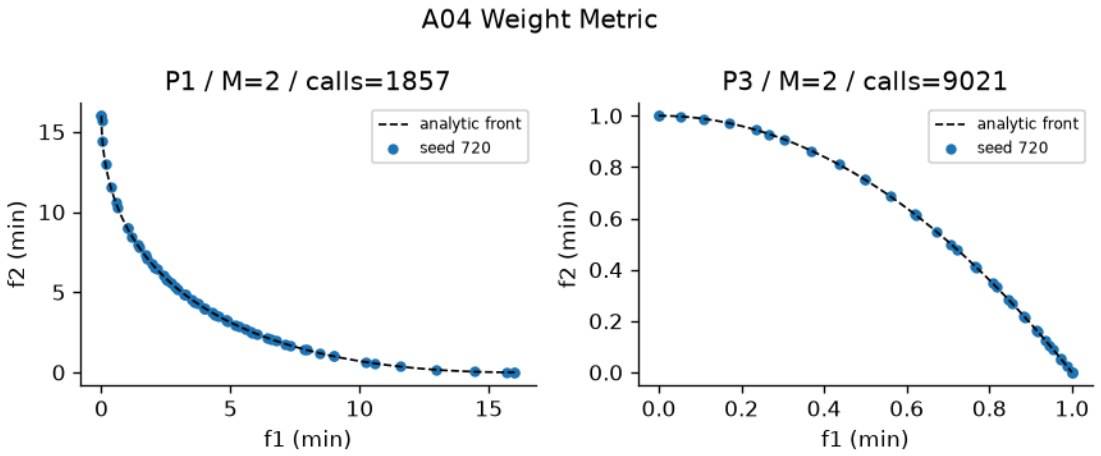


Figure A04. Recorded output on the designated problems.

Observation P1: HV 1.035 ± 0, IGD 0.01234 ± 0; P3: HV 0.5237 ± 0, IGD 0.01616 ± 0. Scalar subproblems may consume fewer evaluations than the shared cap; this is not a guarantee of global optimality.

Useful for: Compromise solutions on nonconvex fronts. Less suitable for: Unreliable ideal estimates or many weights.

5 Value Function

Mechanism Declared decision-maker utility

Implementation Value Function: declared quadratic utility to a preferred normalized aspiration. P14 uses [1.8 kg,12 h]; P4 uses three investor risk/return targets.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P14	1	0.329	0.353	0	0.02234	45 / 10,000
P4	1	0.6662	N/A	9.678e-05	0.06295	156 / 10,000

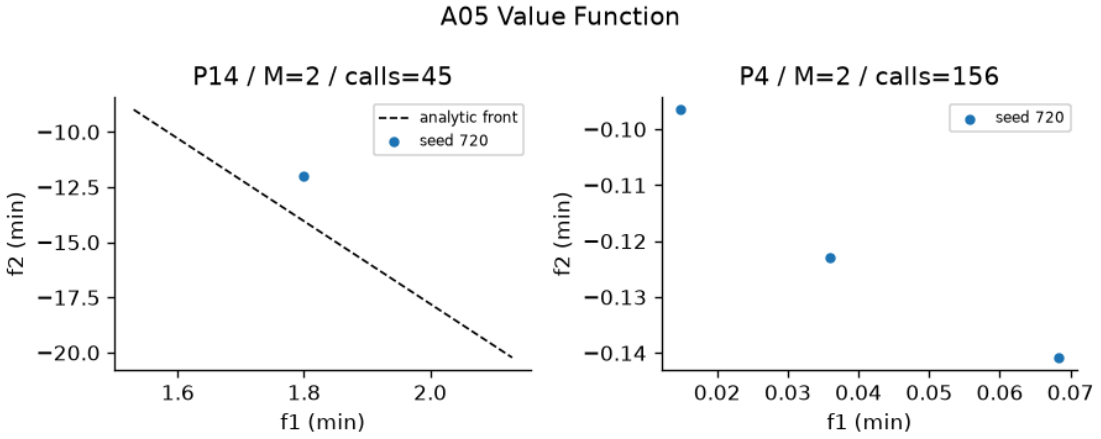


Figure A05. Recorded output on the designated problems.

Observation P14: HV 0.329 ± 0, IGD 0.353 ± 0; P4: HV 0.6662 ± 0, IGD N/A. Scalar subproblems may consume fewer evaluations than the shared cap; this is not a guarantee of global optimality.

Useful for: Laptop or portfolio profile decisions. Less suitable for: Exploration without known preferences.

6 Goal Programming

Mechanism Explicit positive/negative goal deviations

Implementation Goal Programming: goals in natural output, penalize negative deviations only (both goals are lower bounds). Explicit dminus=max(target-achieved,0), dplus=max(achieved-target,0); elimination of linear deviation variables is exact.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P11	1	1.21	9.636e-11	0	0.5277	1,219 / 10,000
P7	1	0.3407	0.3819	0	0.1776	304 / 10,000

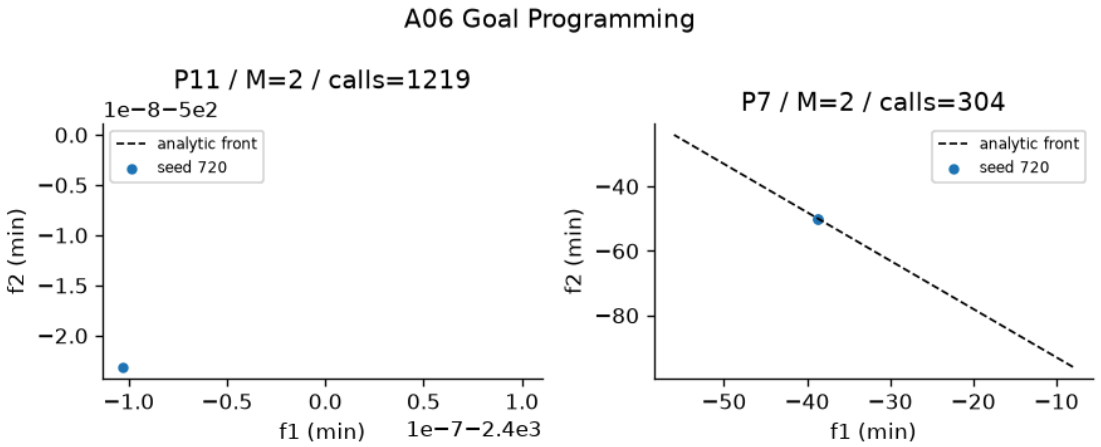


Figure A06. Recorded output on the designated problems.

Observation P11: HV 1.21 ± 0, IGD 9.636e-11 ± 0; P7: HV 0.3407 ± 0, IGD 0.3819 ± 0. Scalar subproblems may consume fewer evaluations than the shared cap; this is not a guarantee of global optimality.

Useful for: Production targets with soft priorities. Less suitable for: Searching an entire front without goals.

7 Lexicographic

Mechanism Sequential strict priorities with tolerances

Implementation Lexicographic: sequential strict-priority scalar minimizations, preserving prior best objective within tolerance=1e-7 in normalized units. Profit first then units on P11; agricultural yield first on P7.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P11	1	1.21	1.631e-11	0	0.08428	219 / 10,000
P7	1	0.11	0.7071	0	0.02369	47 / 10,000

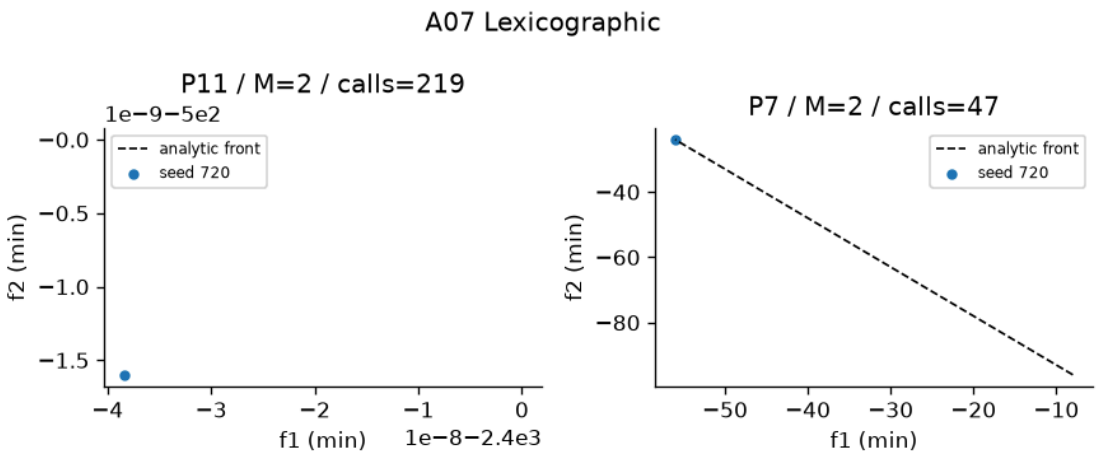


Figure A07. Recorded output on the designated problems.

Observation P11: HV 1.21 ± 0, IGD 1.631e-11 ± 0; P7: HV 0.11 ± 0, IGD 0.7071 ± 0. Scalar subproblems may consume fewer evaluations than the shared cap; this is not a guarantee of global optimality.

Useful for: Profit first and units second. Less suitable for: Balanced objectives with no hierarchy.

8 TOPSIS

Mechanism Closeness to weighted ideal and anti-ideal

Implementation TOPSIS ranking, not an optimizer. Vector normalization, weights, benefit/cost ideal, Euclidean ideal and anti-ideal distance.

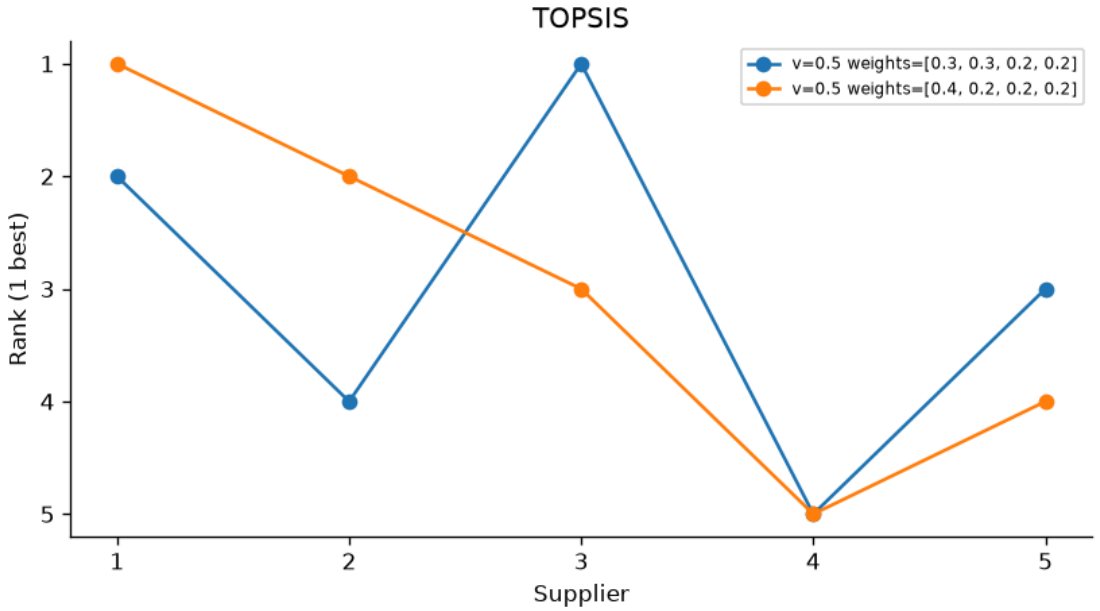


Figure A08. Recorded output on the designated problems.

Observation The method ranks supplied alternatives rather than searching for a continuous Pareto front; see the ranking comparison.

Useful for: Supplier or existing-front ranking. Less suitable for: Optimization without an alternatives table.

9 VIKOR

Mechanism Compromise via group regret and worst regret

Implementation VIKOR compromise ranking with S=sum weighted regret, R=max regret, Q=v*(S-Sbest)/(Sworst-Sbest)+(1-v)*(R-Rbest)/(Rworst-Rbest). Default v=.5.

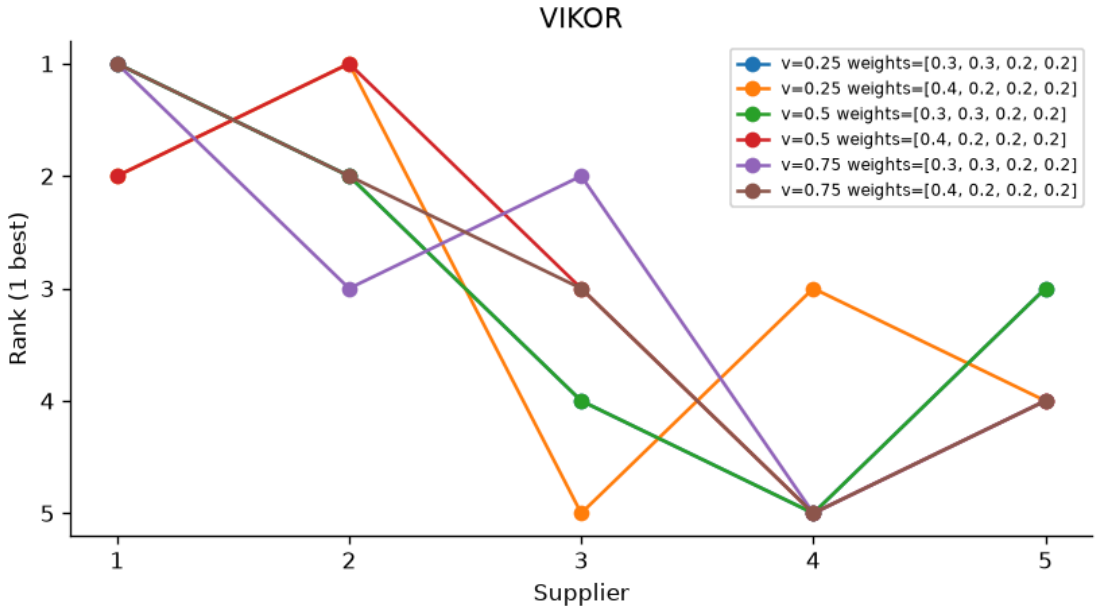


Figure A09. Recorded output on the designated problems.

Observation The method ranks supplied alternatives rather than searching for a continuous Pareto front; see the ranking comparison.

Useful for: Supplier compromise with severe weak criteria. Less suitable for: A Pareto front generator.

10 VEGA

Mechanism Objective-partitioned selection

Implementation VEGA: partition selection by objective, pool subpopulations, crossover and mutation. Tournament by one objective; no parent+child elitism.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P2	10	1.111 ± 0.0009254	0.01436 ± 0.001558	0.017 ± 0.00279	3.039 ± 0.2736	10,000 / 10,000
P3	10	0.173 ± 0.04583	0.3587 ± 0.07168	0.0475 ± 0.02233	3.063 ± 0.2203	10,000 / 10,000

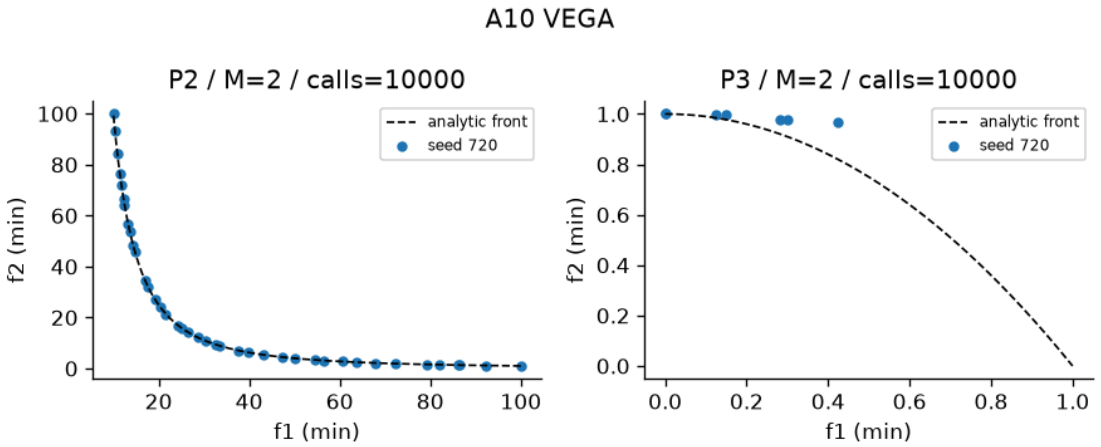


Figure A10. Recorded output on the designated problems.

Observation P2: HV 1.111 ± 0.0009254, IGD 0.01436 ± 0.001558; P3: HV 0.173 ± 0.04583, IGD 0.3587 ± 0.07168.

Useful for: Teaching objective competition. Less suitable for: Reliable uniform front preservation.

11 MOGA

Mechanism Domination-count rank and fitness sharing

Implementation MOGA: domination count rank, rank fitness divided by niche count. Sharing radius sigma=.3; small radius weakens sharing; no elitism.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.5238 ± 0.001085	0.0127 ± 0.0004731	0.01277 ± 0.002314	3.767 ± 0.4383	10,000 / 10,000
P5	10	1.182 ± 0.0001344	0.01403 ± 0.00126	0.02033 ± 0.002006	3.546 ± 0.3147	10,000 / 10,000

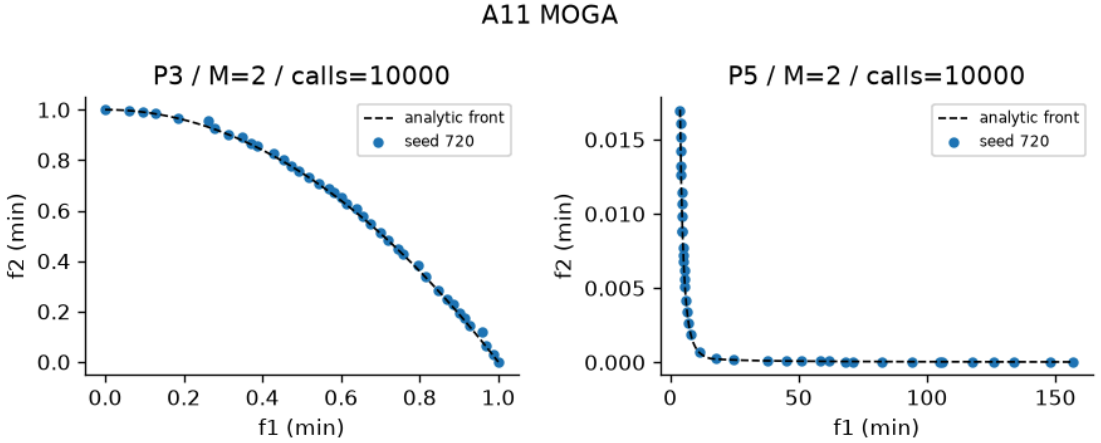


Figure A11. Recorded output on the designated problems.

Observation P3: HV 0.5238 ± 0.001085, IGD 0.0127 ± 0.0004731; P5: HV 1.182 ± 0.0001344, IGD 0.01403 ± 0.00126.

Useful for: Small bi-objective populations. Less suitable for: Many objectives with sparse dominance.

12 NSGA

Mechanism Front layers and within-front sharing

Implementation NSGA: nondominated layers, dummy fitness and within-front sharing. Sharing radius sigma=.3; non-elitist generational replacement.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P5	10	1.182 ± 6.707e-05	0.01358 ± 0.001486	0.01959 ± 0.002904	3.644 ± 0.2729	10,000 / 10,000
P7	10	0.6887 ± 0.001488	0.01228 ± 0.0007609	0.01077 ± 0.001845	3.873 ± 0.3127	10,000 / 10,000

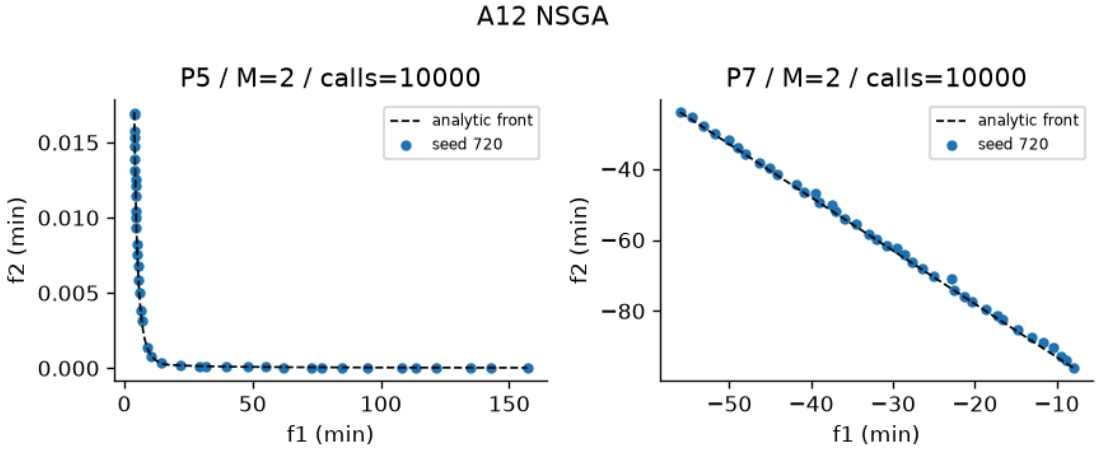


Figure A12. Recorded output on the designated problems.

Observation P5: HV 1.182 ± 6.707e-05, IGD 0.01358 ± 0.001486; P7: HV 0.6887 ± 0.001488, IGD 0.01228 ± 0.0007609.

Useful for: Teaching early Pareto GAs. Less suitable for: Large fast convergence requirements.

13 NPGA

Mechanism Direct contestant dominance then separate comparison-set quadratic niche tie

Implementation Niched Pareto Genetic Algorithm: direct binary tournament with comparison-set crowding. Direct dominance decides; otherwise smaller comparison-set niche count wins.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.4572 ± 0.07144	0.05291 ± 0.06393	0.0329 ± 0.01947	6.464 ± 0.3814	10,000 / 10,000
P5	10	1.181 ± 0.0008078	0.01335 ± 0.00159	0.01876 ± 0.001853	6.78 ± 0.4436	10,000 / 10,000

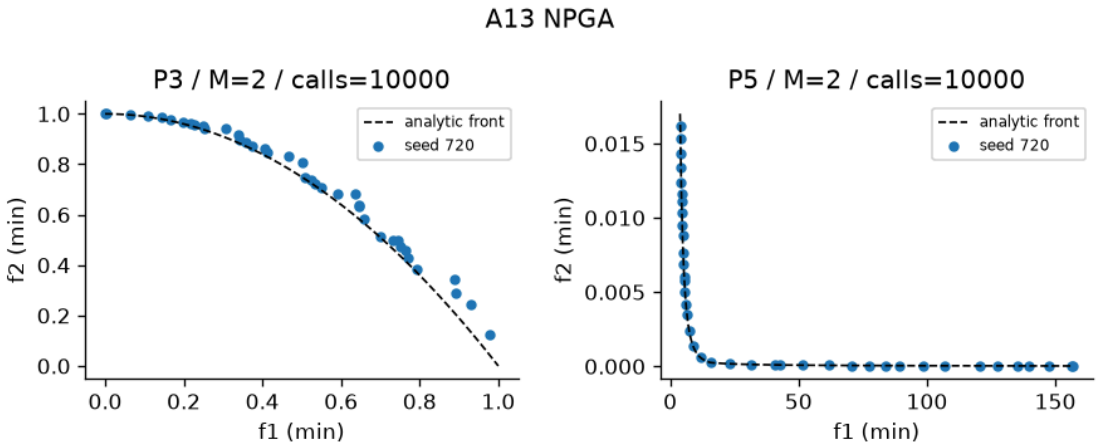


Figure A13. Recorded output on the designated problems.

Observation P3: HV 0.4572 ± 0.07144, IGD 0.05291 ± 0.06393; P5: HV 1.181 ± 0.0008078, IGD 0.01335 ± 0.00159.

Useful for: Bi-objective diversity experiments. Less suitable for: Many-objective fronts where most tournaments tie.

14 NSGA-II

Mechanism Front rank, crowding, parent-child elitism

Implementation NSGA-II: fast front sorting, crowded binary tournament, SBX and mutation, parent+offspring elitist survival. Permutations use OX and swap/inversion.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.5286 ± 0.001001	0.01208 ± 0.001244	0.01765 ± 0.001747	3.528 ± 0.3362	10,000 / 10,000
P5	10	1.182 ± 3.963e-05	0.01406 ± 0.001287	0.02067 ± 0.003274	3.568 ± 0.2887	10,000 / 10,000

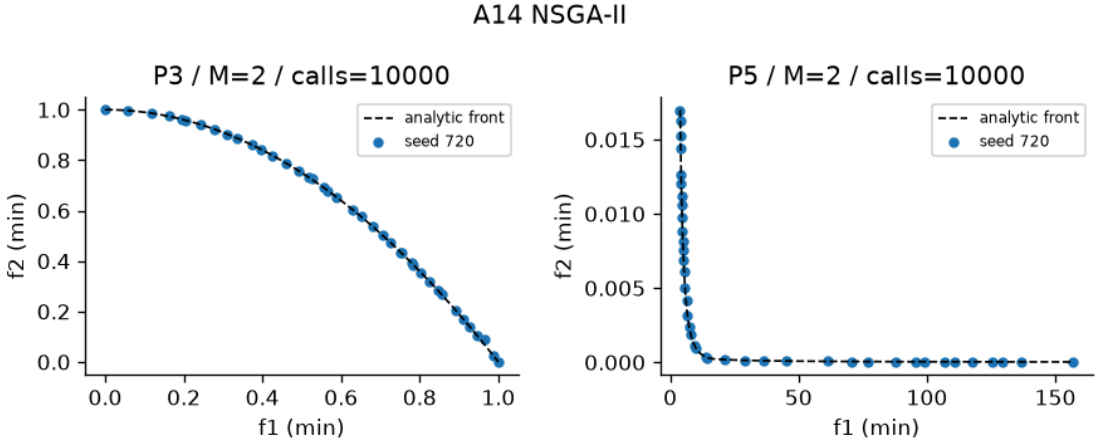


Figure A14. Recorded output on the designated problems.

Observation P3: HV 0.5286 ± 0.001001, IGD 0.01208 ± 0.001244; P5: HV 1.182 ± 3.963e-05, IGD 0.01406 ± 0.001287.

Useful for: General constrained two-objective search. Less suitable for: Very many objectives with weak dominance.

15 NSGA-III

Mechanism Reference-point niching after front rank

Implementation NSGA-III: front ranking, simplex-lattice reference association and niche occupancy survival. Normalize by observed ranges (simplification of intercepts).

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P10 M3	10	0.6611 ± 0.005896	0.108 ± 0.005819	0.09917 ± 0.0101	5.127 ± 0.4593	10,000 / 10,000
P10 M5	10	0.9883 ± 0.01881	0.3083 ± 0.006507	0.2831 ± 0.02087	5.266 ± 0.3087	10,000 / 10,000

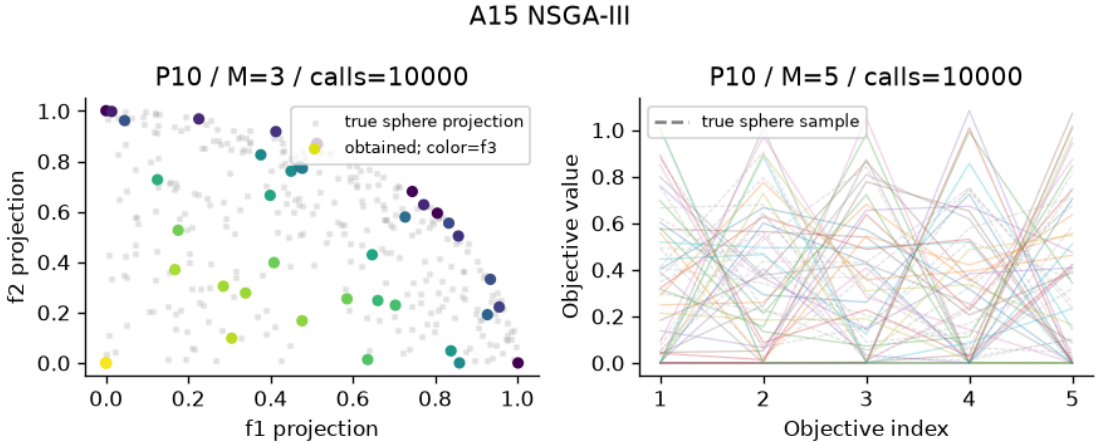


Figure A15. Recorded output on the designated problems.

Observation P10 (M=3): HV 0.6611 ± 0.005896, IGD 0.108 ± 0.005819; P10 (M=5): HV 0.9883 ± 0.01881, IGD 0.3083 ± 0.006507.

Useful for: DTLZ2 with three or five objectives. Less suitable for: Irregular fronts without adapted references.

16 SPEA2

Mechanism Strength, raw fitness and neighbour density

Implementation SPEA2: strength S, raw fitness R=sum strengths of dominators; density 1/(sigma_k+2), fixed archive, lexicographic nearest-distance truncation. Shared defaults: population=40, budget=10,000, optimizer seeds 720–729.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P4	10	0.9332 ± 0.001707	N/A	0.01409 ± 0.001608	6.121 ± 0.088	10,000 / 10,000
P5	10	1.182 ± 0.0001209	0.01608 ± 0.002603	0.01923 ± 0.001586	8.238 ± 0.3582	10,000 / 10,000

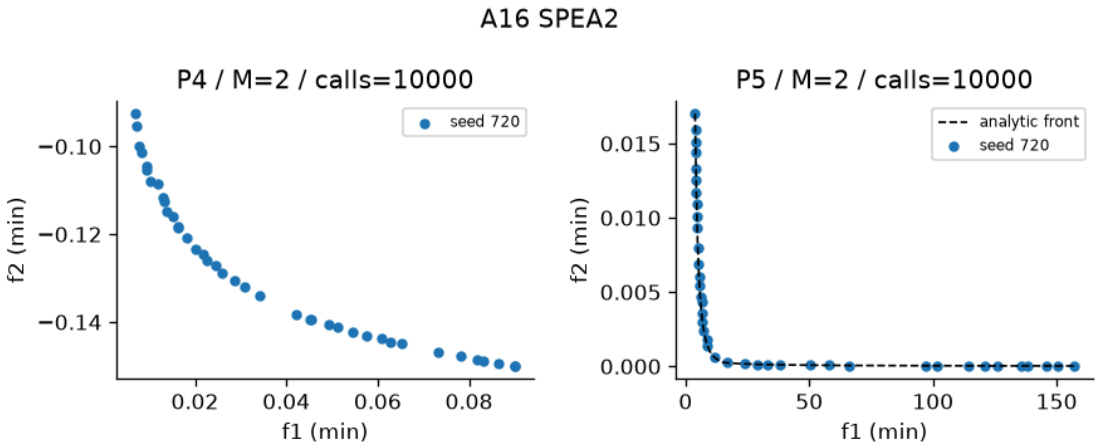


Figure A16. Recorded output on the designated problems.

Observation P4: HV 0.9332 ± 0.001707, IGD N/A; P5: HV 1.182 ± 0.0001209, IGD 0.01608 ± 0.002603.

Useful for: Small smooth front archives. Less suitable for: Large populations with tight memory.

17 PESA-II

Mechanism Archive region occupancy selection

Implementation PESA-II: archive adaptive grid regions; region tournament selects lower occupancy region, then uniform individual. Default grid divisions=8.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.511 ± 0.007104	0.02193 ± 0.003303	0.0234 ± 0.007883	5.236 ± 0.2394	10,000 / 10,000
P5	10	1.181 ± 0.0005926	0.01697 ± 0.002015	0.01689 ± 0.002372	6.328 ± 0.459	10,000 / 10,000

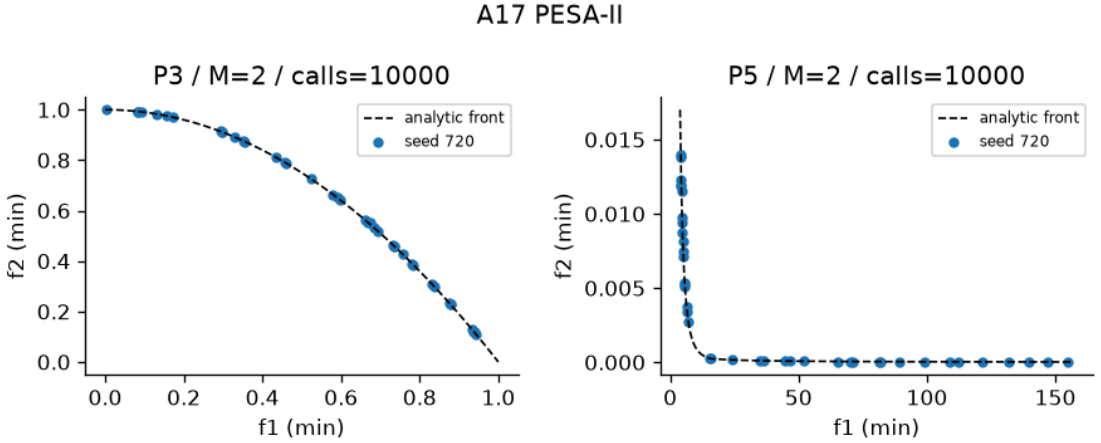


Figure A17. Recorded output on the designated problems.

Observation P3: HV 0.511 ± 0.007104, IGD 0.02193 ± 0.003303; P5: HV 1.181 ± 0.0005926, IGD 0.01697 ± 0.002015.

Useful for: Bi-objective region coverage. Less suitable for: Many objectives with almost unique cells.

18 MOEA/D

Mechanism Neighbouring Tchebycheff subproblems

Implementation MOEA/D: reference weights, neighbourhood mating, running ideal, cooperative Tchebycheff replacement. Neighbourhood size=10.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P10 M5	10	0.9954 ± 0.03858	0.3212 ± 0.02067	0.2613 ± 0.0299	19.58 ± 0.8841	10,000 / 10,000
P3	10	0.5318 ± 8.619e-05	0.009567 ± 0.0001113	0.003944 ± 0.0007786	14.44 ± 0.7004	10,000 / 10,000

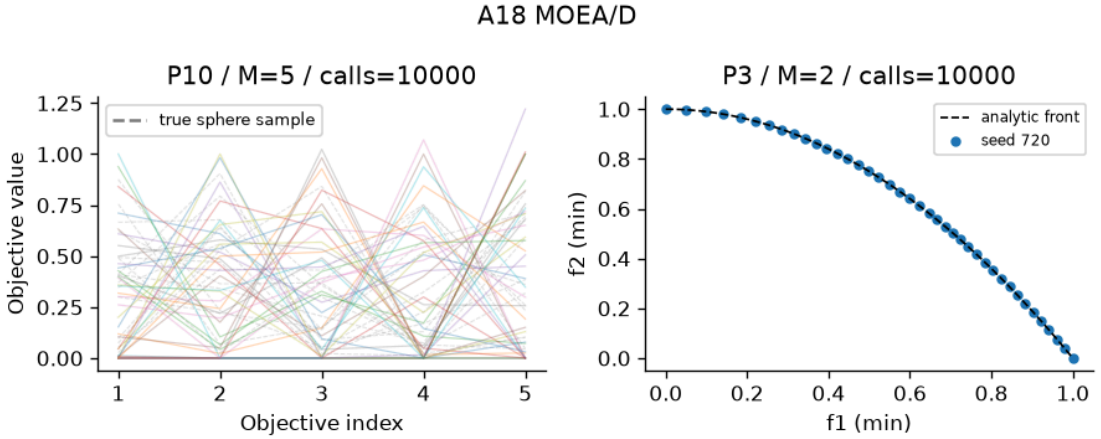


Figure A18. Recorded output on the designated problems.

Observation P10 (M=5): HV 0.9954 ± 0.03858, IGD 0.3212 ± 0.02067; P3: HV 0.5318 ± 8.619e-05, IGD 0.009567 ± 0.0001113.

Useful for: Smooth continuous many-objective design. Less suitable for: Disconnected fronts badly matched by weights.

19 MOEA/D-DE

Mechanism DE variation inside MOEA/D

Implementation MOEA/D-DE: neighbourhood donor x_a+F*(x_b-x_c), binomial crossover CR=.9 with forced coordinate, F=.5; cooperative scalar replacement. Shared defaults: population=40, budget=10,000, optimizer seeds 720–729.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.2124 ± 0.2742	1.042 ± 0.9331	0.001026 ± 0.001503	11.8 ± 2.149	10,000 / 10,000
P5	10	1.182 ± 3.231e-05	0.0491 ± 0.03155	0.02913 ± 0.02089	12.84 ± 0.5549	10,000 / 10,000

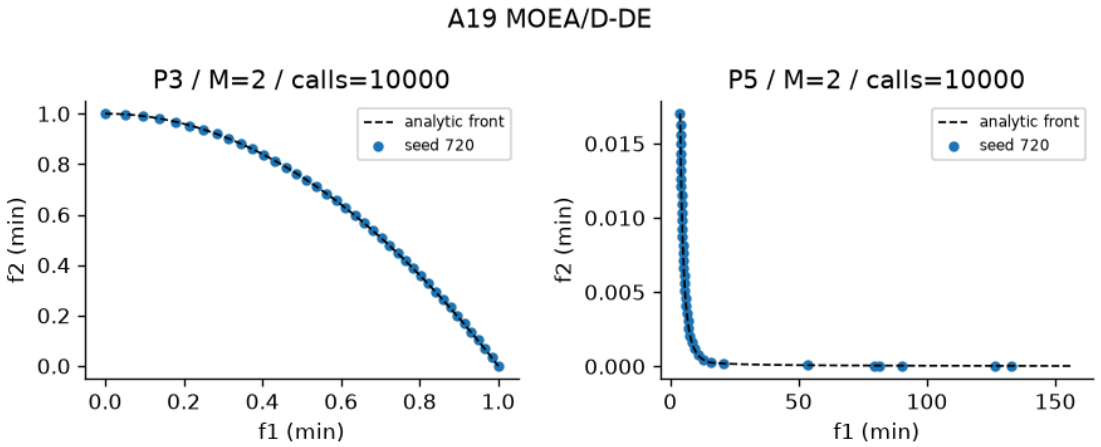


Figure A19. Recorded output on the designated problems.

Observation P3: HV 0.2124 ± 0.2742, IGD 1.042 ± 0.9331; P5: HV 1.182 ± 3.231e-05, IGD 0.0491 ± 0.03155.

Useful for: Continuous real-coded nonconvex design. Less suitable for: Permutation encodings without a new donor operator.

20 MOEA/D-ACO

Mechanism Weight-specific ant pheromones and scalar replacement

Implementation MOEA/D-ACO for tours: each weight has edge pheromone, weighted distance/time heuristic, neighbour Tchebycheff replacement and deposits. Defaults alpha=1 beta=2 evaporation=.1.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P8	10	0.9235 ± 0.0007362	N/A	0.003604 ± 0.001162	26.42 ± 0.9056	10,000 / 10,000

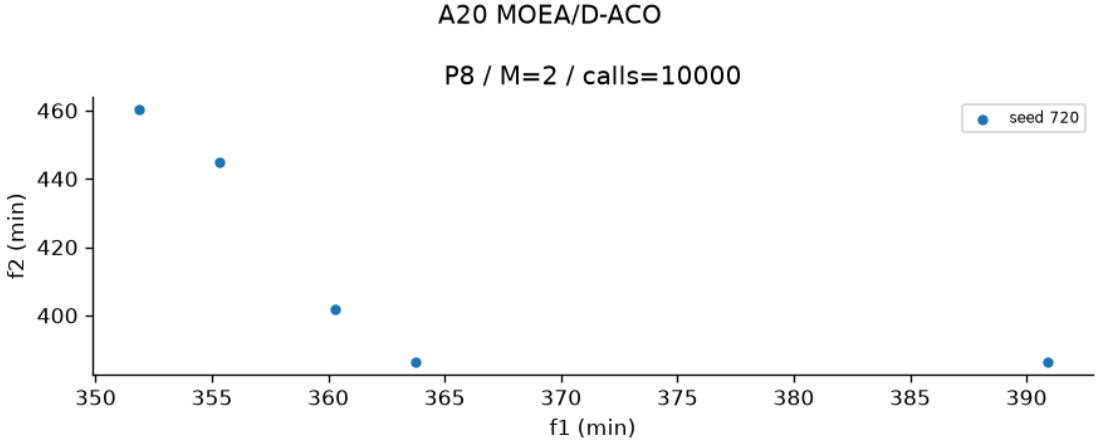


Figure A20. Recorded output on the designated problems.

Observation P8: HV 0.9235 ± 0.0007362, IGD N/A.

Useful for: Bi-objective routing. Less suitable for: Smooth gradient problems.

21 Constrained NSGA-II

Mechanism NSGA-II with Deb feasibility

Implementation Constrained NSGA-II: feasible beats infeasible, lower violation wins among infeasible, Pareto rank then crowding among feasible. NSGA-II in this project also supports Deb rules; this module exposes the rule.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P5	10	1.182 ± 3.963e-05	0.01406 ± 0.001287	0.02067 ± 0.003274	3.317 ± 0.3422	10,000 / 10,000
P6	10	1.011 ± 0.001179	0.01327 ± 0.0007729	0.01679 ± 0.0029	3.39 ± 0.3077	10,000 / 10,000

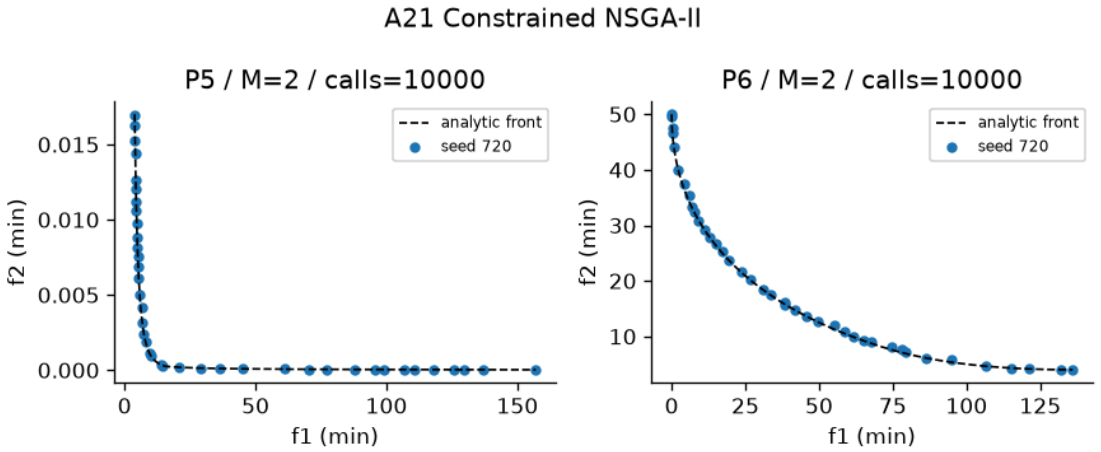


Figure A21. Recorded output on the designated problems.

Observation P5: HV 1.182 ± 3.963e-05, IGD 0.01406 ± 0.001287; P6: HV 1.011 ± 0.001179, IGD 0.01327 ± 0.0007729.

Useful for: Constrained beam or Binh-Korn. Less suitable for: Problems needing stochastic feasibility guarantees.

22 MOPSO

Mechanism Particle velocity with personal and archive leaders

Implementation MOPSO: inertia velocity, personal best, crowding-selected archive leader, occasional mutation. Personal best changes only when a trial dominates it.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P4	10	0.9259 ± 0.00455	N/A	0.01343 ± 0.003755	2.179 ± 0.2485	10,000 / 10,000
P5	10	1.182 ± 0.0006447	0.01421 ± 0.002302	0.02041 ± 0.002691	2.031 ± 0.3814	10,000 / 10,000

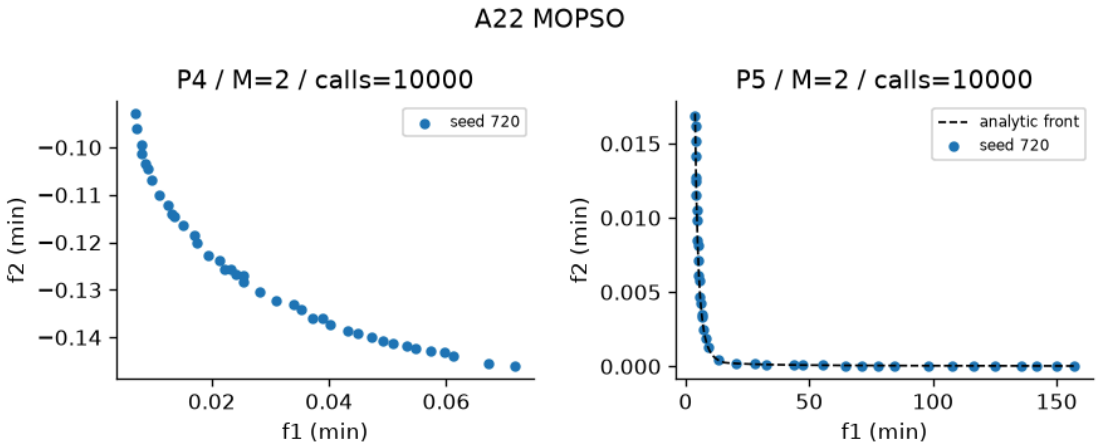


Figure A22. Recorded output on the designated problems.

Observation P4: HV 0.9259 ± 0.00455, IGD N/A; P5: HV 1.182 ± 0.0006447, IGD 0.01421 ± 0.002302.

Useful for: Continuous portfolio and beam designs. Less suitable for: Permutation tours without discrete velocity.

23 MOGWO

Mechanism Three archive leaders and encircling

Implementation MOGWO: alpha, beta, delta archive leaders; decreasing encircling coefficient a=2*(1-used/budget), average three proposed positions. Binary masks use a sigmoid transfer, then at-least-one repair.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P5	10	1.182 ± 0.0001194	0.01341 ± 0.001289	0.02077 ± 0.001969	4.102 ± 0.3726	10,000 / 10,000
P9	10	0.7818 ± 0.04271	N/A	0.04461 ± 0.05219	30.45 ± 2.014	200 / 200

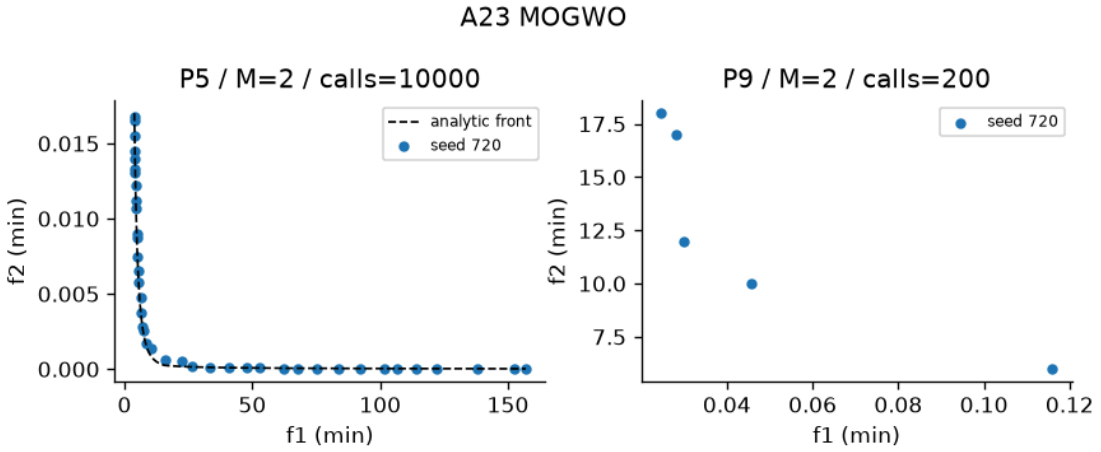


Figure A23. Recorded output on the designated problems.

Observation P5: HV 1.182 ± 0.0001194, IGD 0.01341 ± 0.001289; P9: HV 0.7818 ± 0.04271, IGD N/A.

Useful for: Continuous design or exploratory feature masks. Less suitable for: Required stable high-accuracy feature conclusions on tiny budgets.

24 MOABC

Mechanism Employed moves, crowding-based onlookers and abandonment scouts

Implementation MOABC: employed and onlooker differential-coordinate proposals, Deb dominance greedy replacement, abandoned-source scouts. Limit=20.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.5085 ± 0.008993	0.02247 ± 0.00449	0.02386 ± 0.004992	9.72 ± 0.7229	10,000 / 10,000
P6	10	1.016 ± 0.0002737	0.01101 ± 0.0003537	0.006067 ± 0.001849	13.04 ± 1.245	10,000 / 10,000

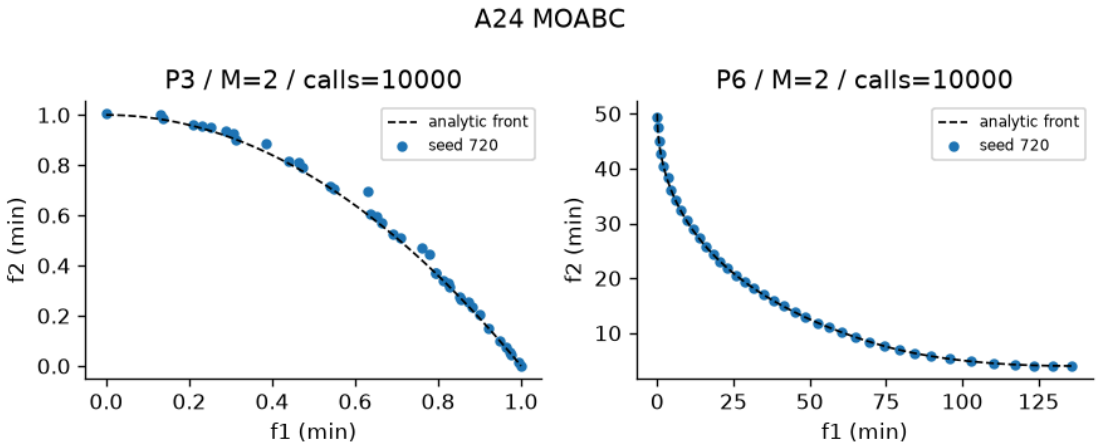


Figure A24. Recorded output on the designated problems.

Observation P3: HV 0.5085 ± 0.008993, IGD 0.02247 ± 0.00449; P6: HV 1.016 ± 0.0002737, IGD 0.01101 ± 0.0003537.

Useful for: Small continuous design with restarts. Less suitable for: Strongly coupled high-dimensional variables.

25 MOACO

Mechanism Two pheromone matrices and route construction

Implementation MOACO: two pheromone matrices, randomized tradeoff exponent, probabilistic valid tour construction, evaporation and archive deposits. Defaults alpha=1 beta=2 evaporation=.1.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P8	10	0.9239 ± 0	N/A	0.003111 ± 0.0008203	18.13 ± 2.17	10,000 / 10,000

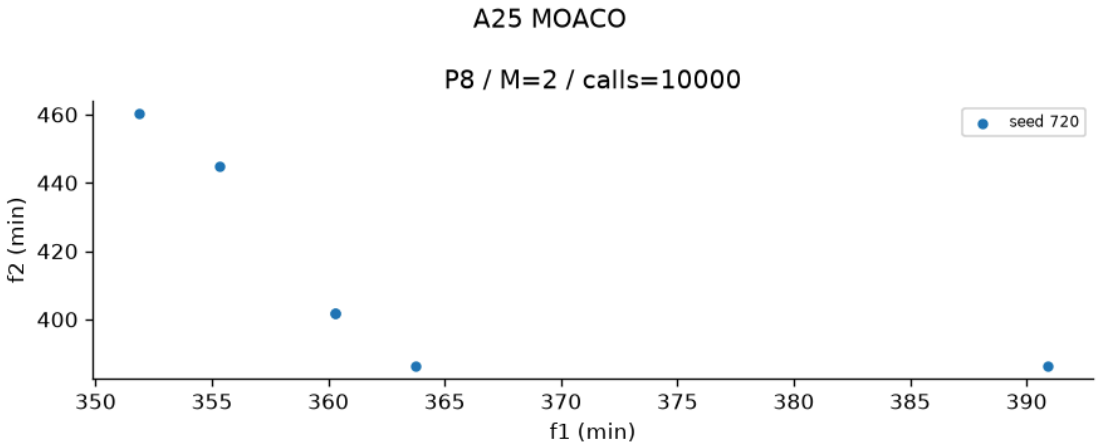


Figure A25. Recorded output on the designated problems.

Observation P8: HV 0.9239 ± 0, IGD N/A.

Useful for: Distance-time routing. Less suitable for: Continuous portfolio allocation.

26 MOSA

Mechanism Incomparable neighbour acceptance and Boltzmann dominated moves

Implementation Multiobjective Simulated Annealing: population of chains, normalized summed worsened-objective energy for dominated moves, Boltzmann acceptance and archive. Temperature T0=.1, cooling=.97 per population sweep, move scale=.1.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.5135 ± 0.006066	0.0204 ± 0.003815	0.03388 ± 0.01308	1.47 ± 0.1377	10,000 / 10,000
P5	10	1.181 ± 0.0003664	0.01318 ± 0.0007081	0.01876 ± 0.001253	1.523 ± 0.1193	10,000 / 10,000

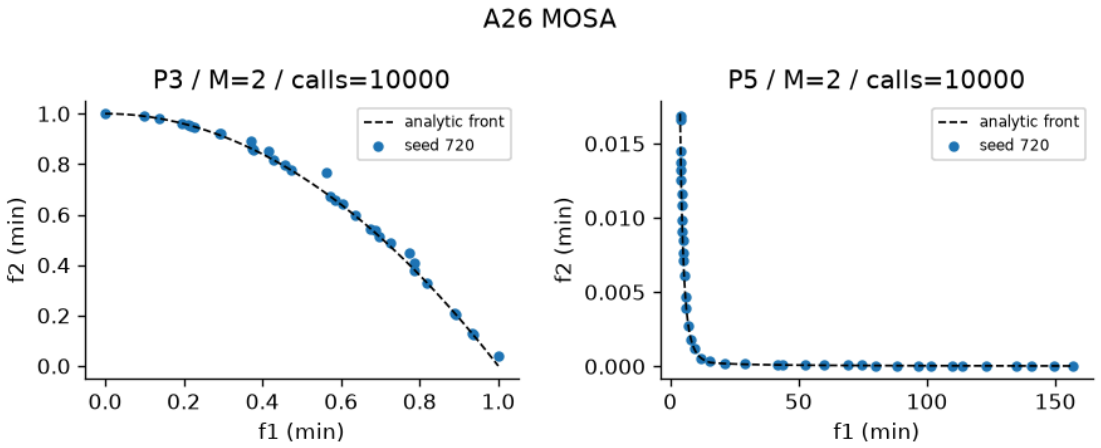


Figure A26. Recorded output on the designated problems.

Observation P3: HV 0.5135 ± 0.006066, IGD 0.0204 ± 0.003815; P5: HV 1.181 ± 0.0003664, IGD 0.01318 ± 0.0007081.

Useful for: Budgeted local exploration on nonconvex fronts. Less suitable for: Uniform global coverage with insufficient chains.

27 MODE

Mechanism DE dominance-only trial replacement

Implementation MODE: DE/rand/1 and forced binomial crossover; trial replaces target only if it dominates under Deb rules. F=.5, CR=.9.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.11 ± 3.023e-10	0.6098 ± 1.147e-10	0 ± 0	1.77 ± 0.06894	10,000 / 10,000
P5	10	1.181 ± 0.002083	0.01381 ± 0.003286	0.01812 ± 0.003533	2.19 ± 0.2773	10,000 / 10,000

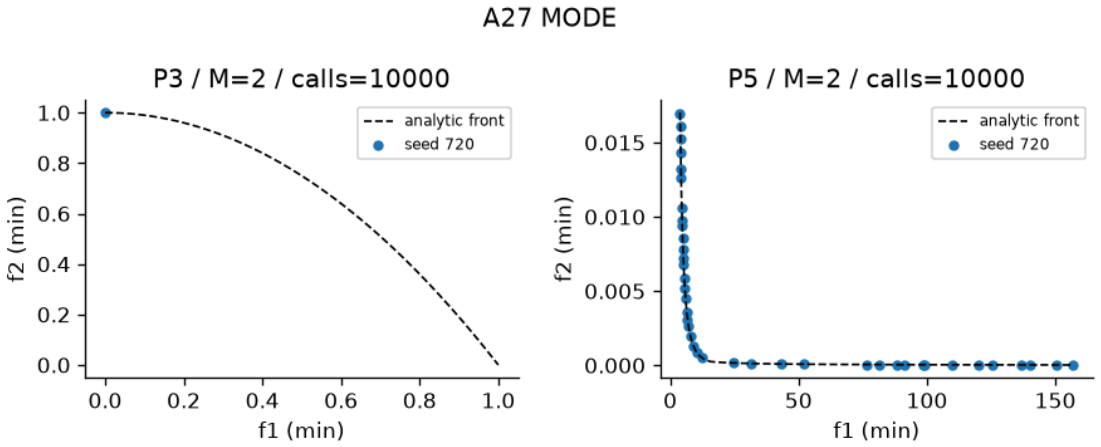


Figure A27. Recorded output on the designated problems.

Observation P3: HV 0.11 ± 3.023e-10, IGD 0.6098 ± 1.147e-10; P5: HV 1.181 ± 0.002083, IGD 0.01381 ± 0.003286.

Useful for: Continuous design with comparable improvements. Less suitable for: Many objectives with almost no domination.

28 MO-TLBO

Mechanism Crowding archive teacher, shrinking TF and nondominated learner moves

Implementation MO-TLBO: teacher phase x+r*(teacher-TF*mean); learner phase towards better peer or away from worse peer. TF shrinks from 2 to 1.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.5318 ± 7.088e-05	0.009509 ± 9.075e-05	0.003671 ± 0.001287	11.39 ± 1.237	10,000 / 10,000
P5	10	1.182 ± 1.659e-05	0.01234 ± 0.0004422	0.02285 ± 0.0009418	11.36 ± 1.012	10,000 / 10,000

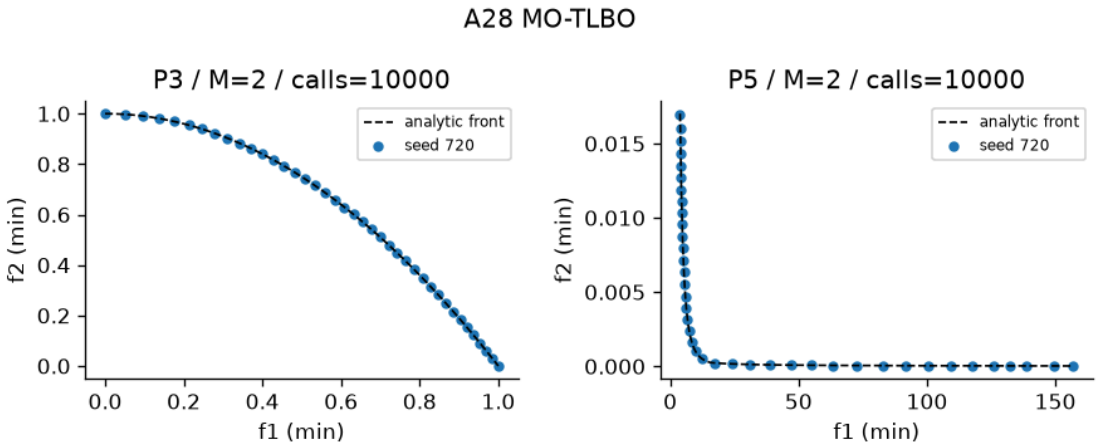


Figure A28. Recorded output on the designated problems.

Observation P3: HV 0.5318 ± 7.088e-05, IGD 0.009509 ± 9.075e-05; P5: HV 1.182 ± 1.659e-05, IGD 0.01234 ± 0.0004422.

Useful for: Small continuous educational examples. Less suitable for: Highly multimodal separated feasible regions.

29 MOBA

Mechanism Frequency velocity, pulse and loudness

Implementation MOBA: frequency velocity toward archive leader, pulse-controlled local random walk, loudness acceptance. fmin=0 fmax=2, loudness*=.95, pulse=1*(1-exp(-.1*g)).

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P3	10	0.3968 ± 0.1443	0.1289 ± 0.1631	0.04146 ± 0.05039	4.604 ± 0.229	10,000 / 10,000
P4	10	0.9332 ± 0.001286	N/A	0.01234 ± 0.002368	4.796 ± 0.3789	10,000 / 10,000

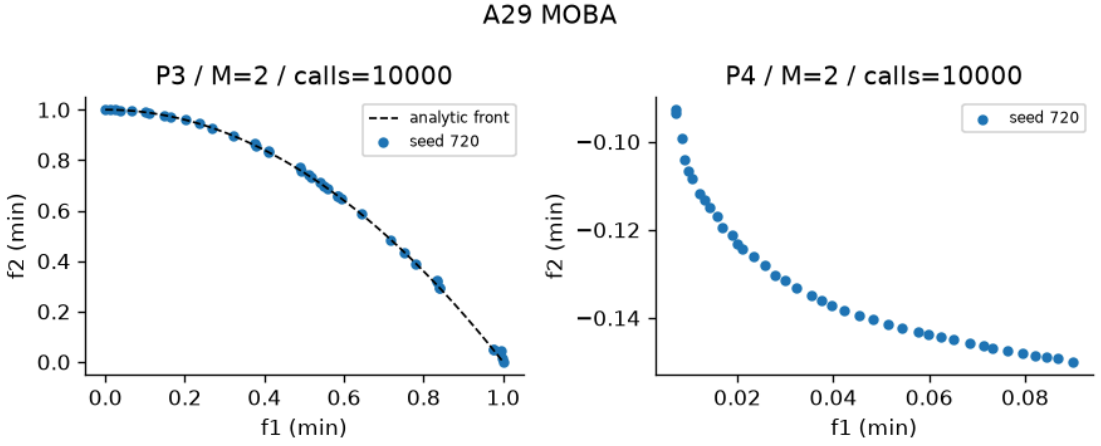


Figure A29. Recorded output on the designated problems.

Observation P3: HV 0.3968 ± 0.1443, IGD 0.1289 ± 0.1631; P4: HV 0.9332 ± 0.001286, IGD N/A.

Useful for: Continuous local refinement. Less suitable for: Robust full-front exploration without schedule tuning.

30 MOMA

Mechanism NSGA-II plus sparse-child combined-score hill climbing

Implementation MOMA: NSGA-II global search and normalized combined-improvement hill-climb on a subset of children. Local fraction=.25, mutation scale=.02.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P5	10	1.182 ± 5.827e-05	0.01291 ± 0.001555	0.01836 ± 0.001705	6.331 ± 0.652	10,000 / 10,000
P9	10	0.8736 ± 0.03212	N/A	0.0402 ± 0.03056	39.46 ± 6.746	200 / 200

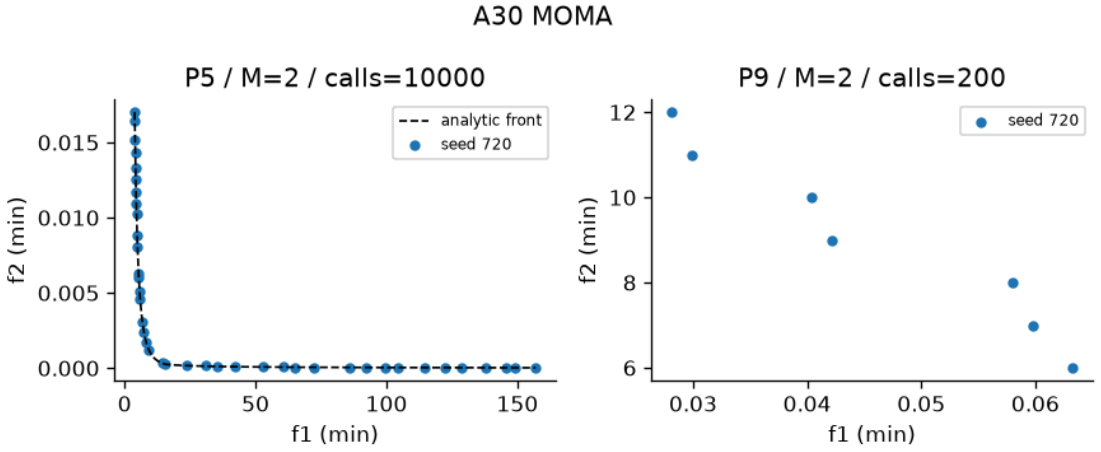


Figure A30. Recorded output on the designated problems.

Observation P5: HV 1.182 ± 5.827e-05, IGD 0.01291 ± 0.001555; P9: HV 0.8736 ± 0.03212, IGD N/A.

Useful for: Beam refinement or routing. Less suitable for: Expensive objectives with no local structure.

31 RVEA

Mechanism Normalized reference vectors and lecture linear-progress APD

Implementation RVEA: reference-vector association and angle-penalized distance (APD), penalty strength alpha=2. Range-adapt vectors every 10 generations.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P10 M3	10	0.6403 ± 0.01143	0.1088 ± 0.003233	0.1006 ± 0.01106	5.642 ± 0.5761	10,000 / 10,000
P10 M5	10	1.012 ± 0.03024	0.3111 ± 0.008554	0.2684 ± 0.02503	4.995 ± 0.5221	10,000 / 10,000

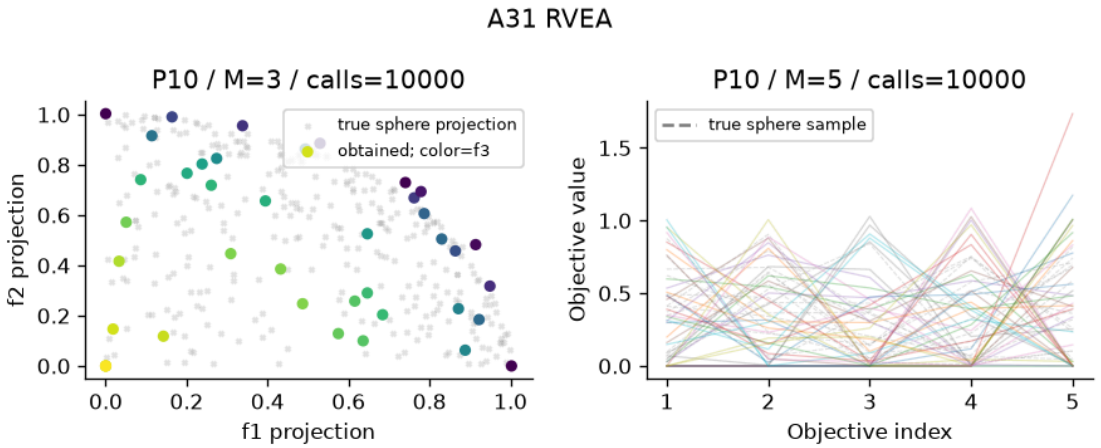


Figure A31. Recorded output on the designated problems.

Observation P10 (M=3): HV 0.6403 ± 0.01143, IGD 0.1088 ± 0.003233; P10 (M=5): HV 1.012 ± 0.03024, IGD 0.3111 ± 0.008554.

Useful for: Five-objective DTLZ2. Less suitable for: Strongly irregular disconnected fronts.

32 IBEA

Mechanism Outgoing additive-epsilon fitness and largest-fitness deletion

Implementation IBEA: additive epsilon indicator I(a,b)=max_j(a_j-b_j); outgoing fitness -sum_j exp(-I(i,j)/kappa), largest removed, iterative worst removal with recalculation. kappa=.05; normalized objective ranges; feasibility penalty before indicator.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P14	10	0.8194 ± 0.01225	0.02618 ± 0.007653	0.02763 ± 0.007352	11.83 ± 0.6681	10,000 / 10,000
P3	10	0.1928 ± 0.01702	0.3581 ± 0.02807	0.00594 ± 0.002949	11.45 ± 1.023	10,000 / 10,000

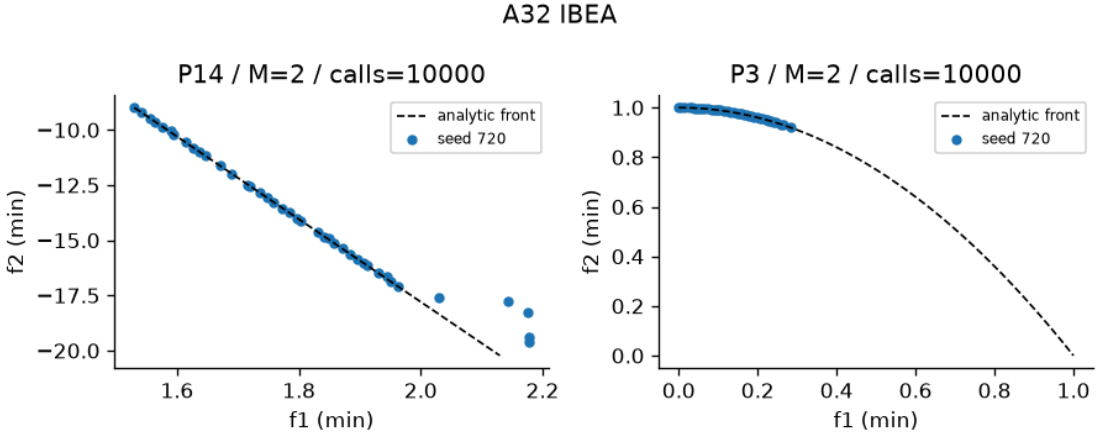


Figure A32. Recorded output on the designated problems.

Observation P14: HV 0.8194 ± 0.01225, IGD 0.02618 ± 0.007653; P3: HV 0.1928 ± 0.01702, IGD 0.3581 ± 0.02807.

Useful for: Small bi-objective indicator experiments. Less suitable for: Large populations with limited CPU.

33 SMS-EMOA

Mechanism Steady-state smallest HV deletion

Implementation SMS-EMOA: one offspring, nondominated sorting, delete least HV contribution in worst front. Exact 2-D exclusive contributions.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P14	10	0.8587 ± 1.992e-05	0.007577 ± 1.879e-05	0.002664 ± 0.0008371	30.59 ± 1.685	10,000 / 10,000
P3	10	0.5318 ± 0.000109	0.009565 ± 0.0001278	0.004121 ± 0.001085	31.09 ± 1.972	10,000 / 10,000

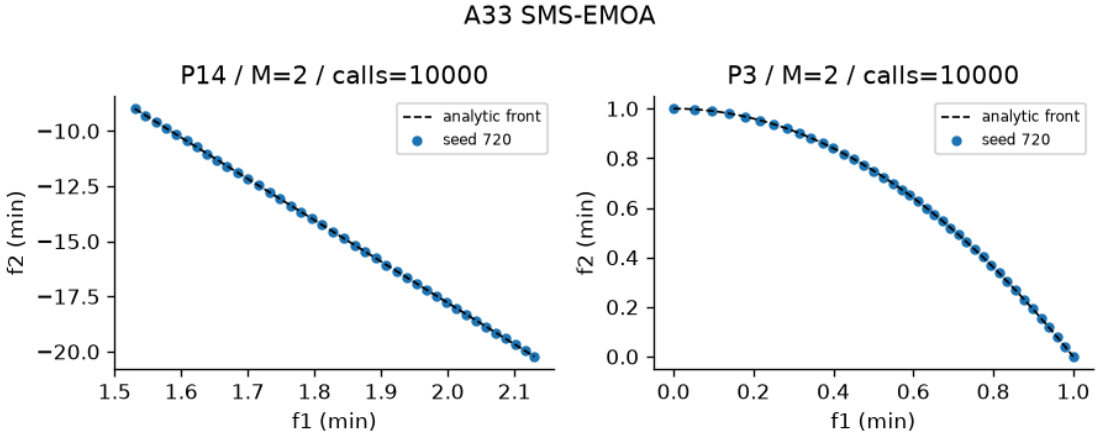


Figure A33. Recorded output on the designated problems.

Observation P14: HV 0.8587 ± 1.992e-05, IGD 0.007577 ± 1.879e-05; P3: HV 0.5318 ± 0.000109, IGD 0.009565 ± 0.0001278.

Useful for: Small bi-objective design. Less suitable for: Many objectives without an HV approximation.

34 HypE

Mechanism Equal-share Monte Carlo HV fitness and repeated deletion

Implementation HypE: Monte Carlo hypervolume fitness with equal shared credits, iterative contribution deletion. Samples=256 selection, final HV uses 8192.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P10 M3	10	0.6611 ± 0.008224	0.1171 ± 0.004252	0.08613 ± 0.0114	20.2 ± 0.6013	10,000 / 10,000
P10 M5	10	0.8057 ± 0.1473	0.4362 ± 0.07524	0.3592 ± 0.07552	23.72 ± 1.897	10,000 / 10,000

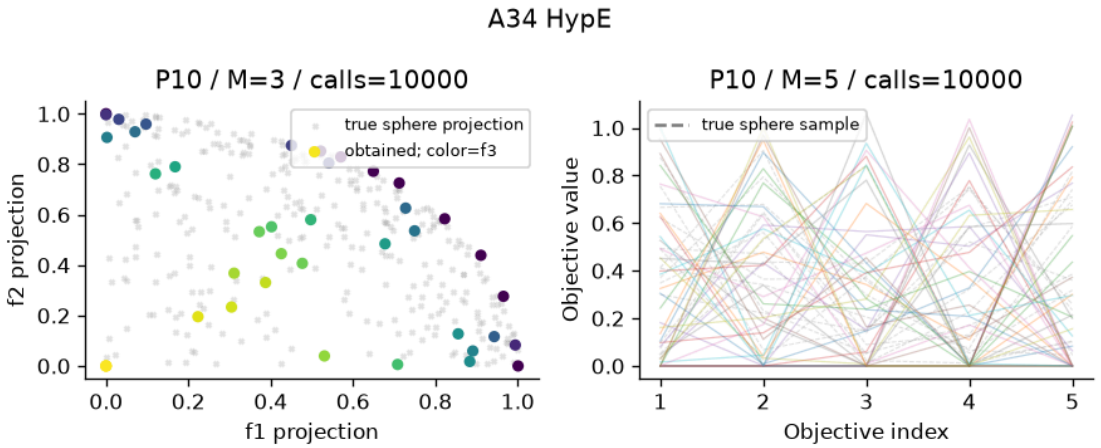


Figure A34. Recorded output on the designated problems.

Observation P10 (M=3): HV 0.6611 ± 0.008224, IGD 0.1171 ± 0.004252; P10 (M=5): HV 0.8057 ± 0.1473, IGD 0.4362 ± 0.07524.

Useful for: Many-objective indicator study. Less suitable for: High-precision survival contribution estimates.

35 R-NSGA-II

Mechanism Reference preference and epsilon clearing

Implementation R-NSGA-II: nondominated fronts, normalized distance to aspiration, epsilon clearing to spread preferred survivors. epsilon=.05; default balanced normalized aspiration [.5,.5], P14 uses natural [1.8,-12]; callers may override.

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P14	10	0.8556 ± 0.0008798	0.01004 ± 0.0008287	0.01306 ± 0.001946	4.152 ± 0.3333	10,000 / 10,000
P5	10	1.176 ± 0.01429	0.01448 ± 0.003167	0.01749 ± 0.003844	4.811 ± 0.3217	10,000 / 10,000

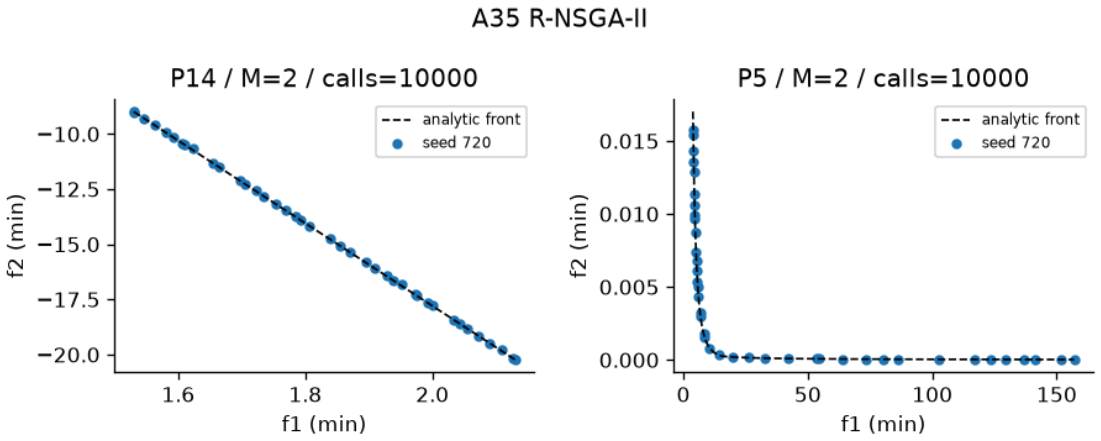


Figure A35. Recorded output on the designated problems.

Observation P14: HV 0.8556 ± 0.0008798, IGD 0.01004 ± 0.0008287; P5: HV 1.176 ± 0.01429, IGD 0.01448 ± 0.003167. The preferred region may trade whole-front coverage for decision relevance.

Useful for: Decision-maker laptop region. Less suitable for: Unknown or frequently changing preferences.

36 MGDA

Mechanism Minimum-norm gradient combination

Implementation MGDA: minimum-norm convex combination of task gradients. For two tasks solve lambda analytically; clamp to [0,1].

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P13	1	0.2839	0.6279	0	2.352	10,000 / 10,000

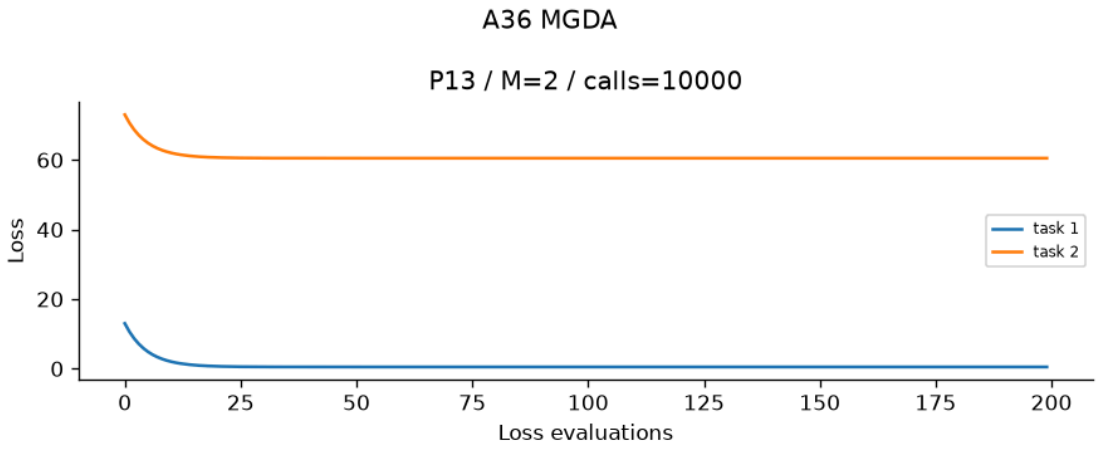


Figure A36. Recorded output on the designated problems.

Observation P13: HV 0.2839 ± 0, IGD 0.6279 ± 0. The final losses describe one training trajectory, not whole-front coverage.

Useful for: Differentiable conflicting tasks. Less suitable for: Discrete designs or a full Pareto-front sweep.

37 GradNorm

Mechanism Gradient-norm adaptive task weights

Implementation GradNorm: adaptive weights minimizing sum |G_i-target_i|. G_i=w_i*||gradient_i||; target=mean(G)*relative_training_rate**alpha, stop-gradient target, weights renormalized to sum(tasks).

Problem	Runs	HV	IGD	Spacing	Time (s)	Calls / cap
P13	1	0.6052	0.433	0	2.667	10,000 / 10,000

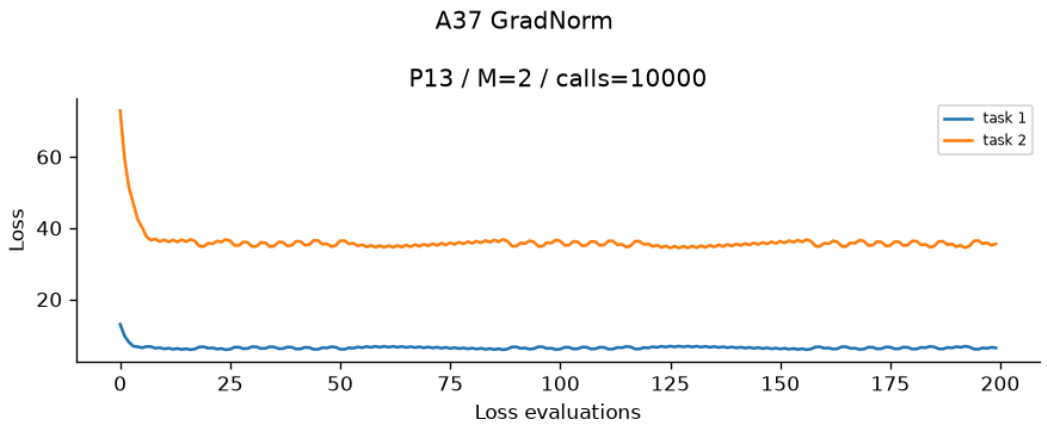


Figure A37. Recorded output on the designated problems.

Observation P13: HV 0.6052 ± 0, IGD 0.433 ± 0. The final losses describe one training trajectory, not whole-front coverage.

Useful for: Shared network with unequal loss gradients. Less suitable for: A guaranteed Pareto descent method.

6 Task C assigned study cards and hand traces

The assignment formulas produce Algorithms 35, 10 and 22. These cards describe the implemented teaching variants and do not claim original-paper equivalence.

35. R-NSGA-II — Reference Point-Based NSGA-II

Historical context: the supplied Bonus 5E lecture identifies Kalyanmoy Deb, J. Sundar, N. Udaya Bhaskara Rao and Shamik Chaudhuri (2006). The method reduces the cost of approximating an entire front when a decision-maker already knows the preferred region.

Motivation and core principle: retain NSGA-II nondominated fronts, normalize objectives, measure Euclidean distance to a decision-maker reference point, and apply ε-clearing so selected points near that reference do not collapse. If clearing leaves too few survivors, backfill in increasing reference-distance order.

Flow: parents + offspring → Deb nondominated fronts → normalized objectives and reference → splitting-front reference distances → ε-clearing → preferred survivors.

Algorithm steps and equations:

1. Combine parents and offspring and form fronts $F_1, F_2, \ldots$ using Deb feasibility rules before objective dominance.
2. Normalize $z_{i\square} = (f_{i\square} - \text{ideal}_{\square})/(\text{nadir}_{\square} - \text{ideal}_{\square})$ using fixed assignment scales. Normalize a natural-unit reference with the same transformation.
3. Compute $d_i = \sqrt{[\Sigma_{\square}(z_{i\square} - r_{\square})^2]}$.
4. Copy complete fronts. In the splitting front, inspect increasing d_i and accept i only when $\min \|z_i - z_q\| \geq \varepsilon$ for every already accepted q.
5. Backfill deferred points by increasing reference distance until the population contains N survivors.

Parameters: a wrong reference focuses on an irrelevant region. An ε that is too small permits crowding, whereas an ε that is too large clears useful neighbours and forces distance-only

backfilling. Small N gives little preferred-region resolution; large N costs more evaluations per generation. Crossover and mutation control exploration.

Convergence, diversity and constraints: nondominated rank supplies convergence pressure before preference distance is used in the splitting front. ε-clearing supplies local diversity. Deb rules handle constraints. Nondominated sorting can follow nonconvex fronts, but an unreachable reference and finite population can bias coverage. A k-dimensional reference extends beyond three objectives, although distance concentration and ε tuning become harder.

Complexity: pairwise sorting is $O(MN^2)$ for M objectives and 2N candidates; reference-distance calculation and straightforward clearing add $O(MN^2)$.

Strengths and limitations: the method directs evaluations toward a decision-relevant region while retaining Pareto ranking and explicit local spacing. Its answer depends on reference scaling and ε, and it does not provide uniform whole-front coverage when preference pressure is strong.

Sensitivity evidence: ε = 0.02, 0.05 and 0.10 on P14, using ten independent optimizer seeds 720–729. Raw rows and aggregate statistics are stored under results/roll72/; the sensitivity figure is Figure C35.

10. VEGA — Vector Evaluated Genetic Algorithm

Historical context: David Schaffer proposed VEGA in 1985. It was an early extension of genetic algorithms to several objectives, but objective-wise selection can duplicate extreme solutions and discard balanced compromises.

Motivation and core principle: evaluate all objectives, allocate equal mating-pool slots to the objectives, select each slot using only its assigned objective, merge and shuffle the subpools, then apply crossover and mutation before nonelitist replacement.

Flow: population → objective-specific tournaments and subpools → merged mating pool → crossover and mutation → nonelitist offspring replacement.

Algorithm steps and equations:

1. Evaluate every objective $f_j(x_i)$ and the constraint violation of each candidate.
2. For mating slot i, use objective j = i mod M; compare two candidates using Deb feasibility and then scalar value f_j when both are feasible.
3. Pool the N winners. Each objective receives equal slots when N is divisible by M.
4. Apply the shared real, binary or permutation-safe crossover and mutation operators appropriate to the problem encoding.
5. Replace the parents with the offspring. The passive output archive records discoveries but does not make the breeding algorithm elitist.

Parameters: small N increases genetic drift; large N improves instantaneous coverage but yields fewer generations under a fixed evaluation budget. Excessive mutation destroys selected structure, whereas insufficient mutation cannot restore lost regions. Strong crossover can mix extremes but provides no diversity guarantee.

Convergence, diversity and constraints: single-objective subpools create pressure toward extremes but provide no explicit Pareto-convergence or diversity pressure. Deb feasibility is used inside every tournament. Search operators can visit nonconvex fronts, but selection does

not protect their interior. With many objectives, each subpool becomes small and balanced tradeoffs receive weaker selection pressure.

Complexity: objective-wise selection requires O(N) comparisons after O(MN) evaluation; variation is O(ND) for D variables. The implementation's passive archive adds pairwise bookkeeping but does not affect parent selection.

Strengths and limitations: VEGA is simple, inexpensive and exposes objective conflicts clearly. Its nonelitist objective partitioning can overrepresent extremes, lose compromises and make final-population coverage unstable.

Sensitivity evidence: population N = 20, 40 and 80 on P3, using ten independent optimizer seeds 720–729. At a fixed budget, this tests simultaneous coverage against the number of generations; see Figure C10.

22. MOPSO — Multi-Objective Particle Swarm Optimization

Historical context: Carlos A. Coello Coello and Maximino Salazar Lechuga proposed MOPSO in 2002. It extends scalar PSO by replacing one global best with guides selected from an external nondominated archive.

Motivation and core principle: maintain particle positions, velocities and personal bests; retain a nondominated external archive; prefer leaders from sparse archive regions; move toward the personal best and leader; mutate selected particles; and update a personal best only when the new point dominates it.

Flow: particles + archive → crowding-weighted leader → bounded velocity and position update → evaluation and mutation → dominance-based personal best → nondominated archive.

Algorithm steps and equations:

1. Initialize x, set velocity v = 0 and pbest = x, then build the nondominated archive.
2. Select an archive leader using crowding-derived probabilities.
3. Update each coordinate using $v' = wv + c_1r_1(\text{pbest} - x) + c_2r_2(\text{leader} - x)$.
4. In the main implementation, clip velocity to 20% of the decision-variable range, set $x' = \text{repair}(x + v')$, and apply occasional mutation. The separate hand-worked teaching trace explicitly uses fixed limits $[-10,10]$.
5. Replace pbest only if x′ dominates it under Deb rules; merge x′ into the nondominated archive and continue until the evaluation budget is exhausted.

Parameters: small inertia w can stagnate quickly, whereas large w may overshoot and weaken exploitation. Small c_1 underuses personal experience; large c_1 can cause oscillation around pbest. Small c_2 underuses the front; large c_2 can crowd particles around leaders. A small archive loses coverage; an oversized archive increases maintenance cost.

Convergence, diversity and constraints: personal bests and archive leaders supply convergence pressure, while crowding-weighted leader choice and mutation support diversity. Deb dominance handles constraints. The swarm can sample nonconvex fronts because it does not scalarize them, but its continuous velocity model is a poor natural fit for permutations. In many objectives, dominance and crowding lose discrimination and archive maintenance becomes costly.

Complexity: velocity updates cost O(ND); straightforward archive dominance and crowding cost $O(MA^2)$ for archive size A, plus problem-evaluation cost.

Strengths and limitations: MOPSO combines memory, an external Pareto archive and gradient-free movement. Performance is sensitive to inertia, acceleration coefficients, velocity limits, mutation and archive-leader pressure.

Sensitivity evidence: inertia w = 0.2, 0.4 and 0.8 on P4, using ten independent optimizer seeds 720–729; see Figure C22. IGD is unavailable because P4 has no exact reference front, and the plot states that limitation.

Core update equations

The following summarize the update rules; the full selection procedures and worked examples follow.

$$z = (f - ideal) / (nadir - ideal)$$

$$d = \|z - r\|_2 \;;\; clearing\ separation \geq \varepsilon$$

VEGA: mating slot i selects by objective j = i mod M.

$$v(next) = \omega v + c_1 r_1 (pbest - x) + c_2 r_2 (leader - x)$$

$$x(next) = repair[x + v(next)]$$

Flow diagram A35 R-NSGA-II

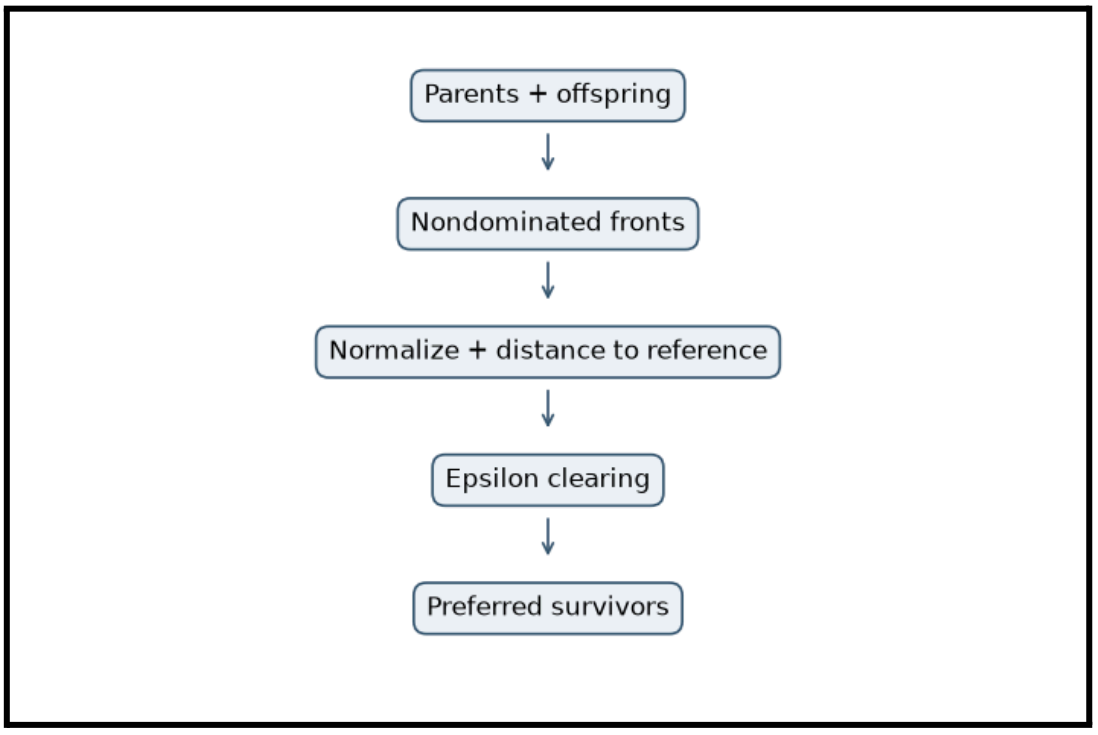


Figure C35 Current implementation control flow

Flow diagram A10 VEGA

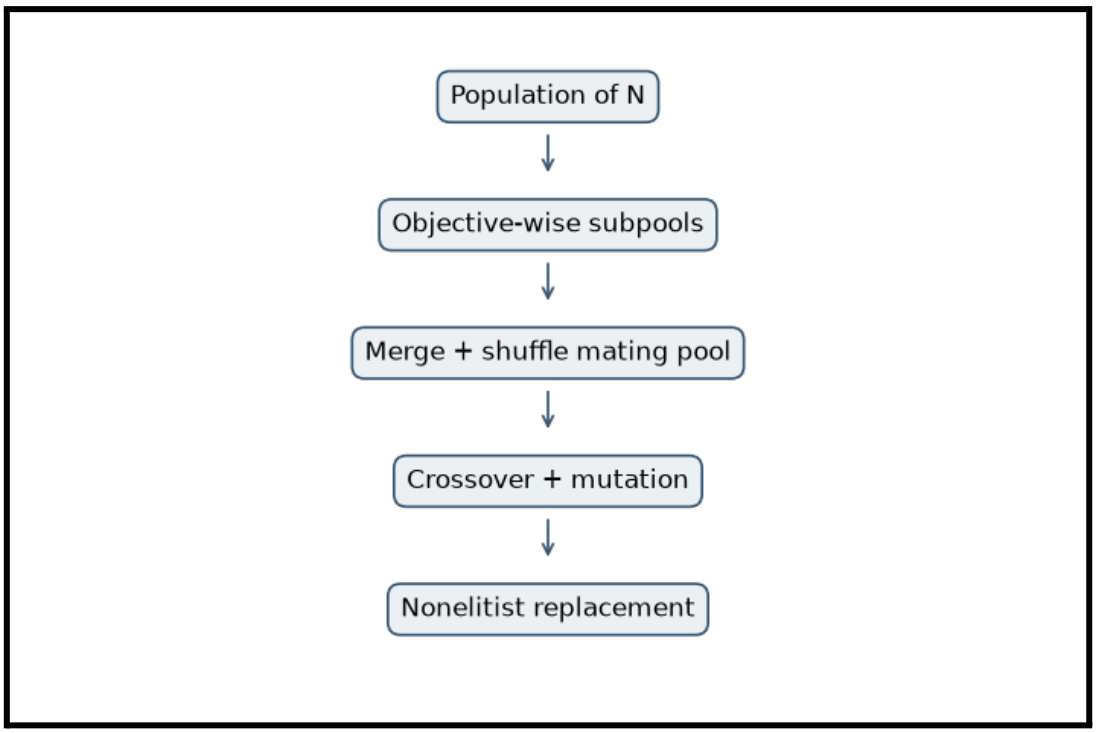


Figure C10 Current implementation control flow

Flow diagram A22 MOPSO

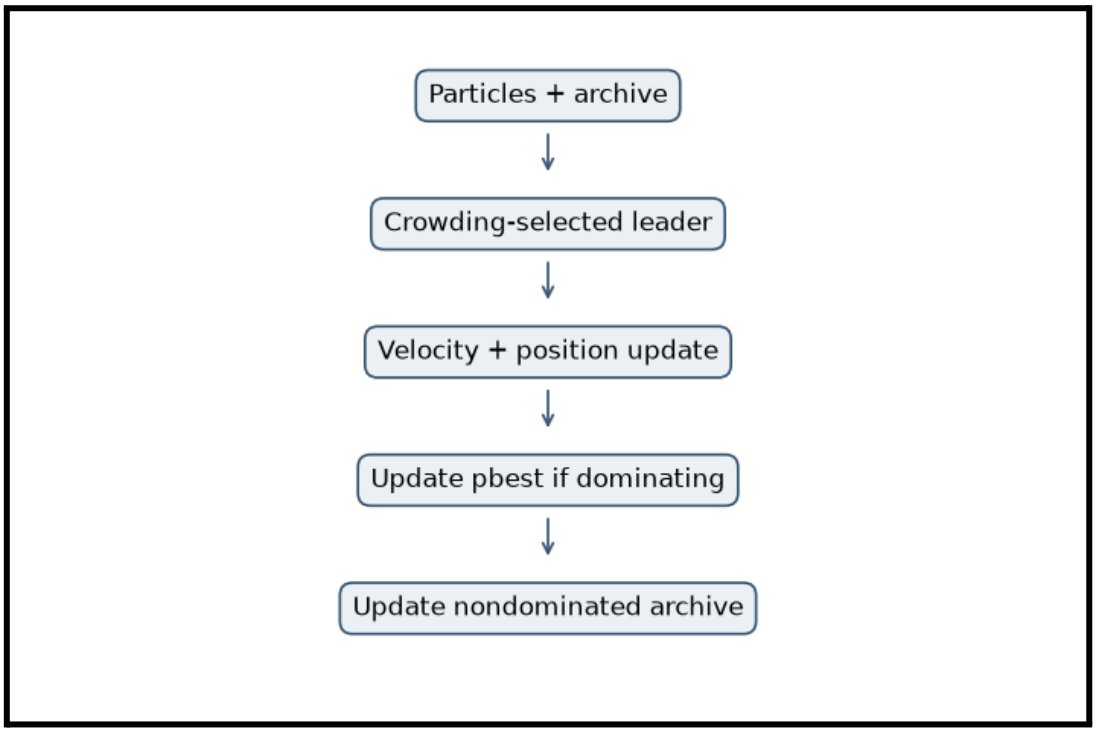


Figure C22 Current implementation control flow

R-NSGA-II two selection iterations

Tiny instance: P1 with f1=(x-3)^2 and f2=(x-7)^2, population N=4, natural-unit reference r=(4,4), fixed ideal=(0,0), nadir=(16,16), and therefore normalized reference (0.25,0.25). Epsilon=0.20. Parents and four fixed offspring form eight candidates. Deb nondominated

sorting is applied first; within the splitting front, candidates are ordered by Euclidean reference distance and epsilon clearing controls local spacing.

R-NSGA-II iteration 1

Parents=[3.0, 4.0, 6.0, 7.0]; offspring=[3.5, 4.5, 5.5, 6.5]; combined indices 0-3 are parents and 4-7 are offspring.

i	Origin	x	f1	f2	z1	z2	d(i,r)
0	P	3	0	16	0	1	0.7906
1	P	4	1	9	0.0625	0.5625	0.3644
2	P	6	9	1	0.5625	0.0625	0.3644
3	P	7	16	0	1	0	0.7906
4	O	3.5	0.25	12.25	0.01562	0.7656	0.5664
5	O	4.5	2.25	6.25	0.1406	0.3906	0.1782
6	O	5.5	6.25	2.25	0.3906	0.1406	0.1782
7	O	6.5	12.25	0.25	0.7656	0.01562	0.5664

Nondominated fronts (zero-based candidate indices): [[0, 1, 2, 3, 4, 5, 6, 7]]. All eight candidates trade off f1 against f2, so the first front is also the splitting front.

Order	i	x	Reference d	Min separation	Decision
1	6	5.5	0.1782	inf	accept
2	5	4.5	0.1782	0.3536	accept
3	1	4	0.3644	0.1888	defer
4	2	6	0.3644	0.1888	defer
5	4	3.5	0.5664	0.3953	accept
6	7	6.5	0.5664	0.3953	accept

Selected survivor indices=[6, 5, 4, 7], hence x=[5.5, 4.5, 3.5, 6.5]. These four survivors become the parents shown in the next iteration.

R-NSGA-II iteration 2

Parents=[5.5, 4.5, 3.5, 6.5]; offspring=[4.25, 4.75, 5.25, 5.75]; combined indices 0-3 are parents and 4-7 are offspring.

i	Origin	x	f1	f2	z1	z2	d(i,r)
0	P	5.5	6.25	2.25	0.3906	0.1406	0.1782
1	P	4.5	2.25	6.25	0.1406	0.3906	0.1782
2	P	3.5	0.25	12.25	0.01562	0.7656	0.5664

i	Origin	x	f1	f2	z1	z2	d(i,r)
3	P	6.5	12.25	0.25	0.7656	0.01562	0.5664
4	O	4.25	1.562	7.562	0.09766	0.4727	0.2698
5	O	4.75	3.062	5.062	0.1914	0.3164	0.08856
6	O	5.25	5.062	3.062	0.3164	0.1914	0.08856
7	O	5.75	7.562	1.562	0.4727	0.09766	0.2698

Nondominated fronts (zero-based candidate indices): [[0, 1, 2, 3, 4, 5, 6, 7]]. All eight candidates trade off f1 against f2, so the first front is also the splitting front.

Order	i	x	Reference d	Min separation	Decision
1	6	5.25	0.08856	inf	accept
2	5	4.75	0.08856	0.1768	defer
3	0	5.5	0.1782	0.08993	defer
4	1	4.5	0.1782	0.2657	accept
5	4	4.25	0.2698	0.0926	defer
6	7	5.75	0.2698	0.1822	defer
7	3	6.5	0.5664	0.4824	accept
8	2	3.5	0.5664	0.3953	accept

Selected survivor indices=[6, 1, 3, 2], hence x=[5.25, 4.5, 6.5, 3.5]. These four survivors become the parents shown in the next iteration.

Code verification: tests/test_traces.py recomputes normalized objectives and Euclidean reference distances, then asserts the recorded clearing decisions and survivor indices for both iterations.

VEGA two objective-wise selections

Tiny instance: the same P1 minimization problem and population N=4. Mating slots 1 and 3 use f1; slots 2 and 4 use f2. Each slot has an explicit two-candidate tournament. Variation uses the implementation's SBX equation c1=0.5[(1+beta)p1+(1-beta)p2], c2=0.5[(1-beta)p1+(1+beta)p2], with u=0.25, beta=(2u)^(1/16), positive sign, and zero mutation increment.

VEGA iteration 1

i	x	f1 min	f2 min
0	3	0	16
1	4	1	9
2	6	9	1
3	7	16	0

Slot	Objective	Tournament indices	Candidate x values	Winner	Winner x
1	f1	[0, 2]	['3', '6']	0	3
2	f2	[1, 3]	['4', '7']	3	7
3	f1	[1, 2]	['4', '6']	1	4
4	f2	[0, 2]	['3', '6']	2	6

Objective-wise mating pool=[3, 7, 4, 6]; parent pairs=[[3, 7], [4, 6]]. With u=0.25, beta=0.9576; mutation increments=[0, 0, 0, 0].

Child	x after SBX	f1	f2	Next population
1	3.085	0.00719	15.33	3.085
2	6.915	15.33	0.00719	6.915
3	4.042	1.087	8.747	4.042
4	5.958	8.747	1.087	5.958

The offspring replace the parents without elitist parent-plus-child survival. This explicit update produces the population used at the start of the next iteration.

VEGA iteration 2

i	x	f1 min	f2 min
0	3.085	0.00719	15.33
1	6.915	15.33	0.00719
2	4.042	1.087	8.747
3	5.958	8.747	1.087

Slot	Objective	Tournament indices	Candidate x values	Winner	Winner x
1	f1	[0, 2]	['3.085', '4.042']	0	3.085
2	f2	[1, 3]	['6.915', '5.958']	1	6.915

Slot	Objective	Tournament indices	Candidate x values	Winner	Winner x
3	f1	[2, 3]	['4.042', '5.958']	2	4.042
4	f2	[2, 3]	['4.042', '5.958']	3	5.958

Objective-wise mating pool=[3.085, 6.915, 4.042, 5.958]; parent pairs=[[3.085, 6.915], [4.042, 5.958]]. With u=0.25, beta=0.9576; mutation increments=[0, 0, 0, 0].

Child	x after SBX	f1	f2	Next population
1	3.166	0.02755	14.7	3.166
2	6.834	14.7	0.02755	6.834
3	4.083	1.173	8.509	4.083
4	5.917	8.509	1.173	5.917

The offspring replace the parents without elitist parent-plus-child survival. This explicit update produces the population used at the start of the next iteration.

Code verification: tests/test_traces.py checks the objective-wise winner indices and independently recomputes every parent and offspring objective used in both population updates.

MOPSO two velocity iterations

Tiny instance: this is an independent scalar MOPSO teaching problem, not assignment P2. It uses x in [1,20], teaching velocity limits [-10,10], and objectives f1=100/x and f2=x^2, both minimized. These fixed teaching limits deliberately differ from the main implementation default, which caps each velocity component at 20% of its decision-variable range. The wider toy domain makes x=12 and x=15 valid. The update is v-new=w v+c1 r1(pbest-x)+c2 r2(leader-x), followed by clipping and x-new=x+v-new. Here w=0.7, c1=c2=1.5, and every r1=r2=0.5. Leaders are explicit archive members. A personal best changes only if the new position dominates it.

MOPSO iteration 1

Archive before update=[5.0, 10.0, 15.0, 12.0]; selected leaders=[15.0, 15.0, 5.0, 5.0]; coefficients w=0.7, c1=1.5, c2=1.5, r1=r2=0.5.

i	x	v	pbest	leader	wv	cognitive	social	v-new	x-new
0	5	0	5	15	0	0	7.5	7.5	12.5
1	10	0	10	15	0	0	3.75	3.75	13.75
2	15	0	15	5	0	0	-7.5	-7.5	7.5
3	12	0	12	5	0	0	-5.25	-5.25	6.75

i	Old objectives	New objectives	pbest decision	pbest after
0	['20', '25']	['8', '156.2']	retain: incomparable	5
1	['10', '100']	['7.273', '189.1']	retain: incomparable	10
2	['6.667', '225']	['13.33', '56.25']	retain: incomparable	15
3	['8.333', '144']	['14.81', '45.56']	retain: incomparable	12

Every move improves one objective and worsens the other, so each trial is incomparable with its pbest and the pbest values remain [5.0, 10.0, 15.0, 12.0]. The nondominated archive becomes [5.0, 10.0, 15.0, 12.0, 12.5, 13.75, 7.5, 6.75] (size 8).

MOPSO iteration 2

Archive before update=[5.0, 10.0, 15.0, 12.0, 12.5, 13.75, 7.5, 6.75]; selected leaders=[15.0, 15.0, 5.0, 5.0]; coefficients w=0.7, c1=1.5, c2=1.5, r1=r2=0.5.

i	x	v	pbest	leader	wv	cognitive	social	v-new	x-new
0	12.5	7.5	5	15	5.25	-5.625	1.875	1.5	14
1	13.75	3.75	10	15	2.625	-2.812	0.9375	0.75	14.5
2	7.5	-7.5	15	5	-5.25	5.625	-1.875	-1.5	6
3	6.75	-5.25	12	5	-3.675	3.938	-1.312	-1.05	5.7

i	Old objectives	New objectives	pbest decision	pbest after
0	['8', '156.2']	['7.143', '196']	retain: incomparable	5
1	['7.273', '189.1']	['6.897', '210.2']	retain: incomparable	10
2	['13.33', '56.25']	['16.67', '36']	retain: incomparable	15
3	['14.81', '45.56']	['17.54', '32.49']	retain: incomparable	12

Every move improves one objective and worsens the other, so each trial is incomparable with its pbest and the pbest values remain [5.0, 10.0, 15.0, 12.0]. The nondominated archive becomes [5.0, 10.0, 15.0, 12.0, 12.5, 13.75, 7.5, 6.75, 14.0, 14.5, 6.0, 5.7] (size 12).

Code verification: tests/test_traces.py checks each inertia, cognitive and social term; verifies x-new=x+v-new; recomputes objectives; validates pbest decisions; and confirms that each updated archive is pairwise nondominated.

Ten trial sensitivity studies

R-NSGA-II

Setting	Trials	HV	IGD	Spacing	Evaluations
Clearing ε = 0.02	10	0.8559 ± 0.0008413	0.00982 ± 0.000661	0.01451 ± 0.002032	10,000
Clearing ε = 0.05	10	0.8556 ± 0.0008798	0.01004 ± 0.0008287	0.01306 ± 0.001946	10,000
Clearing ε = 0.1	10	0.8553 ± 0.001791	0.01025 ± 0.001353	0.01444 ± 0.002119	10,000

Epsilon controls the minimum normalized separation near the reference: increasing it strengthens local clearing but can force distance-based backfill. Largest mean HV occurred at Clearing ε = 0.02 (0.8559 ± 0.0008413); smallest at Clearing ε = 0.1 (0.8553 ± 0.001791). These three settings and ten trials are a finite sensitivity comparison, not a claim of a globally optimal parameter. Overlapping sample SD requires caution.

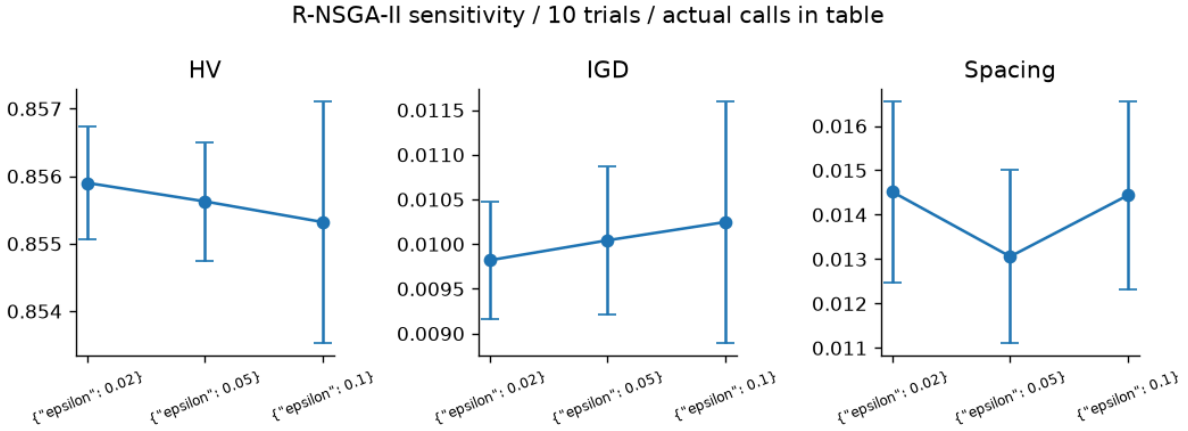


Figure S35 Sensitivity metrics with sample SD across ten seeds

VEGA

Setting	Trials	HV	IGD	Spacing	Evaluations
Population N = 20	10	0.1689 ± 0.05214	0.3754 ± 0.07938	0.0485 ± 0.02062	10,000
Population N = 40	10	0.173 ± 0.04583	0.3587 ± 0.07168	0.0475 ± 0.02233	10,000
Population N = 80	10	0.1681 ± 0.03236	0.3766 ± 0.05733	0.05438 ± 0.04213	10,000

Population size trades simultaneous coverage against the number of generations available under the same 10,000-evaluation budget. Largest mean HV occurred at Population N = 40 (0.173 ± 0.04583); smallest at Population N = 80 (0.1681 ± 0.03236). These three settings and ten trials are a finite sensitivity comparison, not a claim of a globally optimal parameter. Overlapping sample SD requires caution.

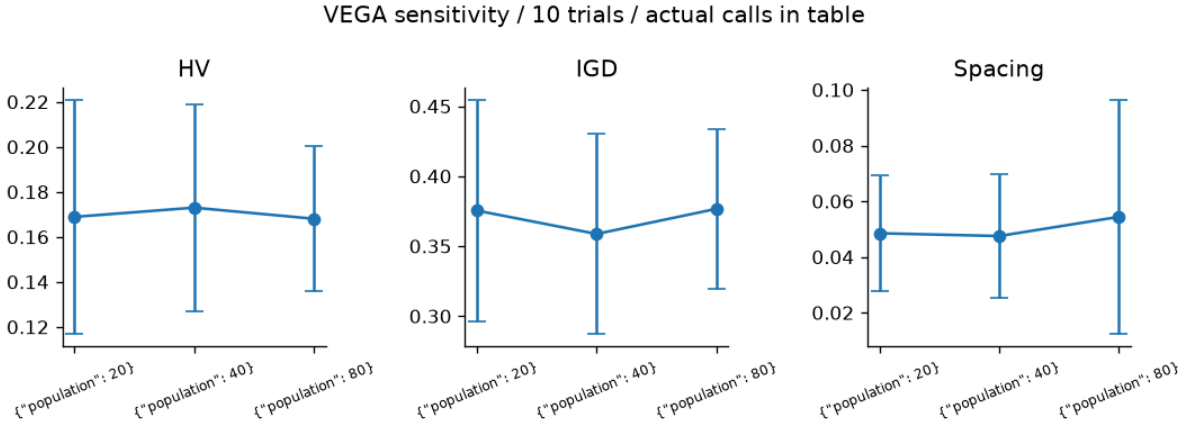


Figure S10 Sensitivity metrics with sample SD across ten seeds

MOPSO

Setting	Trials	HV	IGD	Spacing	Evaluations
Inertia w = 0.2	10	0.9156 ± 0.01095	N/A	0.01346 ± 0.006732	10,000

Setting	Trials	HV	IGD	Spacing	Evaluations
Inertia w = 0.4	10	0.9259 ± 0.00455	N/A	0.01343 ± 0.003755	10,000
Inertia w = 0.8	10	0.9337 ± 0.0006802	N/A	0.01272 ± 0.003053	10,000

Inertia controls retained momentum: low values emphasize current leaders, while high values explore more aggressively and may overshoot. Largest mean HV occurred at Inertia w = 0.8 (0.9337 ± 0.0006802); smallest at Inertia w = 0.2 (0.9156 ± 0.01095). P4 has no exact reference front, so IGD is intentionally unavailable and HV/Spacing carry the interpretation. These three settings and ten trials are a finite sensitivity comparison, not a claim of a globally optimal parameter. Overlapping sample SD requires caution.

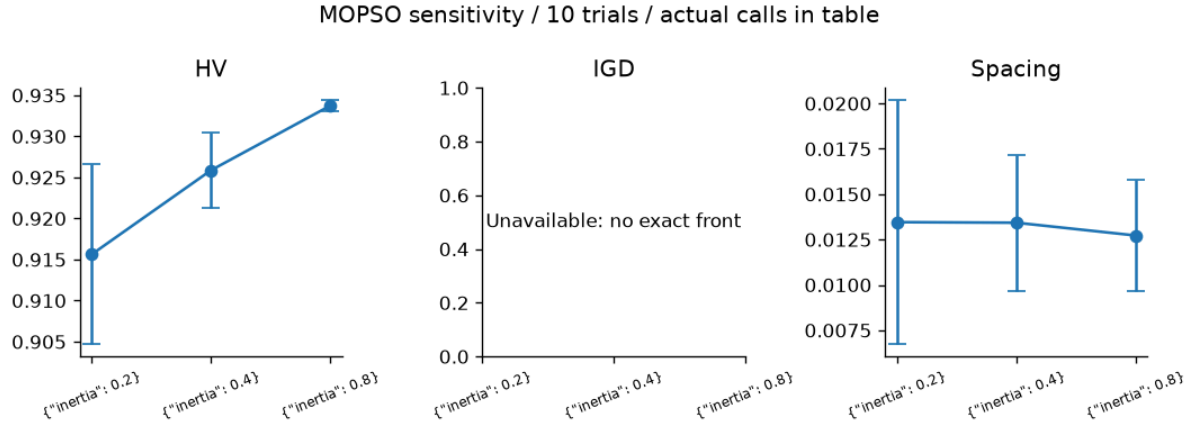


Figure S22 Sensitivity metrics with sample SD across ten seeds

7 Task D head to head comparisons

The continuous comparison includes 25 applicable two-objective optimizer mechanisms. Ranking methods are evaluated separately; ant construction requires tours, gradient methods require differentiability, and many-objective-specific methods are compared on P10. Algorithms 5 to 7 retain their designated utility goals and priorities rather than receiving invented P3/P5 goals. Algorithms 1 to 4 use at least 21 sweep settings, with metric p variants combined.

P3 comparison

Rank	Algorithm	HV	IGD	Spacing	Runtime s
1	SMS-EMOA	0.5318 ± 0.000109	0.009565 ± 0.0001278	0.004121 ± 0.001085	31 ± 2.096
2	MOEA/D	0.5318 ± 8.619e-05	0.009567 ± 0.0001113	0.003944 ± 0.0007786	14.52 ± 0.7945
3	MO-TLBO	0.5318 ± 7.088e-05	0.009509 ± 9.075e-05	0.003671 ± 0.001287	11.37 ± 1.419
4	R-NSGA-II	0.5296 ± 0.0007374	0.01162 ± 0.0005695	0.01701 ± 0.003134	4.502 ± 0.1809
5	SPEA2	0.5292 ± 0.0006002	0.01178 ± 0.0006505	0.01704 ± 0.002577	5.418 ± 0.1784
6	MOMA	0.5291 ± 0.001011	0.01142 ± 0.0006054	0.0158 ± 0.002371	5.549 ± 0.374

Rank	Algorithm	HV	IGD	Spacing	Runtime s
7	NSGA-II	0.5286 ± 0.001001	0.01208 ± 0.001244	0.01765 ± 0.001747	3.562 ± 0.2768
8	Constrained NSGA-II	0.5286 ± 0.001001	0.01208 ± 0.001244	0.01765 ± 0.001747	3.353 ± 0.08979
9	NSGA	0.5252 ± 0.001173	0.01229 ± 0.0005384	0.01471 ± 0.001894	3.53 ± 0.2706
10	MOGA	0.5238 ± 0.001085	0.0127 ± 0.0004731	0.01277 ± 0.002314	3.737 ± 0.4774
11	Weight Metric	0.5215 ± 0.003503	0.02038 ± 0.006272	0.03764 ± 0.006386	2.514 ± 0.1739
12	Epsilon-Constraint	0.5201 ± 0.001464	0.0277 ± 0.003558	0.04664 ± 0.009819	0.5874 ± 0.03399
13	MOSA	0.5135 ± 0.006066	0.0204 ± 0.003815	0.03388 ± 0.01308	1.495 ± 0.1688
14	PESA-II	0.511 ± 0.007104	0.02193 ± 0.003303	0.0234 ± 0.007883	5.214 ± 0.1837
15	MOABC	0.5085 ± 0.008993	0.02247 ± 0.00449	0.02386 ± 0.004992	9.622 ± 0.7525
16	NPGA	0.4572 ± 0.07144	0.05291 ± 0.06393	0.0329 ± 0.01947	6.44 ± 0.3875
17	MOPSO	0.4191 ± 0.03166	0.08923 ± 0.02883	0.01467 ± 0.004929	1.864 ± 0.127
18	MOBA	0.3968 ± 0.1443	0.1289 ± 0.1631	0.04146 ± 0.05039	4.471 ± 0.2249
19	MOGWO	0.277 ± 0.2156	0.371 ± 0.3083	0.006815 ± 0.008808	3.479 ± 1.438
20	MOEA/D-DE	0.2124 ± 0.2742	1.042 ± 0.9331	0.001026 ± 0.001503	11.8 ± 2.105
21	Weighted Sum	0.21 ± 1.426e-11	0.3538 ± 2.065e-11	0.8971 ± 0.3252	0.4034 ± 0.4445
22	Benson reference improvement	0.21 ± 9.846e-12	0.3538 ± 1.133e-11	0.9732 ± 0.08257	0.4763 ± 0.07296
23	IBEA	0.1928 ± 0.01702	0.3581 ± 0.02807	0.00594 ± 0.002949	11.41 ± 0.9504
24	VEGA	0.173 ± 0.04583	0.3587 ± 0.07168	0.0475 ± 0.02233	3.036 ± 0.2107
25	MODE	0.11 ± 3.023e-10	0.6098 ± 1.147e-10	0 ± 0	1.785 ± 0.1596

Ranked by mean HV: SMS-EMOA leads at 0.5318 ± 0.000109, compared with MODE at 0.11 ± 3.023e-10. HV and IGD judge different aspects; this ranking is specific to the fixed scales and reference. All rows have ten trials. Classical actual calls differ despite the common 10,000 cap; see raw_experiments.csv.

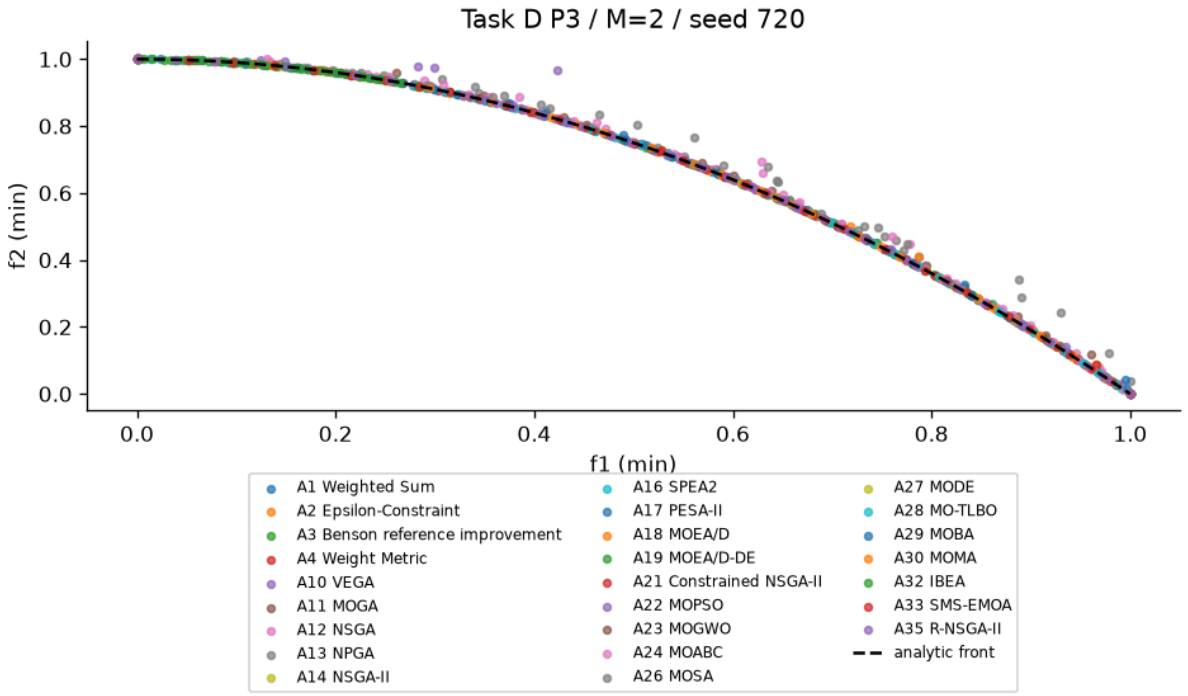


Figure DP3 Combined seed 720 results. The plot shows the recorded nondominated objective vectors for the stated problem and seed.

P5 comparison

Rank	Algorithm	HV	IGD	Spacing	Runtime s
1	MOEA/D	1.182 ± 7.646e-06	0.02074 ± 0.009097	0.02306 ± 0.002372	15.25 ± 0.9376
2	SMS-EMOA	1.182 ± 6.696e-06	0.01369 ± 0.004566	0.02108 ± 0.001477	30.43 ± 1.032
3	MO-TLBO	1.182 ± 1.659e-05	0.01234 ± 0.0004422	0.02285 ± 0.0009418	11.35 ± 0.7589
4	MOEA/D-DE	1.182 ± 3.231e-05	0.0491 ± 0.03155	0.02913 ± 0.02089	12.84 ± 0.5254
5	MOMA	1.182 ± 5.827e-05	0.01291 ± 0.001555	0.01836 ± 0.001705	6.319 ± 0.8465
6	MOBA	1.182 ± 3.443e-05	0.01329 ± 0.002086	0.01959 ± 0.001685	4.263 ± 0.3058
7	NSGA-II	1.182 ± 3.963e-05	0.01406 ± 0.001287	0.02067 ± 0.003274	3.56 ± 0.2764
8	Constrained NSGA-II	1.182 ± 3.963e-05	0.01406 ± 0.001287	0.02067 ± 0.003274	3.286 ± 0.3015
9	SPEA2	1.182 ± 0.0001209	0.01608 ± 0.002603	0.01923 ± 0.001586	8.208 ± 0.3645
10	NSGA	1.182 ± 6.707e-05	0.01358 ± 0.001486	0.01959 ± 0.002904	3.617 ± 0.1725
11	MOGA	1.182 ± 0.0001344	0.01403 ± 0.00126	0.02033 ± 0.002006	3.602 ± 0.2128
12	MOGWO	1.182 ± 0.0001194	0.01341 ± 0.001289	0.02077 ± 0.001969	4.135 ± 0.4022

Rank	Algorithm	HV	IGD	Spacing	Runtime s
13	Weight Metric	1.182 ± 1.515e-06	0.133 ± 5.874e-06	0.09466 ± 0.001492	3.415 ± 0.2956
14	MOPSO	1.182 ± 0.0006447	0.01421 ± 0.002302	0.02041 ± 0.002691	2.027 ± 0.3936
15	MOSA	1.181 ± 0.0003664	0.01318 ± 0.0007081	0.01876 ± 0.001253	1.553 ± 0.1031
16	MOABC	1.181 ± 0.0004952	0.01087 ± 0.0008006	0.01898 ± 0.001757	12.08 ± 0.5502
17	NPGA	1.181 ± 0.0008078	0.01335 ± 0.00159	0.01876 ± 0.001853	6.839 ± 0.5114
18	PESA-II	1.181 ± 0.0005926	0.01697 ± 0.002015	0.01689 ± 0.002372	6.404 ± 0.3942
19	Benson reference improvement	1.181 ± 1.037e-07	0.2131 ± 1.002e-06	0.2052 ± 0.003441	2.257 ± 0.4278
20	Weighted Sum	1.181 ± 1.037e-07	0.2131 ± 1.003e-06	0.2057 ± 0.002616	0.6166 ± 0.08605
21	MODE	1.181 ± 0.002083	0.01381 ± 0.003286	0.01812 ± 0.003533	2.17 ± 0.224
22	VEGA	1.18 ± 0.001236	0.01266 ± 0.001117	0.01672 ± 0.002154	3.383 ± 0.1219
23	R-NSGA-II	1.176 ± 0.01429	0.01448 ± 0.003167	0.01749 ± 0.003844	4.784 ± 0.3045
24	IBEA	1.162 ± 0.01134	0.02208 ± 0.01418	0.01695 ± 0.01816	11.85 ± 0.8034
25	Epsilon-Constraint	1.134 ± 1.97e-08	0.2455 ± 1.942e-07	0.545 ± 0.04585	2.939 ± 0.1895

Ranked by mean HV: MOEA/D leads at 1.182 ± 7.646e-06, compared with Epsilon-Constraint at 1.134 ± 1.97e-08. HV and IGD judge different aspects; this ranking is specific to the fixed scales and reference. All rows have ten trials. Classical actual calls differ despite the common 10,000 cap; see raw_experiments.csv.

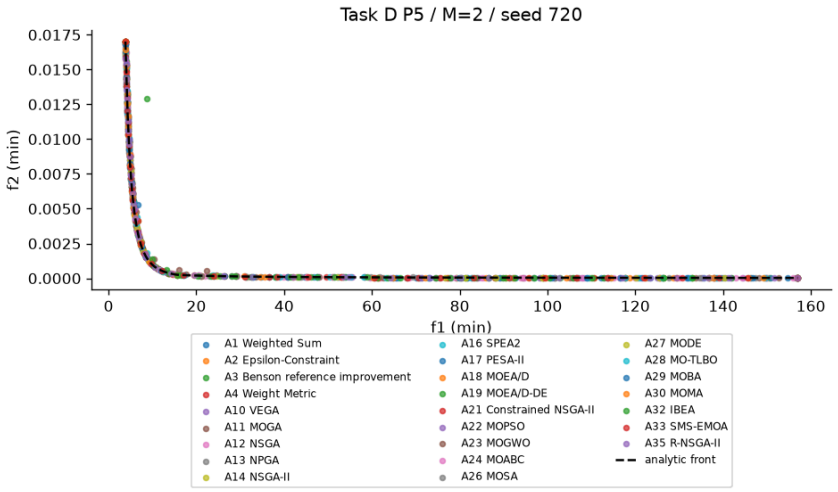


Figure DP5 Combined seed 720 results. The plot shows the recorded nondominated objective vectors for the stated problem and seed.

P10 comparison

Rank	Algorithm	HV	IGD	Spacing	Runtime s
1	RVEA	1.012 ± 0.03024	0.3111 ± 0.008554	0.2684 ± 0.02503	4.987 ± 0.3237
2	MOEA/D	0.9954 ± 0.03858	0.3212 ± 0.02067	0.2613 ± 0.0299	19.64 ± 0.8166
3	NSGA-III	0.9883 ± 0.01881	0.3083 ± 0.006507	0.2831 ± 0.02087	5.311 ± 0.2782
4	HypE	0.8057 ± 0.1473	0.4362 ± 0.07524	0.3592 ± 0.07552	23.95 ± 1.894
5	NSGA-II	0.2132 ± 0.1465	0.5167 ± 0.1017	0.3857 ± 0.04576	4.186 ± 0.2219

Ranked by mean HV: RVEA leads at 1.012 ± 0.03024, compared with NSGA-II at 0.2132 ± 0.1465. HV and IGD judge different aspects; this ranking is specific to the fixed scales and reference. All rows have ten trials. All candidate budgets are 10,000.

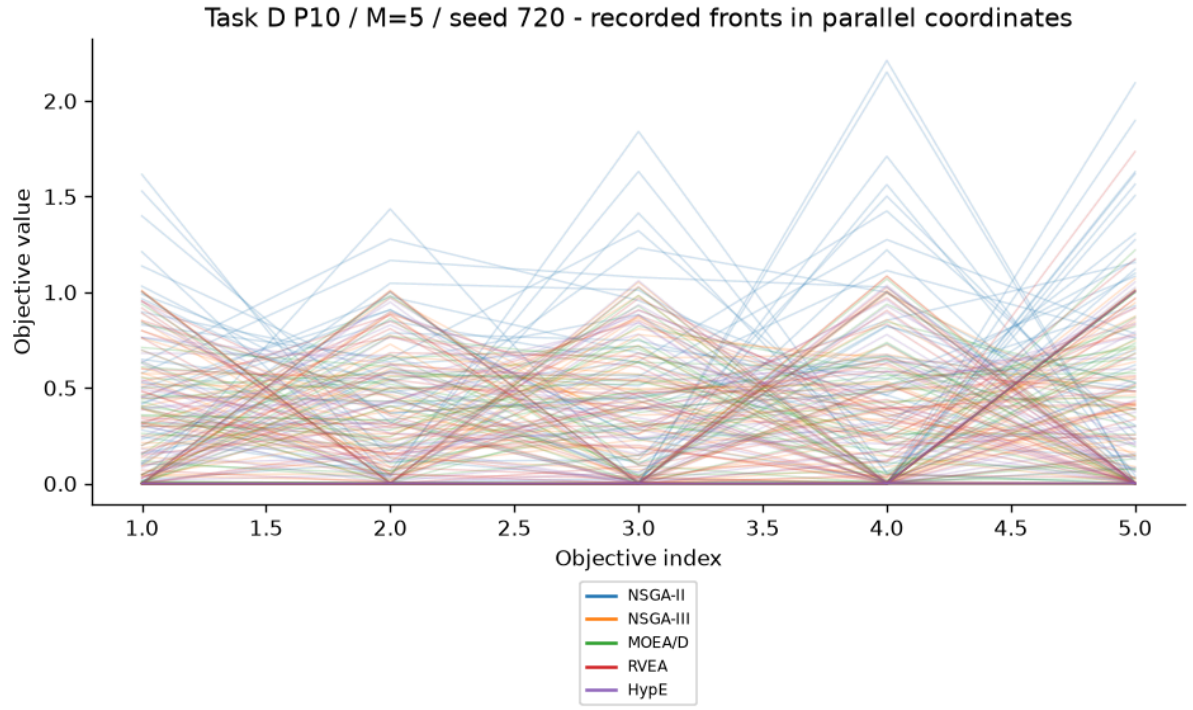


Figure DP10 Combined seed 720 results. The overlaid parallel-coordinate lines show every recorded five-objective front vector; this projection is descriptive rather than a dominance ordering.

M=3 random population nondominated fraction mean=0.318. Increasing nondomination reduces rank discrimination, while vector association adds a directional criterion.

M=5 random population nondominated fraction mean=0.588. Increasing nondomination reduces rank discrimination, while vector association adds a directional criterion.

P8 comparison

Rank	Algorithm	HV	IGD	Spacing	Runtime s
1	MOACO	0.9239 ± 0	N/A	0.003111 ± 0.0008203	17.96 ± 1.854
2	NSGA-II	0.9235 ± 0.001167	N/A	0.003856 ± 0.001973	5.874 ± 0.2541
3	MOEA/D-ACO	0.9235 ± 0.0007362	N/A	0.003604 ± 0.001162	26.15 ± 0.9772

Rank	Algorithm	HV	IGD	Spacing	Runtime s
4	MOMA	0.9235 ± 0.001163	N/A	0.003772 ± 0.003191	7.618 ± 0.5105

Ranked by mean HV: MOACO leads at 0.9239 ± 0, compared with MOMA at 0.9235 ± 0.001163. HV and IGD judge different aspects; this ranking is specific to the fixed scales and reference. All rows have ten trials. All candidate budgets are 10,000.

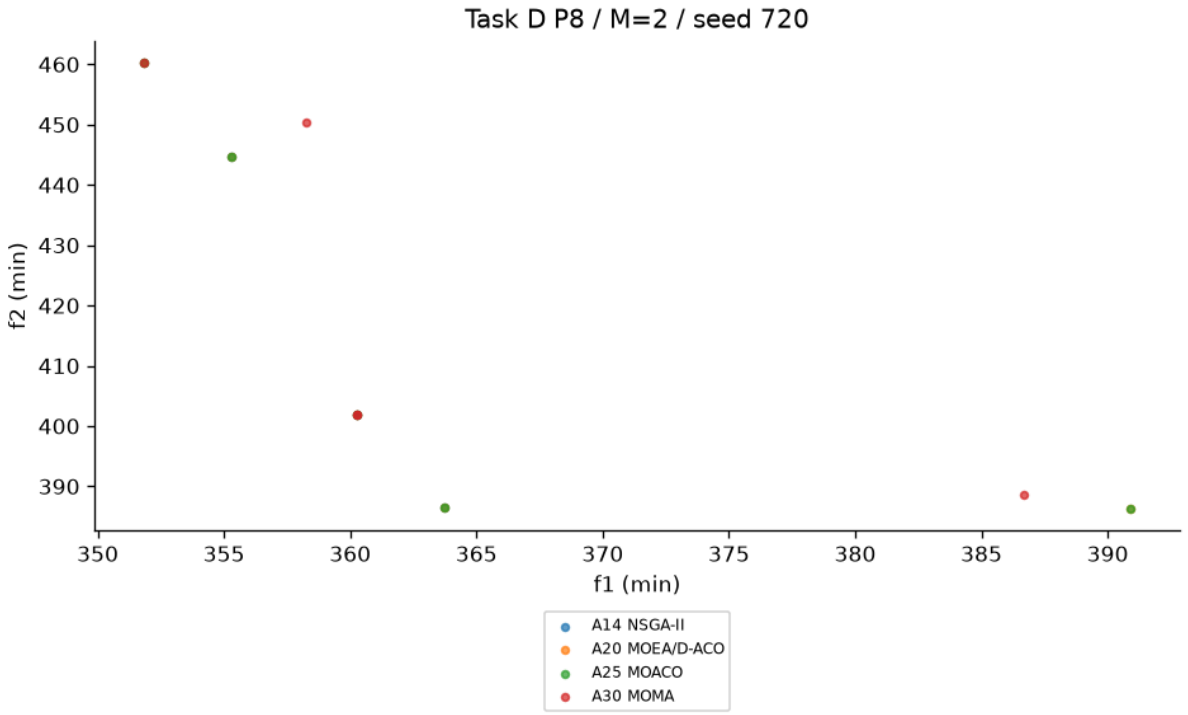


Figure DP8 Combined seed 720 results. The plot shows the recorded nondominated objective vectors for the stated problem and seed.

Nearest neighbour	Distance	Time	Tour
distance	362	521.3	[0, 7, 1, 13, 5, 10, 14, 2, 12, 9, 3, 6, 8, 4, 11]
time	577.5	581.1	[0, 1, 13, 5, 10, 11, 8, 4, 7, 9, 12, 14, 3, 6, 2]

Distance/time baselines use different edge metrics and retain valid 15-city permutations. A good optimized archive can improve one objective while worsening the other, so neither baseline should be assumed to dominate the entire tradeoff set.

P3 population and generation budget investigation

Population	Budget	Full generations	HV	IGD	Spacing
20	1000	49	0.442 ± 0.09599	0.08172 ± 0.09008	0.03294 ± 0.01028
20	5000	249	0.5127 ± 0.001579	0.02481 ± 0.001714	0.03422 ± 0.004984
20	10,000	499	0.5128 ± 0.002392	0.02502 ± 0.002066	0.03765 ± 0.009442
40	1000	24	0.3977 ± 0.1308	0.1217 ± 0.1353	0.03983 ± 0.02274
40	5000	124	0.5285 ± 0.0009619	0.01188 ± 0.0007234	0.01688 ± 0.002303
40	10,000	249	0.5286 ± 0.001001	0.01208 ± 0.001244	0.01765 ± 0.001747

Population	Budget	Full generations	HV	IGD	Spacing
80	1000	11	0.0888 ± 0.0843	0.4429 ± 0.1897	0.1157 ± 0.1474
80	5000	61	0.5358 ± 0.0002164	0.005863 ± 0.0001773	0.008493 ± 0.0009284
80	10,000	124	0.5362 ± 0.0002538	0.005832 ± 0.0001696	0.008031 ± 0.000386

This additional NSGA-II study uses ten seeds for each of nine population and budget settings. At fixed budget, larger populations receive fewer generations; improving population coverage can compete with convergence. More evaluations generally allow more refinement, but the measured IGD values determine the conclusion for each setting rather than an assumed universal trend. Ninety raw fronts and objective metrics are stored under results/roll72/csv/population/.

Ranking existing alternatives and preferred scalar solutions

Input	Method	Weights	v	Top three
P12	topsis	[0.3, 0.3, 0.2, 0.2]	0.5	[3, 1, 5]
P12	topsis	[0.4, 0.2, 0.2, 0.2]	0.5	[1, 2, 3]
P12	vikor	[0.3, 0.3, 0.2, 0.2]	0.25	[1, 2, 5]
P12	vikor	[0.4, 0.2, 0.2, 0.2]	0.25	[2, 1, 4]
P12	vikor	[0.3, 0.3, 0.2, 0.2]	0.5	[1, 2, 5]
P12	vikor	[0.4, 0.2, 0.2, 0.2]	0.5	[2, 1, 3]
P12	vikor	[0.3, 0.3, 0.2, 0.2]	0.75	[1, 3, 2]
P12	vikor	[0.4, 0.2, 0.2, 0.2]	0.75	[1, 2, 3]
P4 NSGA-II front	topsis	[0.5, 0.5]	0.5	[18, 22, 2]
P4 NSGA-II front	vikor	[0.5, 0.5]	0.5	[2, 38, 35]
P4 NSGA-II front	topsis	[0.6, 0.4]	0.5	[32, 8, 3]
P4 NSGA-II front	vikor	[0.6, 0.4]	0.5	[8, 32, 3]

P12 supplier identifiers are 1-based. P4 front identifiers are 0-based rows in P4_nsga2_ranking_input.csv. Moving .1 weight to the first criterion and subtracting .1 from the second preserves the simplex. VIKOR v=.25,.5,.75 and its compromise-set flags remain in rankings.csv.

TOPSIS top-three order changes under the .1 weight shift: [3, 1, 5] -> [1, 2, 3]. This measured local response is not universal preference stability.

VIKOR top-three order changes under the .1 weight shift: [1, 2, 5] -> [2, 1, 3]. This measured local response is not universal preference stability.

Value Function returns declared portfolio profiles or the laptop aspiration. Goal Programming uses production targets 2400 profit and 600 units, and normalized versus unit deviations. Lexicographic profit-first then units selects the profit tie with maximum units, approximately (300,200). Exact decisions and solver success messages are retained in the classical job JSON records; a solver failure is not silently treated as proof of optimality.

P13 gradient methods and equal weights

Method	Scale	BCE initial	BCE final	MSE initial	MSE final
equal	1	0.6962	0.5258 ± 0.01498	14.92	0.1622 ± 0.005318
mgda	1	0.6962	0.2795 ± 0.01688	14.92	14.08 ± 0.3358
gradnorm	1	0.6962	0.3349 ± 0.02439	14.92	0.1584 ± 0.01261
equal	10	0.6962	0.6382 ± 0.02039	1487	83.82 ± 1.539
mgda	10	0.6962	0.6747 ± 0.008354	1487	1486 ± 2.45
gradnorm	10	0.6962	0.5927 ± 0.01521	1487	1108 ± 64.26

The network has a five-input shared tanh layer of width eight and separate classification and regression heads. Ten seeds vary initialization while synthetic observations remain fixed. Regression target scale 10 creates larger gradient norms. Equal weights and MGDA sum to one; GradNorm weights sum to two by definition. Their model learning rates are explicitly recorded, so weight normalization changes effective step size and must be considered when interpreting loss differences.

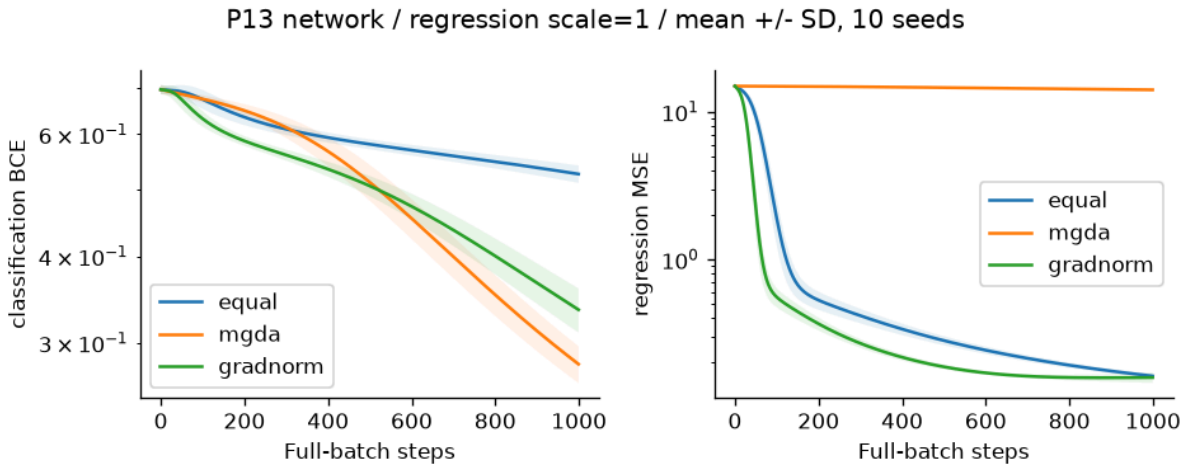


Figure N1 Network task losses with mean and SD from ten initializations

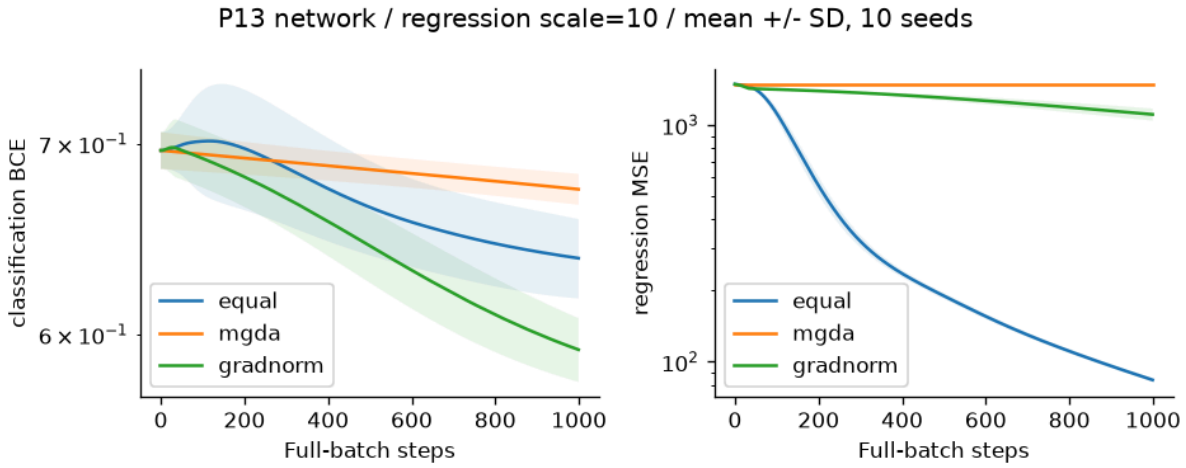


Figure N10 Network task losses with mean and SD from ten initializations

On the toy, fixed equal weights converge to (0,0), while MGDA generally stops at a different Pareto-stationary point and GradNorm changes the task weighting. Individual losses need not

both approach zero: the two target points conflict. The saved 10,000-step histories and toy_equal.csv expose this trajectory rather than claiming full-front coverage from one initial point.

Toy task losses against equal weights

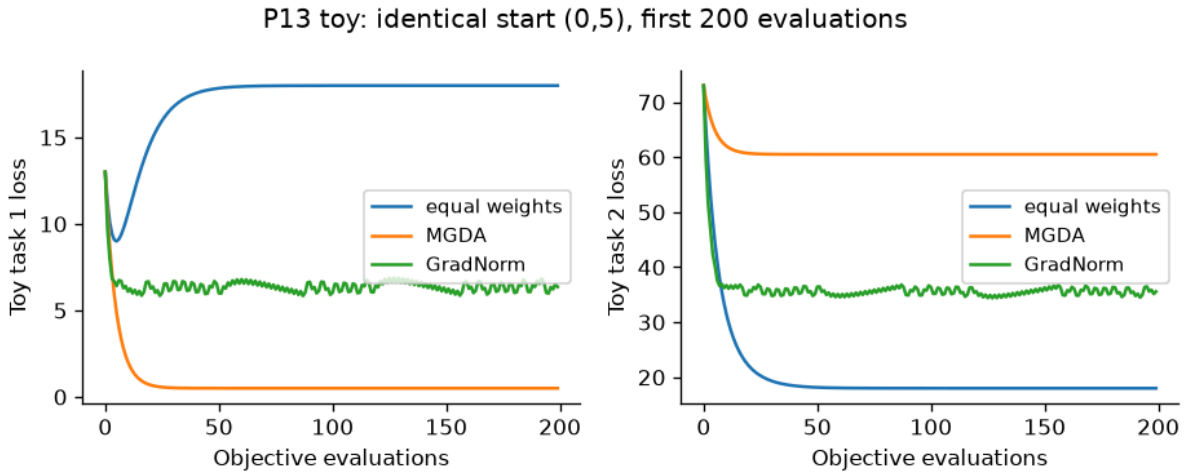


Figure T13 Hand-coded toy loss trajectories against fixed equal weights, with identical starting point and learning rate

The equal-weight toy baseline tends to x=(0,0), giving losses (18,18). MGDA tends to a different point on the Pareto segment, while GradNorm adapts its loss weights. A smaller task 1 loss can coexist with a larger task 2 loss at a Pareto-stationary tradeoff; neither method is expected to make both target distances zero.

8 Task E algorithm selection cases

P1

Suitable: Weighted Sum,Weight Metric. Poor choice: VEGA. Convex one-variable problem; scalarization offers interpretable weights and metric methods expose compromise geometry.

A1 M=2 base: HV=1.0258 ± 0, IGD=0.020135 ± 0; A4 M=2 base: HV=1.0347 ± 0, IGD=0.012339 ± 0

Recommendations combine measured finite-budget outcomes with encoding, constraints and preference reasoning; no universal winner is claimed.

P3

Suitable: SMS-EMOA,MO-TLBO. Poor choice: Weighted Sum. Nonconvex front; hypervolume survival and teacher-learner moves both cover interior tradeoffs in the recorded experiments, while linear scalarization cannot support them.

A28 M=2 D: HV=0.53179 ± 7.0884e-05, IGD=0.0095087 ± 9.0745e-05; A33 M=2 D: HV=0.53181 ± 0.00010904, IGD=0.0095649 ± 0.00012778; A28 M=2 base: HV=0.53179 ± 7.0884e-05, IGD=0.0095087 ± 9.0745e-05; A33 M=2 base: HV=0.53181 ± 0.00010904, IGD=0.0095649 ± 0.00012778

Recommendations combine measured finite-budget outcomes with encoding, constraints and preference reasoning; no universal winner is claimed.

P5

Suitable: Constrained NSGA-II,SPEA2. Poor choice: VEGA. Nonlinear feasibility makes Deb rules useful; archive strength and density support constrained tradeoffs.

A16 M=2 D: HV=1.1818 ± 0.00012094, IGD=0.016079 ± 0.0026026; A21 M=2 D: HV=1.1818 ± 3.9631e-05, IGD=0.014057 ± 0.0012871; A16 M=2 base: HV=1.1818 ± 0.00012094, IGD=0.016079 ± 0.0026026; A21 M=2 base: HV=1.1818 ± 3.9631e-05, IGD=0.014057 ± 0.0012871

Recommendations combine measured finite-budget outcomes with encoding, constraints and preference reasoning; no universal winner is claimed.

P8

Suitable: NSGA-II,MOEA/D-ACO. Poor choice: MGDA. Permutation-safe NSGA-II and ant decomposition have the leading recorded HV means and retain valid tours; gradient descent is undefined for city orders.

A14 M=2 D: HV=0.92354 ± 0.0011666, IGD=N/A; A20 M=2 D: HV=0.92348 ± 0.00073621, IGD=N/A; A20 M=2 base: HV=0.92348 ± 0.00073621, IGD=N/A

Recommendations combine measured finite-budget outcomes with encoding, constraints and preference reasoning; no universal winner is claimed.

P10

Suitable: RVEA,MOEA/D. Poor choice: VEGA. Five objectives weaken dominance; RVEA APD and MOEA/D scalar subproblems retain directional pressure and have the strongest recorded HV means at this budget.

A18 M=5 D: HV=0.99538 ± 0.038577, IGD=0.32121 ± 0.020673; A31 M=5 D: HV=1.0121 ± 0.030242, IGD=0.31113 ± 0.0085541; A18 M=5 base: HV=0.99538 ± 0.038577, IGD=0.32121 ± 0.020673; A31 M=3 base: HV=0.64035 ± 0.011426, IGD=0.10879 ± 0.0032335; A31 M=5 base: HV=1.0121 ± 0.030242, IGD=0.31113 ± 0.0085541

Recommendations combine measured finite-budget outcomes with encoding, constraints and preference reasoning; no universal winner is claimed.

P14

Suitable: R-NSGA-II,Value Function. Poor choice: VEGA. Known manager aspiration makes a preferred region useful; utility or reference clearing expresses it directly.

A5 M=2 base: HV=0.32905 ± 0, IGD=0.35302 ± 0; A35 M=2 base: HV=0.85563 ± 0.00087984, IGD=0.010043 ± 0.00082871

Recommendations combine measured finite-budget outcomes with encoding, constraints and preference reasoning; no universal winner is claimed.

9 Task F viva preparation

1 Why can Weighted Sum miss a dented front?

On P3 the true front is (t,1-t^2). A positive weighted scalar is concave in t, so its minimum occurs at an endpoint; an interior efficient point lacks a supporting linear level set. Changing the weight grid cannot fix this.

2 Why might VEGA drift to extremes?

Each mating-pool segment rewards only one objective. Extreme winners enter repeatedly, while compromise designs receive no Pareto or density protection. Check the final population as well as the passive recording archive before asserting collapse in a particular run.

3 What does elitism guarantee?

Best ranked survivors are selected from parents plus children, so beneficial parents need not disappear just because variation was poor. A finite population still truncates nondominated designs and gives no global-optimality guarantee. NSGA-II/III, SPEA2, PESA-II, decomposition and archive-guided methods retain different forms of memory.

4 Why does crowding weaken with many objectives?

Most candidates are nondominated; many receive boundary bonuses on one of many axes, while marginal nearest-neighbour gaps poorly represent high-dimensional coverage. NSGA-III reference occupancy and RVEA angles replace axis crowding. The measured random-population fractions are in dominance_pressure.csv.

5 Which methods require preferences and what if they are wrong?

Weighted Sum, metric methods and Benson need weights/reference; Value Function needs utility; Goal Programming needs targets; Lexicographic needs priorities; TOPSIS/VIKOR need criterion weights and VIKOR v; R-NSGA-II needs an aspiration. A wrong preference can select a feasible efficient design that is unsuitable for the decision-maker.

6 Explain Deb rules numerically

Lecture A=(3,6),CV=0 and D=(4,5),CV=0 beat B=(1,2),CV=2 and C=(.5,1),CV=5. A,D trade off. B beats C because 2<5. Thus front layers are [A,D],[B],[C], irrespective of the attractive infeasible objectives.

7 How can premature convergence be detected?

Track HV, IGD where exact-front samples exist, final-population spread, archive turnover and decision-variable variance across budget checkpoints and seeds. Flat HV alone may mean saturation, a poor reference box or a single-region stall; inspect more than one diagnostic.

8 Why are TOPSIS and VIKOR ranking methods rather than optimizers?

Their input is an existing alternatives matrix. They compute closeness or regret and reorder rows; they never choose new decision vectors or explore bounds.

Code Walkthrough Preparation

A10 VEGA

In src/algorithms/vega.py, change the population default from 40 to 20 while retaining the 10,000-evaluation cap. Predict twice as many approximate generations but half as many simultaneous candidates. Objective-wise tournaments remain unchanged; coverage can become less stable even if later generations improve convergence. The existing N=20,40,80 sensitivity rows test this tradeoff.

A22 MOPSO

In src/algorithms/mopso.py, change inertia from 0.4 to 0.8 in the velocity call. The retained w*v term doubles, so particles preserve momentum longer, explore farther and can overshoot

leaders or hit clipping more often. Personal-best and archive dominance rules do not change. The stored w=0.2,0.4,0.8 P4 study provides the numerical comparison.

A35 R-NSGA-II

In src/algorithms/r_nsga2.py, increase epsilon in preference selection from 0.05 to 0.10. Candidates must then be farther apart in normalized objective space to survive clearing; more near-reference points are deferred and distance-ordered backfill becomes more likely. Nondominated fronts and reference distances themselves do not change. The stored epsilon=0.02,0.05,0.10 P14 study tests the effect.

10 Findings and limitations

The numerical comparison shows meaningful differences in convergence, coverage and computation across methods, but finite outputs do not establish global optima. Weighted Sum on P3 illustrates a structural unsupported-region limit. Many-objective methods preserve directional niches when ranks tie. Routing methods require valid edge/tour operations. Preference and gradient methods should be judged by their declared decision or learning task rather than by full-front HV alone.

Remaining work is explicit: P9 uses 200 evaluations per run rather than the suggested 10,000, and its smallest observed feature set is an exploratory result. The Task B table contains four argued advantages and disadvantages plus traceable measured metrics, but targeted ablations do not yet establish two experiment-supported benefits and drawbacks for every method. This report does not claim that stricter evidence requirement is fully satisfied. Student explanation and personal conclusions remain necessary for the viva.

Problem	Investigation	Measured value	Interpretation
P2	VEGA real	40	distinct recorded Pareto points at 10,000 evaluations
P2	VEGA integer	10	distinct recorded Pareto points at 10,000 evaluations
P5	random infeasible fraction	0.0255	10,000 independent uniform box designs; same formulas
P5	ignore stress minimum mass	1.791	feasible-front minimum 4.03417; random-design ablation, not an optimizer run
P4	raw equality feasible fraction	0	repair feasible fraction=1.000; penalty=z1+z2+100*violation
P9	all-features 5NN baseline error	0.04042	smallest observed mask within +.01 error: 5; exploratory 200-evaluation runs, not global optimum
P14	aspiration at 12h analytic front weight	1.691	at x1=13, battery=53.4Wh yields 1.6908kg; aspiration (1.8,12h) is dominated
P6	second-constraint minimum left side	18	at (5,0), 9+9=18>7.7; inactive across full box

Problem	Investigation	Measured value	Interpretation
P7	reference feasible allocation	91.67	91.6667>90; exact (40,50) output unattainable

Additional limitations: fixed population 40 limits front resolution; sparse grids and reference lattices can bias irregular fronts; Monte Carlo selection noise is larger than final HV evaluation noise; runtimes measured in a process pool are environment specific; IGD depends on reference sample density; exact fronts for P4 P8 P9 remain unknown here. The shared output archive can mask nonelitist final-population losses, so saved final_f is essential for diagnosing collapse.

11 Reproducibility and repository instructions

Reproduction: keep src/, tests/, results/, figures/, docs/ and data/ together. Install requirements.txt, run python run_all.py test and python run_all.py verify. To rebuild the Word report using saved results without rerunning optimization, use python -m src.experiments.docx_report. The README contains the experimental rerun instructions.

GitHub repository: https://github.com/sofiaabidi/OT_LAB07. The README gives individual algorithm commands and selective reruns. This editable DOCX is the requested report deliverable; no PDF or submission ZIP was generated.

12 References

1. Department of Artificial Intelligence & Data Science (2026). Lab Assignment 7: Multiobjective Optimization - 37 Algorithms, Trade-offs and Applications. Course assignment, issued 5 October 2026.
2. Chapter 5 course lecture materials (provided with this assignment): Multiobjective Optimization, Parts 1-4 and Bonus Module 5E. Source for the teaching variants and lecture-aligned implementations.
3. Schaffer, J. D. (1985). Multiple Objective Optimization with Vector Evaluated Genetic Algorithms. Proceedings of the First International Conference on Genetic Algorithms, pp. 93-100. Publication record
4. Coello Coello, C. A., & Lechuga, M. S. (2002). MOPSO: A Proposal for Multiple Objective Particle Swarm Optimization. Proceedings of the IEEE Congress on Evolutionary Computation, vol. 2, pp. 1051-1056. DOI: 10.1109/CEC.2002.1004388
5. Deb, K., Sundar, J., Udaya, B. N., & Chaudhuri, S. (2006). Reference Point Based Multi-Objective Optimization Using Evolutionary Algorithms. International Journal of Computational Intelligence Research, 2(3), 273-286. DOI: 10.5019/j.ijcir.2006.67
6. Horn, J., Nafpliotis, N., & Goldberg, D. E. (1994). A Niched Pareto Genetic Algorithm for Multiobjective Optimization. Proceedings of the First IEEE Conference on Evolutionary Computation, pp. 82-87. DOI: 10.1109/ICEC.1994.350037
7. Serafini, P. (1994). Simulated Annealing for Multi Objective Optimization Problems. In Multiple Criteria Decision Making: Expand and Enrich the Domains of Thinking and Application, pp. 283-292. Author manuscript

13 Appendix: Task B master comparison

All 37 methods are listed in catalogue order. Each entry identifies its family, defining mechanism, four advantages, four disadvantages, a suitable use and a less suitable setting. The numeric evidence for each implementation is in Task A and the comparative experiments in Task D.

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
1 Weighted Sum (Classical)	Linear scalar solve and a sweep	1. Simple scalar objective; 2. Direct preference weights; 3. Reusable constrained scalar solver; 4. Independent solves can be distributed	1. Unsupported nonconvex points are unreachable; 2. Scaling changes weight meaning; 3. Uniform weights give nonuniform spacing; 4. Local scalar solves lack global guarantees	Convex smooth two-objective design with interpretable weights	Dented fronts and an unknown preference distribution
2 Epsilon-Constraint (Classical)	One objective minimized and others bounded	1. Can represent unsupported efficient points; 2. Epsilon has physical interpretation; 3. Scalar constraints remain explicit; 4. Primary objective stays recognizable	1. Infeasible epsilon values waste solves; 2. Sweeps can duplicate points; 3. Constraint solver quality matters; 4. Many objectives require multidimensional epsilon grids	Nonconvex constrained design with hard limits	High-dimensional objective grids
3 Benson reference improvement (Classical)	Weighted improvement below a reference	1. Reference conveys aspiration; 2. Nonnegative improvement is interpretable; 3. Explicit upper objective bounds; 4. Scalar subproblems are easy to inspect	1. This is the lecture achievement form, not outer approximation; 2. Unrestricted reference sweeps inherit weighted-sum barrier; 3. Reference coordinates require scaling; 4. Weights still encode preferences	Teaching reference improvement on P1/P7	Claiming a general Benson convergence theorem

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
4 Weight Metric (Classical)	Weighted distance to fixed ideal	1. Infinity norm can reach unsupported tradeoffs; 2. Different p values express balance; 3. Ideal point makes distances interpretable; 4. Scalar solver is reusable	1. Ideal estimation affects solution; 2. Max norm is nonsmooth; 3. p and weights require preferences; 4. Finite weight sweep misses narrow regions	Compromise solutions on nonconvex fronts	Unreliable ideal estimates or many weights
5 Value Function (Classical)	Declared decision-maker utility	1. Directly encodes a preferred outcome; 2. Can express nonlinear preferences; 3. Few scalar solves are economical; 4. Natural targets communicate decisions	1. Utility must be elicited; 2. Squared target utility need not be monotone; 3. One preferred design is not a full front; 4. Misspecified preferences bias results	Laptop or portfolio profile decisions	Exploration without known preferences
6 Goal Programming (Classical)	Explicit positive/negative goal deviations	1. Shows unavoidable shortfall; 2. Can normalize incompatible units; 3. Soft goals remain interpretable; 4. Constraint and aspiration roles are separate	1. Goal weights control tradeoffs; 2. Infeasible goals cannot be guaranteed; 3. Units can overwhelm deviations; 4. Satisfied goals may leave efficiency ties	Production targets with soft priorities	Searching an entire front without goals
7 Lexicographic (Classical)	Sequential strict priorities with tolerances	1. Priority order is transparent; 2. Breaks first-objective ties; 3. Uses scalar solvers; 4. Preserves prior objective values to declared tolerance	1. Lower objectives may have little freedom; 2. Tolerance changes strictness; 3. Priority mistakes are expensive; 4. Sequential solves add cost	Profit first and units second	Balanced objectives with no hierarchy

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
8 TOPSIS (Ranking)	Closeness to weighted ideal and anti-ideal	1. Fast finite-table ranking; 2. Uses all criteria; 3. Distances are explainable; 4. Works with benefit and cost columns	1. Does not generate new designs; 2. Vector normalization affects rank; 3. Compensation can hide a bad criterion; 4. Alternatives can change the ideal	Supplier or existing-front ranking	Optimization without an alternatives table
9 VIKOR (Ranking)	Compromise via group regret and worst regret	1. Worst criterion enters R; 2. v exposes group versus individual tradeoff; 3. Compromise conditions avoid forced winners; 4. Finite rankings are inexpensive	1. Requires preference weights; 2. v can change rank; 3. Range normalization changes regrets; 4. Does not generate designs	Supplier compromise with severe weak criteria	A Pareto front generator
10 VEGA (Early evolutionary)	Objective-partitioned selection	1. Very simple selection; 2. No preference weights needed; 3. Handles several objective contests; 4. Cheap scalar tournaments	1. Extremes can dominate mating pool; 2. No parent elitism; 3. Middle tradeoffs may disappear; 4. Objective partition sizes affect pressure	Teaching objective competition	Reliable uniform front preservation
11 MOGA (Early evolutionary)	Domination-count rank and fitness sharing	1. No scalar utility required; 2. Sharing promotes sparse niches; 3. Dominance counts are easy to inspect; 4. Can search unsupported regions	1. No elitism; 2. Sharing radius sensitive; 3. All nondominated members tie in rank; 4. Pairwise density costs quadratically	Small bi-objective populations	Many objectives with sparse dominance
12 NSGA (Early evolutionary)	Front layers and within-front sharing	1. Explicit Pareto hierarchy; 2. Sharing is local to fronts; 3. No objective weights required; 4. Variation supports arbitrary encodings	1. No elitism; 2. Quadratic sorting and sharing; 3. Dummy fitness spacing is arbitrary; 4. Niche radius must be tuned	Teaching early Pareto GAs	Large fast convergence requirements

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
13 NPGA (Early evolutionary)	Direct contestant dominance then separate comparison-set quadratic niche tie	1. No fixed utility weights; 2. Niche tie-break preserves sparse points; 3. Comparison sample controls work; 4. Can visit nonconvex tradeoffs	1. Nonelitist generations lose parents; 2. Small comparison set gives noisy niche counts; 3. Sigma changes density pressure; 4. Lecture tournament differs from original publication	Bi-objective diversity experiments	Many-objective fronts where most tournaments tie
14 NSGA-II (Elitist)	Front rank, crowding, parent-child elitism	1. Elitist survivor retention; 2. Boundary points receive protection; 3. No sharing radius; 4. Permutation and real variation share selection	1. Crowding weakens as objectives grow; 2. Quadratic pairwise sorting; 3. Finite population truncates front; 4. Crowding is not an explicit convergence measure	General constrained two-objective search	Very many objectives with weak dominance
15 NSGA-III (Elitist)	Reference-point niching after front rank	1. Reference niches control coverage; 2. Supports many objectives; 3. Occupancy encourages unused directions; 4. Does not use a sharing radius	1. Range normalization simplifies original intercepts; 2. Reference count grows combinatorially; 3. Poor reference geometry biases coverage; 4. Dominance layers still require quadratic work	DTLZ2 with three or five objectives	Irregular fronts without adapted references
16 SPEA2 (Elitist)	Strength, raw fitness and neighbour density	1. Strength combines domination information; 2. Density distinguishes ties; 3. Fixed archive bounds memory; 4. Lexicographic truncation considers nearest distances	1. Quadratic distance storage; 2. Repeated truncation adds cost; 3. k-neighbour density depends on scale; 4. Archive size limits attainable resolution	Small smooth front archives	Large populations with tight memory

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
17 PESA-II (Elitist)	Archive region occupancy selection	1. Region selection directly rewards sparsity; 2. Archive keeps observed tradeoffs; 3. Grid counts are interpretable; 4. Less dependence on precise distances	1. Grid divisions sensitive; 2. High-dimensional grids become sparse; 3. Archive can shrink to few members; 4. Grid boundaries change selection abruptly	Bi-objective region coverage	Many objectives with almost unique cells
18 MOEA/D (Decomposition)	Neighbouring Tchebycheff subproblems	1. Cooperation shares good designs; 2. Scalar subproblems retain pressure with many objectives; 3. Neighbourhood limits update work; 4. Weights provide coverage plan	1. Reference geometry biases front; 2. Neighbourhood size affects spread; 3. Shared donors can homogenize subproblems; 4. Ideal updates alter scalar values	Smooth continuous many-objective design	Disconnected fronts badly matched by weights
19 MOEA/D-DE (Decomposition)	DE variation inside MOEA/D	1. DE exploits vector differences; 2. Neighbour cooperation; 3. Tchebycheff reaches nonconvex fronts; 4. Forced crossover guarantees a changed coordinate	1. F and CR need tuning; 2. DE requires enough distinct neighbours; 3. Boundary clipping distorts donors; 4. Weight coverage may be uneven	Continuous real-coded nonconvex design	Permutation encodings without a new donor operator
20 MOEA/D-ACO (Decomposition)	Weight-specific ant pheromones and scalar replacement	1. Valid tours by construction; 2. Heuristic directs ant choices; 3. Separate subproblems retain tradeoffs; 4. Pheromone records good edges	1. Pheromone arrays cost N*cities^2; 2. Deposits can lock routes; 3. Alpha beta evaporation matter; 4. Continuous encodings unsupported	Bi-objective routing	Smooth gradient problems

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
21 Constrained NSGA-II (Constrained)	NSGA-II with Deb feasibility	1. Feasible beats infeasible without objective penalty; 2. Lower violation guides infeasible search; 3. Pareto ranking within feasible; 4. Crowding retains feasible diversity	1. Violation normalization affects infeasible order; 2. Feasibility boundary can trap search; 3. Quadratic ranking; 4. Finite crowding resolution	Constrained beam or Binh-Korn	Problems needing stochastic feasibility guarantees
22 MOPSO (Swarm/local)	Particle velocity with personal and archive leaders	1. Memory of personal successes; 2. Few basic coefficients; 3. Archive leader spreads movement; 4. No crossover operator needed	1. Velocities can collapse; 2. Boundary clipping affects dynamics; 3. Leader policy controls diversity; 4. Coefficients require tuning	Continuous portfolio and beam designs	Permutation tours without discrete velocity
23 MOGWO (Swarm/local)	Three archive leaders and encircling	1. Multiple leaders guide movement; 2. Coefficient shrinks automatically; 3. Archive covers tradeoffs; 4. Binary transfer enables masks	1. Averaging leaders may erase diversity; 2. No strict improvement guarantee; 3. Transfer sigmoid biases masks; 4. Binary objective evaluation is costly	Continuous design or exploratory feature masks	Required stable high-accuracy feature conclusions on tiny budgets
24 MOABC (Swarm/local)	Employed moves, crowding-based onlookers and abandonment scouts	1. Scouts restart abandoned sources; 2. Onlooker pressure favours sparse regions; 3. Coordinate moves are inspectable; 4. No gradients needed	1. Dominance-only greed can reject useful tradeoffs; 2. Limit needs tuning; 3. Three phases complicate budget accounting; 4. One-coordinate changes may be slow	Small continuous design with restarts	Strongly coupled high-dimensional variables

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
25 MOACO (Swarm/local)	Two pheromone matrices and route construction	1. Produces valid permutations; 2. Two objectives retain separate memories; 3. Heuristic reduces random route work; 4. Archive deposits reward tradeoffs	1. Strong evaporation forgets edges; 2. Weak evaporation stagnates; 3. Traffic/distance scales matter; 4. No exact TSP optimality guarantee	Distance-time routing	Continuous portfolio allocation
26 MOSA (Swarm/local)	Incomparable neighbour acceptance and Boltzmann dominated moves	1. May accept uphill escape moves; 2. Works without derivatives; 3. Mutation adapts to encoding; 4. Archive records tradeoffs from many chains	1. Cooling schedule sensitive; 2. Local moves can miss distant regions; 3. Incomparable moves remain random at low temperature; 4. Finite cooling gives no global guarantee	Budgeted local exploration on nonconvex fronts	Uniform global coverage with insufficient chains
27 MODE (Swarm/local)	DE dominance-only trial replacement	1. Simple vector donor; 2. No utility weights; 3. CR forces one donor coordinate; 4. Dominating trials preserve target improvement	1. Incomparable trials do not replace targets; 2. F CR sensitivity; 3. Coordinate clipping can stagnate; 4. Only real encoding is implemented	Continuous design with comparable improvements	Many objectives with almost no domination
28 MO-TLBO (Swarm/local)	Crowding archive teacher, shrinking TF and nondominated learner moves	1. No crossover design needed; 2. Teacher encourages exploitation; 3. Learners use peer information; 4. Shrinking TF changes search scale	1. Teacher choice depends on archive; 2. Incomparable moves can wander; 3. Population means blur disconnected regions; 4. TF schedule differs from conventional TLBO	Small continuous educational examples	Highly multimodal separated feasible regions

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
29 MOBA (Swarm/local)	Frequency velocity, pulse and loudness	1. Few interpretable schedules; 2. Local walks exploit leaders; 3. Frequency creates long moves; 4. Archive records candidate tradeoffs	1. Loudness decay suppresses replacement; 2. Velocity can explode then clip; 3. Pulse schedule sensitive; 4. Archive quality may exceed final-chain quality	Continuous local refinement	Robust full-front exploration without schedule tuning
30 MOMA (Hybrid)	NSGA-II plus sparse-child combined-score hill climbing	1. Local moves refine global children; 2. Elitist survival; 3. Works with binary/tour mutation; 4. Local cost explicitly counted	1. Local calls reduce global generations; 2. Combined-score preference changes tradeoffs; 3. Mutation neighbourhood determines refinement; 4. Feature CV makes local search costly	Beam refinement or routing	Expensive objectives with no local structure
31 RVEA (Many-objective)	Normalized reference vectors and lecture linear-progress APD	1. Angular niches aid many objectives; 2. APD shifts toward convergence; 3. Vector adaptation tracks scales; 4. Sparse niches can retain representatives	1. Penalty strength and adaptation period matter; 2. Observed ranges can be unstable; 3. Empty niches require fill policy; 4. Reference alignment can bias coverage	Five-objective DTLZ2	Strongly irregular disconnected fronts
32 IBEA (Indicator)	Outgoing additive-epsilon fitness and largest-fitness deletion	1. No separate sharing operator; 2. Indicator gives selection pressure; 3. Works on nonconvex fronts; 4. Kappa controls discrimination	1. Exponentials can saturate; 2. Kappa changes selection; 3. Recalculated deletion is costly; 4. Fitness depends strongly on scales	Small bi-objective indicator experiments	Large populations with limited CPU

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
33 SMS-EMOA (Indicator)	Steady-state smallest HV deletion	1. Direct quality indicator survival; 2. One-offspring updates are simple; 3. Exact two-objective contribution; 4. Protects coverage within the chosen reference	1. HV reference controls preferences; 2. One-at-a-time sorting costs many calls; 3. Implemented only for two objectives; 4. HV can prioritize extremes unevenly	Small bi-objective design	Many objectives without an HV approximation
34 HypE (Indicator)	Equal-share Monte Carlo HV fitness and repeated deletion	1. HV sharing supports many objectives; 2. Shared samples account for overlapping domination; 3. Re-estimation updates remaining credit; 4. Final evaluation samples are identical across methods	1. Sampling noise changes survivors; 2. Selection sample count is small; 3. Repeated deletion costs CPU; 4. HV still needs a reference box	Many-objective indicator study	High-precision survival contribution estimates
35 R-NSGA-II (Preference)	Reference preference and epsilon clearing	1. Focuses on desired region; 2. Clearing promotes local diversity; 3. Retains nondominated hierarchy; 4. Preference is physically interpretable on P14	1. Wrong aspiration biases search; 2. Epsilon controls local spacing; 3. Full-front quality can drop; 4. Default balanced aspiration is arbitrary	Decision-maker laptop region	Unknown or frequently changing preferences
36 MGDA (Gradient)	Minimum-norm gradient combination	1. Common descent when norm is nonzero; 2. Analytical two-task weights; 3. No objective-scale utility required; 4. Can converge to Pareto stationarity	1. Requires gradients; 2. Stationarity is not full-front coverage; 3. May stop while each task has gradient; 4. Nonconvex network optima are local	Differentiable conflicting tasks	Discrete designs or a full Pareto-front sweep

Algorithm / family	Main mechanism	Four advantages	Four disadvantages	Suitable	Avoid
37 GradNorm (Gradient)	Gradient-norm adaptive task weights	1. Balances task gradient magnitudes; 2. Training rates influence targets; 3. Hand-derived update is inspectable; 4. Task-scale stress can be evaluated	1. Requires gradients and initial losses; 2. Alpha weight_lr matter; 3. Does not guarantee common descent; 4. Toy final point is not an entire front	Shared network with unequal loss gradients	A guaranteed Pareto descent method